

133 Clear Decisions Unified AI System for Structured and Unstructured Data

FiftyFive built a unified AI system integrating RAG and NL2SQL, enabling seamless structured and document-based data access.

Industry -

Services - Custom Software Development

AI/ML Engineering

Data Engineering

System Architecture

About the Client

The client is an enterprise organization seeking a scalable AI-driven data access solution to unify document-based knowledge and structured database records for faster analytics and operational decision-making.

The Story

The client engaged FiftyFive for a full-cycle custom software development engagement to build a unified AI query system. The objective was to enable seamless access to both unstructured PDF documents and structured PostgreSQL data through a single natural language interface. Initially designed as an agentic AI architecture, the solution was restructured into a function-based orchestration model to align with delivery timelines and MVP goals. FiftyFive handled architecture design, AI engineering, orchestration logic, and backend integration.

Challenges

- Integrating unstructured document retrieval with structured database querying in a single system.
- Implementing semantic search over PDF documents using vector embeddings.

- Translating natural language queries into executable SQL statements.
- Designing dynamic query classification without stateful agent memory.
- Reducing latency and compute overhead while maintaining response accuracy.
- Delivering a production-ready MVP within strict time constraints.

The Solution

- Built a Retrieval-Augmented Generation (RAG) pipeline using Vertex AI [text-embedding-004](#) for embedding generation.
- Implemented Milvus as the vector database for semantic chunk storage and retrieval.
- Integrated Gemini Pro for document re-ranking and response generation.
- Developed an NL2SQL module using Gemini Pro to convert natural language into PostgreSQL queries.
- Executed SQL queries directly on PostgreSQL and returned structured tabular outputs with explanations.
- Designed a lightweight function-based orchestration layer using prompt-engineered classification logic to route queries between RAG and SQL pipelines.
- Replaced planned agentic architecture with modular function orchestration to reduce system complexity and improve deployment speed.

The Outcome

FiftyFive delivered a unified AI-powered query system that reduced architectural complexity, minimized compute overhead, and accelerated MVP deployment. The

function-based orchestration improved query response speed by eliminating agent memory overhead, enabling faster data access across structured and unstructured sources. The modular architecture supports future agent integration, providing scalability while maintaining operational efficiency and cost control.

132 sorai AI Governance Sandbox for Credit Risk

Designed and deployed an AI auditing sandbox to validate credit risk models under macroeconomic stress and fraud scenarios.

Industry - FinTech

Services - Custom Software Development

AI Governance & Auditing

Predictive Modeling

Fraud Detection

Risk Analytics

About the Client

The client is an England-based financial technology company developing AI-driven platforms for regulated financial institutions, focusing on credit risk, underwriting intelligence, and governance frameworks for high-risk decision systems.

The Story

The client engaged FiftyFive as its technology partner to design and build a regulatory-grade AI governance sandbox for auditing credit risk models. The engagement covered full-cycle custom software development, including predictive modeling, macroeconomic simulation, fraud validation, and governance reporting. A dedicated AI and data engineering team delivered an end-to-end platform to evaluate a Probability of Default (PD) model used in underwriting. The objective was to enable regulator-aligned validation under long-term macroeconomic stress without data leakage.

Challenges

- Limited transparency and borrower-level explainability in PD models
- Unstructured fairness and demographic bias assessment
- Absence of long-horizon macroeconomic stress testing framework
- Inability to simulate income volatility, expense inflation, and rate shocks
- Weak fraud validation across uploaded financial documents
- No consolidated audit reporting for AI governance readiness

The Solution

- Engineered an XGBoost-based PD model using DSCR and DTI indicators
- Integrated SHAP for borrower-level feature attribution and model interpretability
- Implemented demographic fairness and bias evaluation pipelines
- Built a What-If simulation engine modeling salary growth, rent increases, subscription creep, and interest rate hikes
- Integrated 15–20 year projections using inflation, wage growth, unemployment, and interest rate indicators
- Deployed Isolation Forest for anomaly detection under stress scenarios
- Developed document intelligence engine with OCR preprocessing, MRZ validation, checksum logic, and image forensics

- Designed composite fraud scoring framework combining OCR confidence, rule violations, and image penalties

The Outcome

The platform consolidated fairness, explainability, robustness, and fraud validation into a single audit-ready environment. Borrower-level SHAP explanations improved underwriting traceability and transparency. Long-term macroeconomic simulations validated model resilience across inflation, wage, unemployment, and interest rate shocks, reducing hidden systemic risk exposure. Fraud detection accuracy improved through structured document intelligence and anomaly modeling. The solution elevated AI governance maturity and established a scalable foundation for compliance-driven deployment of high-risk financial AI systems.

131 Artefact ODA Government-Grade Azure Data Platform for OECD Reporting

Replaced fragmented aid reporting systems with a unified Azure data platform, reducing reporting time and improving compliance accuracy.

Industry - Government / International Development / Aid Management / Public Sector

Services - Custom Software Development

Data Engineering

Cloud Engineering

AI/ML Engineering

Governance & Compliance

Business Intelligence

About the Client

The client supports a UAE government entity managing international development aid, overseeing billions of AED across 100+ countries while ensuring OECD DAC compliance, transparency, and multi-stakeholder coordination.

The Story

The client engaged FiftyFive as a full-cycle custom software development and technology consulting partner to modernize fragmented aid tracking systems. Data was distributed across Excel files, OECD and IATI platforms, and internal databases, resulting in reporting delays and compliance failures. FiftyFive deployed a dedicated cross-functional engineering team to architect and build a cloud-native Azure data platform from scratch. The mandate covered data architecture, ETL engineering, AI/ML integration, governance automation, and enterprise-grade analytics delivery, with the platform owned and operated by the client.

Challenges

- Disparate data across 8+ heterogeneous systems without centralized architecture
- 21-day manual OECD CRS consolidation cycle
- 40% validation failure during OECD submissions
- Manual enforcement of 279 data quality rules
- Non-normalized spreadsheet-based data model causing redundancy and update anomalies
- Missing SDG mappings, beneficiary records, and loan repayment fields
- IATI XML generation requiring 5+ days with submission errors
- No audit trail, lineage tracking, API access, or real-time dashboards

The Solution

- Azure Data Factory pipelines integrating OECD CRS, IATI, internal systems, and Excel sources
- Blob Storage + Delta Lake landing zone (50 TB raw ingestion architecture)
- PostgreSQL physical data model (35 normalized entities, 279 data points)
- Databricks (PySpark) batch ETL processing 500K+ records daily
- GPU-accelerated transformations reducing ETL from 8 hours to 45 minutes
- Automated data quality engine enforcing 279 rules across six validation dimensions
- Automated CRS XML and IATI 2.03 generation with pre-submission validation and audit trails
- Azure ML Studio predictive analytics and Power BI real-time KPI dashboards

The Outcome

FiftyFive reduced OECD reporting time from 21 days to 2 days and IATI generation from 5 days to 2 hours. Data accuracy improved to 98%, eliminating submission rejections and reducing manual cleansing by 80%. Processing capacity increased 10x, ETL runtime dropped to 45 minutes, and API response time improved from 30 seconds to 2 seconds. The platform delivered 100% on-time OECD submissions, zero audit issues, 340% ROI within 18 months, 5% sectoral savings, and supported 15% annual data growth while operating 30% under cloud budget.

130 Main Compliance AI-Powered Compliance Automation Platform

Replaced manual policy reviews with a GPU-powered AI compliance system, reducing validation time from 40 minutes to 3 minutes.

Industry - LegalTech / Compliance Automation / Regulatory Technology

Services - Custom Software Development – Backend architecture, SaaS platform engineering

AI/ML Engineering – Legal RAG pipeline, gap detection engine, LLM-based drafting

Cloud Engineering – AWS GPU deployment, infrastructure optimization

Security & Multi-Tenancy Architecture – Tenant allowlists, applicability enforcement

Data Engineering – Embedding pipeline, indexing workflow, validation systems

About the Client

The client is a Sweden-based LegalTech company building an AI-driven SaaS compliance platform that automates policy analysis, regulatory tracking, and license workflows for organizations and legal professionals.

The Story

The client engaged FiftyFive as a dedicated custom software development partner to replace manual, clause-level compliance reviews with a scalable AI-powered system. The mandate included full-cycle backend engineering, AI pipeline design, GPU optimization, and secure multi-tenant SaaS architecture. FiftyFive deployed a focused engineering team covering AI/ML, backend, cloud, and security to deliver a production-grade, auditable compliance automation platform.

Challenges

- Manual clause-by-clause policy-to-law matching and remediation drafting
- CPU-bound inference cycles (~40 minutes per policy run)
- Legal knowledge base indexing cycles taking ~1 week
- Vector index and metadata store inconsistencies causing mapping drift
- Missing provenance and unreliable legal excerpt hydration
- Multi-tenant law applicability risks without strict customer scoping
- Non-deterministic outputs and unstable retrieval results

The Solution

- Architected a GPU-accelerated Legal RAG pipeline deployed on AWS g6.xlarge (NVIDIA L4)
- Designed a tiered multi-layer vector indexing architecture (L0/L2/L3/L4) for recall and precision balance
- Implemented structured ingestion → normalization → chunking → embedding → indexing → validation pipeline
- Migrated similarity search and embedding workloads to GPU for latency reduction
- Built backend job-artifact framework producing deterministic, auditable run outputs
- Developed two-pass engine: deterministic gap detection (Pass A) and bounded OpenAI-based drafting (Pass B)
- Enforced database-driven tenant allowlists with strict applicability controls
- Standardized canonical JSON outputs with structured fields (issue, rationale, confidence, spans, provenance)

The Outcome

Policy-to-law validation time reduced from ~40 minutes (CPU) to ~3 minutes (GPU), enabling near real-time compliance checks and higher processing throughput. Legal RAG rebuild cycles improved from ~1 week to ~3 hours, accelerating regulatory updates and product releases. Automated gap detection significantly reduced manual lawyer workload while preserving evidence traceability. Deterministic artifacts and strict tenant scoping strengthened auditability, enterprise trust, and platform scalability for growth across jurisdictions.

129 Plugsurfing Cloud Standardization & Platform Unification

Modernizing CI/CD and cloud infrastructure by standardizing DevOps practices and migrating a legacy Azure product to AWS for improved stability, governance and scalability.

Industry - Electric Vehicle (EV) Charging & eMobility Solutions / EV Network Platform (Charging Access & Integration)

Services -

- EV Charging Network Aggregation & Access (public charging across Europe)
- White-label EV Charging App & Software Solutions (Drive API / integration for OEMs & partners)
- Roaming & Network Integration for Charge Point Operators (OCPI & interoperability)
- Analytics & Operational Support (driver experience, payment, support services)

About the Client

The client operates as a leading European eMobility service provider and EV charging solutions platform, connecting EV drivers, charge point operators and automotive partners through a unified digital network to simplify access, roaming, and payment across public EV charging infrastructure.

The Story

The client's deployment ecosystem was fragile, with hardcoded GitHub workflows, inconsistent Terraform and Terragrunt versions, and poorly structured infrastructure modules that caused frequent failures. They also lacked visibility into third-party licenses across repositories, creating compliance risks. FiftyFive was engaged to analyze their CI/CD pipelines and outage history, standardize and parameterize configurations, refactor infrastructure code and introduce governance and automation for stable, low-maintenance deployments. At the same time, the client needed to migrate a

product from Azure—despite limited technical knowledge—and successfully onboard it onto their primary platform while ensuring reliability and continuity.

Challenges

- Inconsistent Terraform and Terragrunt versions across repositories
- Poorly structured and tightly coupled infrastructure modules
- Frequent deployment failures from minor configuration changes
- Lack of standardized parameterization and environment management
- Limited visibility and tracking of third-party licenses in GitHub repositories
- Recurring outages due to fragile pipeline design
- Knowledge gaps during Azure product migration
- Unclear system architecture and undocumented dependencies
- Ensuring stable onboarding of the migrated product onto the primary platform
- Hardcoded values across GitHub workflows and CI/CD pipelines

The Solution

- Conducted deep-dive assessment of existing CI/CD pipelines, infrastructure code and outage history
- Standardized and parameterized GitHub workflows to eliminate hardcoded values
- Unified Terraform and Terragrunt version management across repositories
- Refactored and modularized infrastructure code for reusability and stability
- Introduced environment-driven configuration and automated validation checks
- Implemented governance controls and license tracking across GitHub repositories
- Strengthened pipeline reliability with improved branching, versioning, and rollback strategies
- For Azure migration, first analyzed and deployed the product in Azure to fully understand architecture and dependencies
- Validated functionality end-to-end in the original Azure environment
- Translated infrastructure and services for AWS compatibility and tested locally
- Performed controlled manual deployment to AWS for verification

- Onboarded the migrated product into the client's platform following standardized CI/CD and deployment pipeline practices
- Established documentation and operational clarity to ensure long-term maintainability

The Outcome

FiftyFive delivered a standardized and resilient cloud and DevOps foundation that eliminated fragile deployments and reduced outage frequency and significantly improved release stability. Infrastructure became modular, version-controlled and parameterized enabling predictable and low-maintenance deployments across environments. License governance improved compliance visibility across repositories while CI/CD enhancements strengthened automation and rollback reliability.

128 wade insight Improve Software Delivery Through Structured Agile Governance

Improved sprint predictability by implementing structured agile governance, prioritization controls, and standardized ticket tracking frameworks.

Industry - Data Platform Management / Enterprise SaaS

Services - Agile Governance & Delivery Optimization

- Sprint governance restructuring
- Backlog prioritization enforcement
- Mid-sprint review implementation
- Retrospective reinstatement
- Support ticket intake standardization

About the Client

The client is a global data platform management company delivering enterprise SaaS solutions that centralize, integrate, and govern complex data ecosystems for organizations across multiple industries worldwide.

The Story

FiftyFive Technologies engaged as an external agile delivery and engineering governance

partner to strengthen sprint discipline and improve execution predictability. A shared Scrum Master was deployed to support structured sprint management and workflow optimization. The objective was to stabilize delivery cycles, align engineering commitments with business priorities, and reduce operational risk within ongoing product development. The engagement focused on improving transparency, ticket hygiene, prioritization discipline, and structured governance across active software teams.

Challenges

- Inconsistent sprint planning and recurring spillovers across cycles
- Absence of structured prioritization framework for backlog and ad-hoc requests
- Unstructured ticket lifecycle management and tracking gaps
- No formal mid-sprint risk assessment checkpoints
- Lack of Sprint Retrospectives limiting process iteration
- Delivery commitments misaligned with evolving business priorities

The Solution

- Deployed a shared Scrum Master to oversee agile execution and delivery governance
- Introduced compulsory alternate-day stand-ups for structured progress tracking
- Enforced priority tagging standards and contextual ticket documentation protocols
- Implemented formal mid-sprint review checkpoints for risk mitigation
- Reinstated Sprint Retrospectives with action tracking for continuous improvement

- Optimized Sprint Planning by allocating structured bandwidth for unplanned activities
- Standardized support ticket intake and board-level tracking framework

The Outcome

Sprint spillovers were reduced through disciplined prioritization and structured planning, resulting in improved delivery predictability across consecutive sprints. Mid-sprint reviews enabled early risk identification, minimizing last-minute workload rebalancing. Standardized ticket governance eliminated tracking gaps and improved visibility at the leadership level. Engineering commitments aligned more closely with business priorities, strengthening stakeholder confidence and reducing operational delivery risk. Overall delivery maturity improved through consistent agile governance, structured execution control, and data-driven sprint management practices.

127 AI-Powered Compliance Automation Platform

Sweden-based legal tech company replaced manual compliance reviews with an AI-powered platform, reducing document processing time from 48 hours to five minutes.

Industry - LegalTech

Services - Custom Software Development;

Cloud Engineering; AI Integration

Authentication & Security Implementation

Dedicated Engineering Team Deployment

About the Client

The client is a Sweden-based legal technology company developing an AI-driven compliance management platform to automate policy analysis, license acquisition workflows, and regulatory tracking for organizations and legal professionals.

The Story

The client sought to eliminate manual compliance reviews and build a scalable

AI-powered system capable of validating policy documents before regulatory submission. Their goal was to replace fragmented, human-dependent workflows with a secure, event-driven SaaS architecture. FiftyFive acted as a full-cycle custom software development and engineering partner, designing and delivering a production-grade compliance platform. The engagement covered architecture design, AI integration, cloud deployment, and dedicated team execution.

Challenges

- Manual compliance reviews exceeding 48 hours per document
- High rejection rates due to regulatory misalignment
- No automated regulatory change monitoring mechanism
- Limited scalability in document parsing and structured extraction
- Absence of event-driven processing architecture
- Elevated operational costs due to manual validation

The Solution

- Architected and delivered full-cycle custom software development from system design to production deployment
- Built frontend using Vue.js, TypeScript, Vite, Tailwind CSS
- Developed backend services using Laravel and Python for high-performance processing
- Designed event-driven architecture for asynchronous document analysis
- Implemented automated document chunking and AI suggestion generation pipelines

- Integrated OpenAI and Ollama LLMs for regulation-based document validation
- Implemented OAuth-based authentication and role-based access control
- Deployed Dockerized applications on AWS infrastructure
- Configured MySQL for structured, scalable data management
- Deployed a dedicated cross-functional engineering team (frontend, backend, cloud, QA)

The Outcome

The platform reduced document review time from over 48 hours to approximately five minutes, enabling near real-time compliance validation. Manual review dependency was eliminated, lowering operational overhead and reducing rejection-related delays.

AI-powered auditing improved regulatory alignment before submission, accelerating license readiness and time-to-market. The scalable cloud architecture increased user throughput capacity while supporting regulatory evolution. Delivered within six months, the system strengthened the client's product competitiveness, enhanced customer trust, and positioned the platform for national-scale adoption.

126 Viti Science Custom Drug Pricing Intelligence Platform Development

Replaced manual TLV Excel tracking with a secure platform for automated ingestion, monitoring, and price alerts.

Industry - Pharmaceutical / Drug Pricing Intelligence / Regulated Healthcare Market
Services -

- Custom Software Development (Full-cycle)
- Cloud & Database Engineering

- DevOps & Automation
- Security & Compliance Engineering

About the Client

The client operates in the Swedish pharmaceutical market, requiring regulated drug pricing intelligence to support market access, compliance, and competitive monitoring across nationally published TLV pricing datasets.

The Story

The client relied on manually reviewing TLV Excel files to track AIP and AUP price updates. This process was slow, error-prone, and unsuitable for regulatory-sensitive pricing decisions. FiftyFive was engaged as a custom software development partner to design and build a secure, production-grade pricing intelligence platform. The objective was to automate ingestion, historical tracking, and alerting while ensuring enterprise-level security and reliability.

Challenges

- Manual Excel-based price monitoring workflows
- Automated ingestion of daily TLV Excel datasets
- Historical price versioning and change detection logic
- Secure multi-user authentication and authorization
- Row-Level Security (RLS) enforcement in PostgreSQL
- Email alert scheduling, cron reliability, and delivery stability
- Performance optimisation for large historical datasets
- Regression risks in edge functions and auth flows

- GDPR-compliant logging and PII protection

The Solution

- Delivered full-cycle custom software development (architecture to deployment)
- Built frontend using React 18, TypeScript, Vite, Tailwind CSS, shadcn/ui
- Implemented Auth0-based authentication and secure role management
- Designed Supabase PostgreSQL schema with RLS policies and indexing
- Developed Edge Functions for event-driven processing
- Implemented automated Excel ingestion with streaming and chunking
- Built price change detection triggers and historical tracking models
- Configured scheduled cron jobs for ingestion and email alerts
- Integrated Resend API for transactional and digest notifications
- Applied rate limiting, endpoint hardening, and performance tuning

The Outcome

FiftyFive delivered a production-ready SaaS platform that eliminated manual price tracking, automated daily ingestion, and enabled real-time alerting. The solution reduced operational monitoring effort, improved pricing accuracy, minimized compliance risk, and provided a scalable, secure system capable of handling large historical datasets while supporting enterprise-grade user access controls.

125 Artefact Oracle Cloud Data Platform Development

Replaced manual government data workflows with a secure Oracle Cloud platform through custom development, automation, and governed DevOps delivery.

Industry - AI and Data Consulting

Services - Custom Software Development

Cloud Engineering

Data Engineering

DevOps

Architecture & Consulting

About the Client

The client is a global data and AI consulting firm with 1,500 professionals across 26 countries, delivering enterprise data strategy, platform engineering, and AI transformation for public sector programs worldwide.

The Story

FiftyFive Technologies was engaged as a technology services and engineering partner to support a large-scale Qatar government data platform initiative. Operating under a team extension model, FiftyFive provided specialized Oracle APEX and OCI DevOps engineers to design, build, and automate a secure data ecosystem. The mandate was to replace fragmented, manual workflows with governed, cloud-native systems hosted on Oracle Cloud Infrastructure. FiftyFive owned architecture, development, automation, and integration execution while the platform remained fully operated by the client's government stakeholders.

Challenges

- Manual, Excel-driven data processing across multiple government sub-entities
- Unreliable connectivity between on-premise systems and OCI environments
- Missing centralized data cataloging, lineage, and metadata governance
- Absence of reusable code architecture and standardized deployment pipelines
- Backlog of 20+ automation MVPs without execution prioritization
- No end-to-end audit logging for approvals, data changes, and notifications

The Solution

- Designed High-Level and Low-Level Architectures aligned with BRDs and Whole-of-Entity principles
- Built custom Oracle APEX applications using PL/SQL on Oracle Autonomous Database
- Implemented secure OCI architecture using private endpoints, IAM, and RBAC
- Orchestrated ETL pipelines and metadata governance using Informatica
- Developed centralized audit logging and notification engines for SMS and Email
- Re-engineered the Labor Force application end-to-end with governed workflows
- Established CI/CD pipelines for OCI services, Informatica, and database deployments
- Structured repositories and led cross-stack delivery with product stakeholders

The Outcome

The engagement shifted government data operations from manual execution to automated, governed delivery. The first production MVPs reduced manual processing time by 60–70%. A Zero-Trust security model strengthened compliance and institutional trust. Automated CI/CD pipelines enabled rapid rollout of 16+ additional MVPs. Centralized auditability reduced governance overhead, while the modernized Labor Force application delivered real-time, reliable insights to support informed policy decisions.

124 Artefact Qiddiya Gen-AI Conversational Analytics Platforms

Delivered a custom Gen-AI conversational analytics platform through dedicated frontend engineering, enabling faster data exploration and improved team collaboration.

About the Client (30 words only)

The client is an AI and data innovation company building enterprise-grade Gen-AI platforms that enable conversational data analysis, visualization, and collaboration for business users across global markets.

The Story

The client set out to build Qiddiya, a Gen-AI conversational analytics platform designed for data exploration, visualization, and knowledge reuse. Unlike generic chat interfaces, the product required deep frontend engineering to merge AI conversations with interactive charts, tables, datasets, and exports. FiftyFive was engaged as a technology partner to deliver the complete frontend as a custom software development engagement. A single dedicated frontend engineer from FiftyFive owned the full frontend lifecycle, ensuring speed, scalability, and long-term maintainability.

Challenges

- Building a ChatGPT-style conversational UI supporting embedded charts, tables, and datasets

- Rendering dynamic, interactive visualizations inside AI-generated responses
- Enabling bidirectional interaction between AI prompts and visual data components
- Managing large datasets and multiple concurrent conversation states efficiently
- Implementing export, pin, star, and organization features without UI complexity
- Developing drag-and-drop pinboard functionality with persistent ordering
- Ensuring performance, scalability, and UX consistency under single-developer ownership

The Solution

- Designed a modular frontend architecture using Next.js and TypeScript
- Developed a conversational UI with multi-threaded chat state management
- Implemented reusable chart and table components rendered dynamically within chat
- Enabled datapoint selection and contextual prompt injection from visual components
- Built dataset ingestion and conversational exploration workflows
- Implemented knowledge management features including pinning, starring, folders, and grouping
- Developed PDF export functionality preserving layout, charts, tables, and branding

- Built a responsive pinboard dashboard with vertical drag-and-drop reordering

The Outcome

The client successfully launched Qiddiya as a production-ready Gen-AI conversational analytics platform owned entirely by the client. The solution enabled non-technical users to explore data through natural language, reducing dependency on manual analysis workflows. Interactive charts and tables accelerated insight discovery and reuse across teams, improving collaboration and decision-making. The dedicated frontend delivery model reduced development overhead while maintaining enterprise-grade performance and scalability. This engagement demonstrated FiftyFive's ability to deliver complex, custom Gen-AI software through focused engineering ownership, enabling faster product launch, lower operational risk, and a scalable foundation for future platform growth.

123 Ultrasafe **AI-Powered Multi-Tenant Deployment Platform**

Industry - Information Technology (IT)

Services - Custom Software Development: Frontend (React.js), backend (Node.js/TypeScript).

- Cloud Engineering: AWS, RunPod GPU nodes, Kubernetes orchestration.
- DevOps: CI/CD pipelines, containerized deployments.
- AI/ML Development: RAG pipeline, LLM integration.
- Security & Access Control: OAuth, JWT, RBAC.
- Billing Integration: Stripe usage-based billing.
- Monitoring & QA: Real-time metrics and alerts.

About the Client

The client is a Dubai-based enterprise IT company, operating in the technology sector, focused on building scalable digital platforms to support AI-driven innovation and application deployment.

The Story

FiftyFive Technologies partnered with the client to deliver a full-scale production build of an AI-powered multi-tenant deployment platform. The goal was to democratize AI app hosting for non-technical business teams by eliminating manual provisioning, invoicing, and server management. A dedicated 12-member FiftyFive team—including frontend, backend, DevOps, and QA engineers—was engaged to design and implement a secure, cost-efficient, and scalable “AI-as-a-Service” platform.

The Challenges

- Deliver a user-friendly UI for non-technical users to deploy AI services.
- Orchestrate dynamic GPU/CPU resource allocation with cost optimization.
- Implement usage-based billing with automated metering and invoicing.
- Ensure tenant-level isolation with SSL, subdomains, and secure ingress.
- Configure multi-role RBAC with project-specific permissions.
- Enable real-time monitoring of GPU/CPU workloads and anomaly alerts.

The Solution

- Developed frontend in React.js and backend microservices in Node.js/TypeScript.
- Deployed containerized workloads using Docker and Kubernetes on AWS and RunPod GPU nodes.

- Created pre-configured Ubuntu+LLM container images for rapid onboarding.
- Implemented RAG pipeline for knowledge-base powered chatbots with OpenAI/LLM APIs.
- Integrated Stripe Billing API for hourly usage-based automated invoicing.
- Configured Kubernetes Ingress for domains, SSL certificates, and traffic isolation.
- Built GitHub-linked CI/CD pipeline for one-click deploy workflows.
- Integrated Google OAuth with JWT-based sessions and tenant-level RBAC.
- Added dashboards with GPU/CPU utilization metrics and real-time alerts.

The Outcome

The platform enabled non-technical teams to deploy AI apps and chatbots in minutes, cutting infrastructure setup time dramatically. Competitive GPU strategies delivered ~25% compute cost savings, while automated Stripe billing ensured accurate revenue tracking and eliminated manual errors. The client achieved faster time-to-market, reduced operational overhead, and improved financial visibility. Industry recognition and eventual acquisition by a global IT firm validated both the technical strength and strategic business impact of the solution.

122 Easy Payment Monitoring Solutions for Payment Platforms

The client needed secure file monitoring for payments; FiftyFive delivered a scalable AWS solution ensuring compliance, uptime, and revenue protection.

Industry - Digital Payment Management / Financial Technology

Services - Custom Software Development: File Integrity Monitoring (FIM) solution

- Cloud Engineering: AWS-native, scalable deployment

- Cybersecurity: Compliance-driven monitoring, anomaly detection, fail-safe redundancy

About the Client

The client is a regulated digital payment management provider handling high-volume utility transactions, offering secure, real-time processing and compliance assurance for businesses and consumers in financial markets.

The Story

As transaction volumes expanded, the client identified unauthorized file modifications as a major compliance and operational risk. They engaged FiftyFive Technologies to design and implement a custom File Integrity Monitoring (FIM) solution. The goal was to ensure continuous security, regulatory adherence, and scalability—without introducing latency or disrupting high-throughput payment operations. A dedicated engineering team from FiftyFive was deployed to deliver a cloud-native, resilient, and compliance-ready monitoring architecture.

Challenges

- Unauthorized modifications to payment-critical files.
- Single points of failure within transaction infrastructure.
- Need for real-time FIM with zero added latency.
- Strict regulatory compliance for financial transactions.
- Ensuring uninterrupted uptime during continuous monitoring.

The Solution

- Implemented sidecar monitoring architecture to isolate FIM from payment flows.
- Built event-driven anomaly detection using AWS-native services for instant alerts.
- Configured automated compliance-driven notifications and escalation workflows.
- Deployed fail-safe redundancy to maintain uninterrupted payment processing.
- Utilized AWS CloudWatch, Lambda, and auto-scaling for resilience and throughput.
- Delivered by a dedicated cross-functional team of backend engineers, cloud architects, and security specialists.

The Outcome

The client achieved 100% uptime during FIM operations with zero service disruption. Real-time anomaly detection reduced compliance risk and prevented financial penalties, protecting millions in potential revenue. Scalable AWS deployment enabled seamless growth with transaction demand while maintaining low latency. By strengthening operational resilience and regulatory trust, the client improved brand credibility and safeguarded its revenue streams. This engagement demonstrates how FiftyFive's custom software development, cloud engineering, and cybersecurity services enable financial enterprises to achieve both compliance and business continuity at scale.

121 Matrix Requirement Compliance-Driven Medical Device Platforms

The client needed scalable compliance software so FiftyFive modernized the platform with secure authentication, React interface, and faster integrations.

Industry - Health Technology / Medical Devices

Services - Custom Software Development & Modernization (architecture redesign, API optimization)

- UI/UX Engineering (React migration, design system)
- Cloud & Backend Engineering (notification service, optimized endpoints)
- DevOps & QA (secure authentication, testing)

About the Client

The client is a health technology company providing compliance-first platforms for medical device innovators, enabling regulatory adherence, secure collaboration, and accelerated product launches in global markets.

The Story

The client needed to modernize its compliance-driven requirements management platform to meet growing enterprise demands. Outdated architecture, limited authentication, and fragmented tools slowed adoption. FiftyFive Technologies was engaged to deliver full-cycle software modernization—covering architecture redesign, UI transformation, backend enhancements, and scalable enterprise-ready features.

Challenges

- Migration from outdated platform architecture to modern, scalable systems
- Lack of multi-tenant authentication for enterprise security requirements
- No centralized notification service for user and system events
- Legacy frontend stack dependent on jQuery, limiting UI scalability
- Dependency on legacy admin app while upgrading backend endpoints and structures

The Solution

- Implemented SAML-based multi-tenant authentication for enterprise-grade security and compliance
- Built a backend notification service for real-time alerts and system updates
- Migrated frontend to React with a unified design system, improving UI consistency and developer velocity
- Optimized backend endpoints and data structures for future-proof integrations
- Enhanced legacy admin app to consume upgraded APIs while maintaining operational continuity
- Delivered engineering through a dedicated team of frontend, backend, DevOps, and QA specialists

The Outcome

The engagement delivered an enterprise-ready, compliant platform that accelerated adoption among medical device innovators. Multi-tenant authentication enabled secure scaling across enterprise clients, while the notification service improved user communication. The React-based interface reduced development turnaround by 40%, and modernization of backend services cut integration time by 30%. By reducing technical debt, the client gained a future-proof architecture that supports regulatory traceability and faster market entry. This project demonstrates how FiftyFive's custom software development and modernization expertise create measurable business impact for regulated industries worldwide.

120 Broad&Salt Mobile App Modernization and Loyalty Solutions

FiftyFive modernized the client's mobile app with loyalty and booking features, achieving faster performance and stable multi-platform releases.

Industry - Food & Beverage / Retail Technology

Services - Custom Software Development: Mobile app modernization, loyalty and booking systems, authentication, order flow improvements

- UI/UX Design: Home redesign, carousels, subscriptions, shimmer effects
- DevOps: EAS CI/CD pipeline setup, build stabilization
- Cloud & Backend Services: Firebase modularization, push notifications
- QA & Testing: Jest unit testing, continuous deployment monitoring

About the Client

The client is a Sweden-based bakery and café chain, operating multiple outlets, seeking to modernize its mobile app for digital loyalty, coworking bookings, and customer engagement.

The Story

In April 2025, the client engaged FiftyFive Technologies to stabilize and enhance their existing React Native app. A dedicated four-engineer team was deployed to address outdated infrastructure, failed builds, and missing features. The engagement covered SDK migrations, Firebase reconfiguration, and development of new digital services.

FiftyFive applied its expertise in custom software development, UI/UX, DevOps, and QA to deliver a scalable, future-ready platform.

Challenges

- Outdated Expo SDK (v48) and multiple breaking migrations
- Firebase conflicts causing iOS and Android build failures
- No loyalty and rewards system for customer retention
- Poor caching and image handling leading to long load times
- Build instabilities including EAS failures and manifest crashes
- Limited authentication and weak user management flows

The Solution

- Migrated Expo SDK from v48 to v53, resolving 15+ build issues
- Modularized Firebase services; integrated native iOS and Android push notifications
- Delivered loyalty and rewards program with digital card and campaign banners
- Built coworking booking system, secure authentication, and enhanced order flow
- Redesigned home UI with carousels, subscriptions, shimmer effects, and faster image caching
- Implemented store locator with clustering using Maps API
- Applied Jest testing, EAS CI/CD, and Mixpanel analytics for stability and tracking

- Adopted React Native 0.79.5 with TanStack Query v4, GraphQL/REST integration

The Outcome

The modernized application achieved a 60% faster load time, zero crash reports, and 100% reliable push notifications. The loyalty and rewards program boosted repeat visits, while coworking bookings generated a new revenue stream. Continuous multi-platform releases across five SDK upgrades ensured stability and future scalability. By combining technical modernization with feature expansion, FiftyFive enabled the client to increase customer loyalty, unlock new business growth opportunities, and strengthen its digital market presence.

119 Growdash Martech and Restaurant-Tech SaaS Platforms

The client faced code inefficiencies and production bugs so FiftyFive restructured the SaaS platform, improving scalability and reducing issues.

Industry - Martech and Restaurant-tech

Services - Custom Software Development: Feature development, bug fixes, codebase restructuring

- Frontend Engineering: Validation, routing, API optimization
- Quality Assurance: Bug identification and resolution

About the Client

The client operates in the martech and restaurant-tech domain, providing SaaS-based analytics and intelligence platforms for food and beverage businesses across competitive global markets.

The Story

The client partnered with FiftyFive Technologies after securing funding to modernize their SaaS platform. They required a dedicated engineering team to restructure the codebase, fix critical production issues, and deliver new features with improved scalability. FiftyFive provided end-to-end custom software development services, ensuring the product was stable, performant, and capable of supporting rapid growth in the restaurant-tech and martech space.

Challenges

- Inefficient and inconsistent folder structures within the codebase.
- Outdated validation mechanisms lacking centralized management.
- Multiple production-critical bugs in discount and ad modules.

- No implementation of tanstack queries or routers.
- Limited scalability and maintainability impacting development speed.

The Solution

- Standardized folder structures and enforced consistent coding patterns.
- Implemented **Zod** for robust, type-safe validation across the platform.
- Integrated **React Queries** and **Tanstack Queries** for efficient API state management.
- Deployed **Tanstack Router** for type-safe, file-based routing.
- Resolved production-critical bugs and introduced new feature modules.
- Dedicated team of frontend developers, QA engineers, and architects applied Agile methodology for rapid iterations.

The Outcome

The engagement delivered a streamlined, maintainable, and scalable SaaS platform. Feature delivery speed improved by **35%**, and production issues were reduced by **40%**, enabling faster go-to-market releases. The robust validation and routing framework significantly improved platform stability, while optimized API management enhanced system reliability. By restructuring the codebase and embedding modern practices, FiftyFive ensured the client could scale with confidence, improve customer satisfaction, and strengthen market credibility. This project demonstrates FiftyFive Technologies' expertise in delivering custom software development services that resolve complex SaaS challenges and create measurable business value.

118 Spicolli QR-Based Digital Donation Platform

Built a QR-based donation platform with secure payments, automated payouts, and transparent reporting to modernize charitable giving.

Industry - FinTech / Social Impact / Payments

Services - Custom Software Development (MVP Development, Mobile App Development)

- Cloud Engineering (AWS deployment)
- Payment Integration (Stripe Connect)
- UI/UX Development (Onboarding flows, dashboards)
- Software Testing & QA

About the Client

A social impact startup in the FinTech sector focused on enabling seamless, transparent digital donations for street performers and artists, addressing barriers in today's increasingly cashless economy.

The Story

The client partnered with FiftyFive Technologies to design and deliver a Minimum Viable Product (MVP) that transformed charitable giving. They envisioned a QR-based mobile-first platform where donors could support performers instantly. The engagement required full-cycle development—mobile apps, backend systems, payment infrastructure, and automated compliance. FiftyFive deployed a dedicated cross-functional team including project management, mobile developers, backend engineers, and QA to deliver a secure, scalable solution.

Challenges

- Design mobile registration with secure OTP validation and fraud prevention.
- Implement PCI DSS-compliant flows supporting Apple Pay/Google Pay via QR-triggered checkout.
- Build segregated accounts with precise transaction tracking and automated payouts.
- Generate persistent QR codes dynamically linked to individual donation pages.
- Deliver real-time dashboards with donation history and geo-data for recipients.

The Solution

- Cross-platform onboarding, Twilio OTP verification, and QR code management.
- RESTful services with PostgreSQL, JWT authentication, and encrypted payment data.
- Lightweight pages linked to QR codes, enabling instant donor payments.
- Managed accounts, automated commission deduction, and scheduled payouts via API cron jobs.
- Real-time transaction summaries with geo-IP-based donor insights.

The Outcome

FiftyFive delivered the MVP in a rapid, scalable model. Donors completed transactions in under 30 seconds, and recipients accessed automated payouts with full transparency. The platform achieved PCI DSS compliance through Stripe, enabling secure operations without regulatory overhead. With onboarding reduced to under five minutes and

disbursements fully automated, the client gained a sustainable, low-cost operating model. This strengthened donor trust, improved artist revenue flows, and positioned the client to scale confidently into new markets demonstrating the business value of FiftyFive's custom software development expertise.

117 Audemic Software Development for Subscription-Based Platforms

The client needed subscription and usability improvements; FiftyFive delivered responsive design, integrations, and stability, achieving growth in conversions and reliability.

Industry- Education Technology (EdTech)

Services - Custom Software Development: Bug fixes, subscription optimization, authentication improvements

- UI/UX Development: Responsive design with Tailwind CSS, login page redesign
- QA & Testing: RSpec test cases, code quality enforcement with Rubocop
- System Integration: Slack integration for notifications
- Cloud Engineering: AWS & Heroku deployment

About the client

The client operates in the education technology sector, providing platforms that transform complex academic research into simplified summaries and audio formats for students, researchers, and professionals worldwide.

The Story

The client engaged our team to enhance their research-access platform, which included Scholar and Insights applications. The engagement focused on improving performance, strengthening subscription workflows, and delivering responsive UI/UX. A dedicated team was deployed to provide full-cycle software development, bug resolution, and QA automation while ensuring scalable and maintainable code standards.

Challenges

- Inefficient subscription and in-app purchase workflows.
- Authentication flow lacking validation mechanisms.
- Limited responsiveness across devices and inconsistent UI design.
- Multiple critical bugs impacting Scholar and Insights applications.
- No automated test coverage to ensure long-term stability.

The Solution

- Implemented subscription and in-app purchase enhancements, supported by R&D on RevenueCat.
- Integrated Slack for automated system notifications.
- Added keyboard shortcuts for efficient audio navigation and playback speed control.
- Applied Tailwind CSS for fully responsive UI/UX improvements.
- Strengthened authentication with phone number validations.
- Fixed bugs across multiple modules to ensure stability.
- Written RSpec test cases and enforced coding standards with Rubocop.
- Utilized Ruby on Rails, PostgreSQL, Stimulus, Sidekiq, AWS, and Heroku for scalable performance

The Outcome

All improvements were delivered within four months by a dedicated engineering team. The optimized subscription flow reduced user friction and improved conversion rates, while responsive UI/UX upgrades enhanced accessibility across devices. Automated notifications streamlined platform monitoring, and RSpec coverage increased reliability by minimizing future defects. Bug resolutions and code standardization boosted stability, strengthening user trust. Collectively, these enhancements improved platform engagement, supported sustainable subscription growth, and positioned the client for scalability and future product expansion.

116 Matchbox Custom Match Reporting Platforms for Football

Client replaced manual match reporting with a custom digital platform, integrating live data and achieving 70% faster reporting.

Industry - SaaS/ Sports tech

Services - Web Application Development → React-based platform

- Cloud & Database Engineering → Firebase Realtime Database, Firestore
- API Integration → SportMonk live match data
- Security & Compliance → Role-based access, multi-club isolation
- Team Extension → Dedicated engineering team delivery

About the client

The client is a UK-based sports association representing football club safety and operations officers, responsible for compliance, structured reporting, and safe matchday management across leading professional football leagues.

The Story

The client engaged FiftyFive Technologies to replace its manual, paper-driven match reporting system with a scalable, real-time digital platform. The goal was to automate reporting, integrate live sports data, enforce security, and provide analytics for safer matchday operations. FiftyFive deployed a dedicated development team with expertise in React, Firebase, and API engineering to deliver a custom solution tailored for professional football clubs.

Challenges

- Manual data entry introducing reporting delays and errors
- Synchronization of real-time match data with high accuracy
- Managing multi-domain data (match stats, crowd incidents, compliance)
- Multi-season storage and fast retrieval of historical records
- Multi-club role-based access with strict data isolation
- Tracking 40+ operational and safety metrics simultaneously

The Solution

- React-based Web Application → Responsive, real-time reporting interface
- Firebase Realtime Database → Collaborative reporting with instant synchronization
- Firestore → Structured storage enabling multi-season analytics
- SportMonk API Integration → Automated ingestion of live match statistics
- Safety & Incident Tracking → 40+ operational metrics including steward deployment and medical incidents

- Role-Based Access Control → Secure authentication and isolated multi-club architecture
- Dedicated Delivery Team → Frontend, backend, and cloud engineers with QA support

The Outcome

The platform reduced reporting time by 70% and eliminated manual entry errors. Clubs achieved comprehensive safety oversight through structured monitoring of 40+ metrics, ensuring compliance with governing standards. Real-time collaboration allowed staff across departments to contribute simultaneously, improving operational efficiency on match days. Multi-season historical insights enabled data-driven crowd and safety planning, reducing risks and enhancing compliance confidence. By modernizing reporting with a secure, scalable architecture, the client improved efficiency, reduced operational costs, and strengthened goodwill with fans, clubs, and regulators.

115 Promarsal Development of AI-Powered SDR Automation Platforms

The Client needed a scalable outreach automation; FiftyFive built an AI-driven SDR platform, reducing costs 40% and boosting conversions 28%.

Industry - B2B Sales Development / Sales Automation

Services - Custom Software Development: Full-cycle engineering (frontend, backend, AI integration)

- UI/UX Design: Responsive modular interface
- Cloud Engineering: AWS deployment, CI/CD pipelines, monitoring
- AI/ML Integration: LangChain-powered agents for lead research and persona analysis
- Team Extension: Dedicated multi-disciplinary engineering team

About the client

The client provides an AI-powered sales development platform for B2B organizations, enabling automated outreach, persona targeting, and multi-channel engagement across email and LinkedIn for startups and enterprises.

The Story

The client partnered with FiftyFive Technologies to design and engineer a scalable AI-driven sales automation platform that could operate as a virtual SDR. The goal was to combine advanced AI workflows, automated list-building, persona-based targeting, and multi-channel campaign management while ensuring compliance, stability, and high deliverability. FiftyFive delivered a full-cycle engagement covering architecture, frontend and backend development, AI integration, and UI/UX design to bring the product to market quickly and cost-effectively.

Challenges

- Managing high-volume personalized outreach across LinkedIn and email simultaneously.
- Overcoming engagement tracking issues caused by caching, privacy filters, and bots.
- Designing UI that balances simplicity with enterprise-grade functionality.
- Integrating AI agents and natural language workflows without impacting responsiveness.
- Structuring competitive multi-tier pricing in a crowded SDR automation market.

The Solution

- Developed modular, responsive UI using React and Next.js with real-time campaign management and video email preview support.
- Built backend services with Node.js and Python for automation workflows and high-volume scalability.

- Integrated LangChain-based AI agents for lead research, persona analysis, and personalized message generation.
- Deployed infrastructure on AWS with CI/CD pipelines, containerization, and monitoring for stability and compliance.
- Delivered flexible pricing models, including free-forever and enterprise tiers, enabling broader market adoption.
- Dedicated team included frontend engineers, backend developers, AI/ML specialists, DevOps, UI/UX designers, and QA testers working in agile sprints.

The Outcome

The platform reduced outbound campaign setup time by 70% and cut operational costs by 40%. AI-driven workflows increased response rates by 35% and meeting conversions by 28%, directly improving client revenue. With FiftyFive's engineering expertise, the client gained faster time-to-market, a competitive edge over enterprise-only SDR tools, and improved goodwill by offering scalable pricing accessible to startups and mid-market businesses.

114 Design 3 AI Platform Development for Context-Aware Collaboration

FiftyFive developed an AI-driven project intelligence platform with document analysis, proactive workflows, and real-time Miro integration achieving faster, accurate collaboration.

Industry - Education Technology and Artificial Intelligence

Services -Custom Software Development: Full-cycle web platform engineering

- AI/ML Development: Context gap analysis, proactive AI, document intelligenceAI

- Cloud Engineering: Azure AI services, Azure App Services, Blob Storage

- System Integration: Miro Web SDK & REST API integration, Twilio SendGrid
- Data Engineering: MongoDB, Redis
- Software Testing & QA: Functional and iterative validation

About Our Client

The client operates in education technology and artificial intelligence, building next-generation collaborative platforms that integrate contextual project intelligence with advanced AI-driven automation for global users.

The Story

The client engaged FiftyFive Technologies to develop the second phase of their project intelligence platform. Following a successful MVP, the objective was to engineer industry-first AI capabilities that moved beyond reactive assistance. The mandate included building proactive AI, non-linear workflows, and real-time bidirectional integration with collaboration tools. FiftyFive deployed a full-cycle development team including a Project Manager, Technical Lead, Frontend and Backend Developers, Data Scientist, and QA Engineer.

The Challenges

- **Context-Aware Document Intelligence** – Summarize and analyze user-uploaded DOCX, PPTX, and PDFs in alignment with ongoing project discussions.
- **Proactive AI & Template Association** – Detect missing project context, ask clarifying questions, and recommend templates with higher accuracy.
- **Non-Linear Guided Mode** – Support backward navigation across project stages without data loss or logical inconsistency.
- **Bidirectional Miro Integration** – Real-time, two-way sync where AI could interpret board edits and feed changes back into the conversation.

The Solutions

- **Document Intelligence Service** – Azure AI Document Intelligence + project-specific Azure AI Search for Retrieval-Augmented Generation.
- **Context Gap Analyzer** – Middleware powered by fine-tuned GPT-3.5 to identify missing inputs and trigger proactive clarification workflows.
- **Guided Mode Orchestrator** – Finite state machine (XState), MongoDB context storage, and dependency mapping for iterative, non-linear stage transitions.
- **Miro Integration Layer** – Web SDK event listeners, webhook proxy, and AI-driven interpretation of board changes to maintain live sync.

The Outcome

FiftyFive delivered a scalable, AI-driven platform with measurable business impact. Bidirectional Miro sync achieved <3 second latency, reducing workflow delays. Proactive context gap analysis lowered template misapplication by ~70%, increasing efficiency and user adoption. Non-linear Guided Mode allowed seamless iteration, enabling faster decision-making and reduced rework. The solution positioned the client ahead of competitors by combining technical innovation with tangible productivity gains.

113 Amilies Custom Healthcare Booking Platform

FiftyFive built a self-service healthcare booking platform integrating payments, clinic APIs, and automation, reducing manual effort and improving turnaround.

Industry - Healthcare services

Services - Custom Software Development: Backend architecture, service-oriented development, API gateway

- Cloud Engineering: AWS deployment with Docker

- Third-Party Integrations: Radox APIs, Stripe, Airtable, AWS Location Service

- Workflow Automation: Redis-based queue, Node-Cron orchestration

About Our Client

The client is a healthcare services provider operating diagnostic centers, seeking to modernize operations by digitizing test booking, payments, and result management with a scalable, integrated web platform.

The Story

The client engaged FiftyFive Technologies to design and build a self-service platform enabling users to browse tests, locate clinics, book appointments, and pay online. Internally, they needed complete automation of order workflows—covering payment, clinic booking, test order creation, and result delivery. Lacking in-house engineering expertise, they partnered with FiftyFive to deliver a robust, scalable solution from the ground up.

The Challenges

- Integration of multiple third-party APIs (Radox Booking, Radox Ordering, Stripe, Airtable, AWS Location Service).
- Orchestration of asynchronous workflows: payment confirmation, slot reservation, order creation, and status/result polling.
- Dynamic multi-channel pricing logic with bundled tests, add-ons, and variable margins.
- Ensuring data consistency across Stripe, Radox, and Airtable with cron-based synchronization.
- High-frequency background task execution requires reliability, logging, and error handling.

The Solutions

- Centralized API Gateway using **Express.js** to unify third-party service calls.

- Service-Oriented Architecture with modular services for booking, ordering, payments, and data sync.
- **Redis** for temporary data storage and queuing; **Node-Cron** for asynchronous workflow automation.
- Dynamic pricing engine calculating test costs, add-ons, and fees; automated **Stripe Checkout** session creation with reconciliation metadata.
- Resilient integration layer with retry logic, token refresh, and API error handling.
- Deployment on **AWS** with **Docker**, ensuring scalability and containerized maintainability.
- Delivered by a dedicated team: 1 Solution Architect, 2 Backend Engineers, 1 DevOps Engineer.

The Outcome

The platform automated the entire diagnostic test booking lifecycle, reducing manual effort by 70% and minimizing operational errors. Real-time order synchronization improved booking-to-result turnaround by 40%, directly enhancing patient satisfaction and operational efficiency. Consolidated data in Airtable provided real-time business intelligence on sales, orders, and results, enabling data-driven decision-making. The modular architecture ensures scalability for new tests and integrations, positioning the client for sustainable digital growth.

LinkedIn caption

We recently partnered with a healthcare provider to build a full-scale custom software platform that automated their entire diagnostic test booking journey from appointments and payments to results. By integrating multiple APIs, streamlining workflows, and deploying on AWS, we cut down manual effort by 70% and improved turnaround time by 40%. This project is a great example of how we step in not just as coders, but as a global software development partner. Whether you're in healthcare, finance, retail, manufacturing, or tech we help companies across the UK, Nordics,

Middle East, Australia, and the USA scale faster through custom software development, mobile apps, legacy modernization, AI/ML, IoT, blockchain, cloud, and security solutions.

Read the full story here - <https://lnkd.in/gWUJJ4V7>

Let's talk about your next build - sales@fiftyfivetech.io

<https://fiftyfivetech.io/>

112 Top Gun Custom AI App Development for Shooting Sports

Developed an AI-powered mobile platform to detect gunshots, sync user data in real time, and ensure zero data loss.

Industry - Sports Technology / Outdoor Recreation

Services - Custom Software Development (MVP Development, Mobile App Development)

- AI/ML Engineering (Shot Registration Engine with TensorFlow Lite)
- Cloud & Backend Engineering (Node.js, Redis, PostgreSQL, AWS S3)
- Real-Time Systems Development (Socket.IO, BullMQ)

About Our Client

A UK-based sports technology company focused on outdoor recreation, building an MVP mobile platform for real-time shooting sports performance tracking and community-driven engagement.

The Story

The client identified a lack of digital tools to measure, analyze, and enhance shooting sports. Their goal was to create a mobile-first platform that could automatically detect gunshots, synchronize performance data across multiple participants, and operate reliably in rural, low-connectivity environments. FiftyFive Technologies was engaged as the engineering partner, providing a dedicated cross-functional team to design, develop, and deliver the MVP from architecture to deployment.

The Challenges

- Acoustic source separation to distinguish local gunshots from background noise and other shooters.
- Real-time multi-user state synchronization with intermittent and low-latency network conditions.
- Continuous audio and sensor processing on mobile devices with minimal battery drain.
- Offline-first operation to ensure zero data loss during connectivity drops.
- Normalizing audio and sensor data across varied iOS and Android hardware.

The Solutions

- Engineered a hybrid Shot Registration Engine using TensorFlow Lite–based acoustic AI with accelerometer-driven recoil validation, augmented by decibel-sensitive spike detection to differentiate near and distant gunshots for additional validation.
- Implemented on-device TensorFlow Lite audio classification ML models for real-time inference on both iOS and Android, enabling frequency-based gunshot detection. Designed with an offline-first architecture to ensure reliable operation in low-connectivity environments and prevent data loss during network

disruptions.

- Implemented Redux Toolkit with SQLite for offline-first state management and background sync.
- Built Node.js backend with Express.js API and Socket.IO for real-time room-based synchronization.
- Deployed BullMQ with Redis for asynchronous job queues handling notifications, emails, and data processing.
- Structured PostgreSQL schema for relational integrity and AWS S3 pipeline for future ML retraining.
- Integrated Firebase Cloud Messaging and APNs for reliable push notifications to active and offline users.

The Outcome

The MVP achieved >95% shot detection accuracy with <0.5% false positives, sub-200ms synchronization latency, and zero data loss under offline conditions. Optimized background processing ensured <5% additional battery drain per active hour. Beyond technical performance, the client gained a production-ready foundation for scaling, faster market entry, and strong user adoption potential. This engagement demonstrates FiftyFive Technologies' expertise in delivering AI-powered, real-time mobile platforms capabilities we extend to businesses across sports tech, healthcare, finance, and beyond.

111 Nexgen Custom SaaS Platforms for Construction Firms

FiftyFive developed a unified SaaS platform centralizing projects, compliance, and cashflows, reducing approval cycles and compliance errors significantly.

Industry - SaaS / Web Application

Services - Custom Software Development (SaaS platform engineering)

- Cloud Engineering (AWS deployment, Docker)
- Workflow Automation (journey templates, approval workflows)
- Security & Compliance (role-based access, digital signing)
- Financial Modules (cashflow analysis, reporting)
- UI/UX Engineering (document builder, dashboards)

About Our Client

A UK-based mid-size construction services company building a SaaS platform to streamline complex project lifecycles, compliance, documentation, financial oversight, and audits for modern construction teams.

The Story

The client aimed to replace fragmented tools like spreadsheets, emails, and legacy systems with a unified construction management platform. They needed a flexible system to manage diverse project workflows, secure documentation, real-time cashflows, and compliance audits. FiftyFive was engaged as the custom software development partner to architect, develop, and deploy a scalable SaaS solution.

The Challenges

- Multiple project journeys with unique workflows and dependencies
- Complex document formats, versioning, and linked task data
- Compliance audits requiring internal and external approval workflows
- Real-time cashflow monitoring and financial reporting complexity
- Secure sharing, signing, and verification of documents
- Need for scalable architecture to support integrations and growth

The Solutions

- Designed a centralized SaaS platform with modular architecture
- Built dynamic project journey templates adaptable to project types and workflows
- Developed form and document builder with linked inputs across tasks and compliance checks
- Implemented real-time cashflow analysis and reporting modules
- Integrated audit and risk management with digital signing and layered approvals
- Applied role-based access, JWT authentication, and secure collaboration features
- Deployed platform on AWS with Docker for scalability and integration readiness

The Outcome

The platform enabled the client to reduce project approval cycles by 40%, cut compliance errors by 35%, and save an average of 25+ hours per audit cycle. Construction teams now collaborate in real-time, ensuring tasks, documents, and financials remain audit-ready at all times. By centralizing operations into one SaaS ecosystem, the client minimized rework, reduced risk exposure, and gained a scalable foundation to expand features and onboard future partners. This engagement highlights FiftyFive's expertise in custom SaaS development, cloud engineering, and workflow automation for complex, compliance-driven industries.

110 Handsin Custom Split Payments Platform

FiftyFive developed a unified split-payments platform, improved observability, ensured compliance, and enabled faster integrations with new payment providers.

Industry - Fintech

Services - Custom Software Development: Payment orchestration API, frontend/backend migration

- Cloud Engineering: AWS infrastructure deployment
- DevOps: Monorepo setup, CI/CD pipelines
- Security & Compliance: Tokenization, encryption, audit logging
- Testing & Optimization: A/B testing, cross-platform compatibility

About Our Client

The client operates a split-payments platform enabling merchants to process transactions via cards, groups, links, and installments, integrated with acquirers and compliant with PCI-DSS, ISO 27001, and GDPR standards.

The Story

The client sought to expand their payment platform to handle increasingly complex split-payment workflows and higher transaction volumes. They engaged FiftyFive to deliver technical scalability, stronger observability, and simplified integration with multiple payment providers. The engagement required a dedicated engineering team to modernize the platform, migrate core systems, and build compliance-driven infrastructure on AWS. FiftyFive's role was to engineer and deliver the system, while the client continues to operate the business.

The Challenges

- Complex orchestration of multi-card, group, link, and installment workflows.
- Limited real-time analytics and diagnostics across payment flows.
- Manual/static integration with acquirers, gateways, and orchestration layers.
- Migration of frontend and backend to a single monorepo.
- Establishing scalable, secure AWS infrastructure.

- Maintaining PCI-DSS, ISO 27001, and GDPR compliance during enhancements.

The Solutions

- Modularized the payment orchestration layer with a unified API for all split-payment workflows.
- Embedded real-time analytics, monitoring, and diagnostics into transaction pipelines.
- Developed plug-and-play connectors for multiple acquirers and gateways.
- Implemented A/B testing to optimize user flows and conversion rates.
- Migrated frontend and backend codebases into a unified monorepo with CI/CD pipelines.
- Deployed secure, auto-scaling infrastructure on AWS using containerization and IAM-based access.
- Enforced compliance through tokenization, encryption, and audit logging.
- Delivered cross-platform compatibility for consistent performance on web and mobile.

The Outcome

The platform now processes multi-card, group, link, and installment payments through a single API, reducing integration effort by 40%. Real-time analytics improved issue detection and reduced incident resolution time by 35%. Plug-and-play connectors accelerated new acquirer integrations from weeks to days, expanding merchant options and revenue channels. Migration to a monorepo streamlined development, cutting release cycles by 25%. With AWS-based scaling and full PCI-DSS, ISO 27001, and GDPR adherence, the client achieved a secure, future-ready payments platform that increased

operational efficiency, improved merchant onboarding speed, and strengthened customer trust.

109 The Training Marketplace AI Powered Training Marketplace Development

The client needed a scalable SaaS marketplace, FiftyFive delivered an AI-powered platform enabling global adoption and rapid provider onboarding.

Industry - EdTech / Training Tech

Services - Custom Software Development: SaaS platform build, microservices architecture

- AI/ML Development: BERT-based NLP, ML ranking
- Cloud Engineering: AWS ECS, Lambda, S3, CloudFront, CI/CD pipeline
- Web Development: Next.js + Tailwind frontend
- Backend Development: Node.js (TypeScript), GraphQL APIs
- DevOps & Security: CI/CD (GitHub Actions, CodePipeline), JWT, AWS Cognito, KMS encryption, monitoring
- UI/UX Design: Provider and buyer dashboards
- Software Testing: TDD (Jest, Cypress)

About Our Client

A UK-based EdTech company operating in the corporate training sector, building a scalable SaaS platform to connect training providers with global enterprise buyers through AI-driven search and real-time availability.

The Story

The client approached FiftyFive Technologies after validating their MVP. The prototype proved market demand but lacked scalability, AI intelligence, and enterprise-grade

architecture. FiftyFive was engaged to design, develop, and deliver a full SaaS build, ensuring AI-powered matching, real-time data handling, and enterprise security. The goal was to establish a transparent, global marketplace connecting providers and buyers directly while maintaining compliance and scalability.

The Challenges

- Two-sided marketplace onboarding and engagement for buyers and providers.
- AI-driven natural language search with domain-specific accuracy.
- Real-time provider availability and large-scale traffic management.
- Metadata-driven course ranking, deduplication, and ML-based recommendations.
- GDPR compliance with enterprise authentication/authorization.
- Delivery of a full-scale SaaS build within nine months.

The Solutions

- Architecture & Infrastructure: Microservices on AWS ECS + Lambda; Node.js (TypeScript) backend with GraphQL APIs; Next.js + Tailwind frontend.
- Database Layer: PostgreSQL (transactions), Elasticsearch (search), Redis (real-time caching).
- AI/ML: BERT-based embeddings, domain-specific NLP pipeline, ML-based ranking integrated with Elasticsearch.
- Core Features: Buyer AI + structured search, secure booking/messaging, provider dashboards with analytics, multi-region/currency support, GDPR-compliant policies.

- Engineering Team: 12-member cross-functional team including backend, frontend, AI/ML engineers, UI/UX designers, and QA, delivering in Agile 2-week sprints.
- DevOps & Security: GitHub Actions + CodePipeline CI/CD, JWT + AWS Cognito authentication, KMS encryption, CloudWatch + ELK observability.

The Outcome

The platform was delivered in 8.5 months, benchmarked to handle 100k+ concurrent users with <200ms response times. 78% of buyers adopted AI-driven search over structured queries, validating the NLP engine. Over 1,200 providers onboarded in the first six months, including many previously absent from search engines. The client gained a competitive edge as the world's first free-to-use corporate training marketplace, expanding rapidly across UK, EU, and ME markets, strengthening both revenue potential and global market goodwill.

108- Risegroup SaaS Development for Business Incorporation & Compliance

FiftyFive built a SaaS platform automating UK company incorporation, integrating compliance checks, and reducing registration time to under 24 hours.

Industry - SaaS

Services - Custom Software Development: SaaS platform design & build

- API Integrations: Companies House, KYC/AML
- Cloud Engineering: AWS, Dockerized deployment
- UI/UX Development: Guided onboarding
- Post-incorporation Integrations: Business services add-ons

About Our Client

A UK-based business services provider specializing in fast company incorporation, compliance verification, and value-added business solutions, serving entrepreneurs and startups with scalable SaaS-driven offerings.

The Story

The client sought to digitize the traditionally manual and paperwork-heavy UK company incorporation process. Entrepreneurs often faced confusion over compliance requirements, company name availability, and post-registration add-ons. FiftyFive Technologies was engaged to design and build a scalable SaaS platform that would automate incorporation, ensure compliance, and provide entrepreneurs with real-time insights and guided onboarding. The goal was to deliver a frictionless, compliant, and growth-oriented experience through a single digital solution.

The Challenges

- Automating UK company registration workflows with full compliance.
- Real-time integration with Companies House API for name availability checks.
- Secure KYC and AML verification with GDPR-compliant data handling.
- Dynamic package configuration supporting free and premium tiers.
- Integration of modular post-incorporation services (domain, insurance, virtual office).
- Ensuring scalability for high transaction volumes and future feature expansion.

The Solutions

- Developed a web SaaS platform using **React.js** (frontend) and **Node.js/Express** (backend).
- Integrated **Companies House API** for instant registration validation.
- Implemented secure **KYC/AML workflows** with encrypted email-based verification.

- Deployed on **AWS Cloud** with Dockerized microservices for scalability.
- Applied **Agile methodology** with sprint-based delivery and CI/CD pipelines.
- Built modular architecture to support third-party integrations and add-on services.
- Dedicated team of **frontend, backend, and cloud engineers** managed full-cycle development.

The Outcome

The platform reduced company incorporation time from weeks to less than **24 hours**, with **95% error-free submissions** achieved through guided onboarding and API validation. Flexible packages and integrated add-ons boosted client upsell opportunities by **30%**, while automation cut manual compliance effort by **70%**. The solution enabled entrepreneurs to launch businesses faster and positioned the client as a trusted digital-first incorporation provider. For FiftyFive, this engagement showcased expertise in SaaS engineering, API-driven compliance, and scalable cloud deployment, reinforcing our role as a technology partner for business-critical platforms.

107- Grat tracker/RoleDrop Custom Recruitment Platform Development with AI & Automation

FiftyFive built an AI-powered recruitment platform to solve fragmented job discovery, streamline tracking, and deliver faster hiring outcomes.

Industry - Recruitment Technology

Services - Custom Software Development (web portals, dashboards).

- AI/ML Development (semantic search, OpenAI integration).
- Cloud Engineering (AWS, Dockerized deployments).
- Third-Party Integrations (ATS APIs, Zyte scraping).

About Our Client

A UK-based recruitment technology company focused on graduate hiring, building scalable platforms to help STEM and Finance students discover, apply, and track opportunities with AI-driven personalization and real-time application management.

The Story

The client needed to eliminate inefficiencies in graduate job discovery and tracking. With fragmented sources, no personalization, and missed deadlines, students struggled to manage applications effectively. The client partnered with FiftyFive Technologies to design, develop, and deliver a scalable AI-powered recruitment ecosystem. The engagement covered end-to-end custom software development, AI/ML integration, and cloud deployment.

The Challenges

- Fragmented job data sources across multiple platforms
- No AI-powered personalized recommendations for students
- Lack of centralized tracking for job applications
- Missed deadlines due to absence of real-time alerts
- High manual workload for administrators and recruiters
- Inefficient job data enrichment and inconsistent formats

The Solution

- Custom software development of two portals: Candidate and Admin
- Secure onboarding using Google OAuth and OTP authentication

- AI/ML-powered semantic job search with OpenAI-driven NLP models
- Kanban-style tracker for structured application management
- Real-time alerts and notifications integrated via Mail
- Admin dashboard with ATS API integrations (Workday, Greenhouse, SmartRecruiters)
- Automated job sourcing through webhooks, newsletters, and Zyte scraping
- Data enrichment and normalization pipelines using OpenAI APIs
- Tech stack: React.js, NestJS, MongoDB, Dockerized AWS deployments
- Agile methodology with CI/CD pipelines; delivered by a team of 4 developers and 1 lead

The Outcome

The platform enabled **80% faster job discovery** through AI-driven recommendations and automated alerts. Students improved application success rates with structured Kanban tracking and deadline reminders. Recruiters reduced manual effort through automated scheme management and integrated ATS workflows, cutting administrative overhead significantly. With its modular, cloud-native foundation, the system is positioned for future SaaS scaling, mobile expansion, and employer-focused dashboards—delivering clear business value, operational efficiency, and long-term scalability.

106- The Money Cloud Building Fintech Platforms with API Integrations

Client achieved CoP compliance, stronger API integrations, and modernized dashboard through FiftyFive's secure fintech development and deployment.

Industry - Fintech

Services - Custom Software Development: API integration, dashboard refactoring, bug fixing

Cloud/Platform Engineering: Zero downtime deployment, sandbox & production testing

Security Engineering: OAuth 2.0, HTTPS, rate-limiting

Communication Solutions: SendGrid integration for secure notifications

About Our Client

A UK-based fintech platform delivering international money transfer and foreign exchange services, operating within a highly regulated ecosystem and focused on compliance, secure API integrations, and customer trust.

The Story

The client engaged FiftyFive Technologies to modernize their money transfer platform under strict regulatory timelines. The objectives included ensuring PayUK Confirmation of Payee (CoP) compliance, strengthening Equals Money API integrations, removing outdated providers from the dashboard, and improving security and communication systems. The engagement required a sprint-based delivery approach with a dedicated fintech engineering team, ensuring zero service disruption while upgrading critical workflows.

The Challenges

- Meeting PayUK's regulatory deadline for CoP validation
- Ensuring reliable Equals Money API collaboration and integration stability
- Safe removal of outdated dashboard providers without impacting FX features
- Fixing legacy bugs in recipient verification and transaction workflows
- Strengthening API and platform security to financial-grade standards
- Maintaining uninterrupted operations during upgrades and deployments

The Solutions

- Implemented sprint-based agile delivery with backend engineers, API specialists, QA testers, and DevOps support
- Enhanced Equals Money API integration with new endpoints for balance retrieval and CoP validation
- Redesigned recipient onboarding workflows to handle CoP match/non-match cases with user-friendly error handling
- Refactored dashboard codebase, safely removing outdated integrations while preserving FX functionality
- Integrated SendGrid for secure, logged, and monitored customer communications
- Enforced OAuth 2.0, HTTPS, and rate-limiting to meet fintech-grade security compliance
- Fixed legacy bugs in transaction and verification flows with full regression testing

- Deployed changes via staged releases in sandbox and production, ensuring zero downtime

The Outcome

The client achieved full CoP compliance ahead of regulatory deadlines, avoiding penalties and ensuring trust with financial institutions. API latency was reduced by 35%, improving transaction speed and customer satisfaction. Legacy issues were resolved, and outdated services were safely retired, modernizing the platform while preserving core FX functions. Secure, SendGrid-powered notifications enhanced communication reliability. Overall, the platform achieved greater stability, regulatory compliance, and operational efficiency—strengthening customer trust and supporting sustainable business growth.

105- Ground Water Level Custom Web App Development for Groundwater Analysis on AWS Serverless

Client needed faster groundwater analysis, FiftyFive built a Python-based AWS serverless app delivering secure, automated results within minutes.

Industry - Groundwater and sediment research, environmental technology

Services - Custom Software Development (web application development)

- Cloud Engineering (AWS Lambda, API Gateway, S3)

- Automation (scheduled cleanup, role-based authentication)

About Our Client

A Sweden-based organization specializing in groundwater and sediment research, leveraging digital transformation to simplify scientific analysis and deliver scalable, cloud-based solutions for environmental professionals and researchers.

The Story

The client needed to convert a complex scientific script for groundwater level calculations into a scalable web application. The goal was to deliver fast, accurate results with clear visualizations and downloadable reports while ensuring secure access, cost optimization, and future-ready scalability. FiftyFive was engaged to provide custom cloud engineering and software development services, deploying a team with expertise in Python and AWS serverless technologies.

The Challenges

- Restrict execution of compute services to authenticated users only.
- Execute computationally heavy groundwater analysis without latency.
- Prevent API Gateway request timeouts during data retrieval.
- Optimize cloud resource consumption to avoid idle infrastructure costs.
- Automate scheduled cleanup of outdated generated reports from storage.

The Solutions

- Developed three interconnected services exposed through Python-based APIs.
- Deployed scientific computation scripts on AWS Lambda for event-driven execution.
- Integrated AWS API Gateway for secure, scalable user requests.
- Implemented authentication to enforce role-based access control.
- Optimized Lambda code execution paths to eliminate flow breaks and reduce runtime.

- Configured AWS S3 for report storage and automated deletion using Lambda cron jobs.
- Adopted serverless architecture patterns to ensure resources are consumed only when triggered, achieving high cost-performance efficiency.

The Outcome

The application now generates groundwater analysis results for any selected area within 2–3 minutes, reducing manual processing from several hours to near real-time. Users can visualize layered sediment polygons, analyze groundwater levels, and download structured reports containing charts, graphs, and data tables. The solution cut idle infrastructure costs by leveraging serverless execution and automated cleanup, ensuring scalability without resource waste. This engagement highlights how FiftyFive's custom cloud engineering and software development expertise enabled the client to achieve faster insights, improved cost efficiency, and a future-ready digital platform for environmental data analysis.

104- GECO Jobs Custom Recruitment Platform Development with AI, Subscriptions, and Scheduling

Built a scalable recruitment platform with AI job matching, subscription billing, and scheduling to improve efficiency and revenue growth.

Services - Custom Software Development (MVP to SaaS scaling)

AI/ML Development (job matching engine)

Mobile App Development (Flutter-based candidate app)

Cloud Engineering (AWS + Docker deployment)

Payment Integration (Stripe subscriptions, coupons)

Third-Party Integrations (Google Meet, Microsoft Teams)

Industry - Recruitment Technology / SaaS

About the Client

A UK-based recruitment technology company focused on modernizing job discovery and hiring workflows, building scalable SaaS platforms to support growing candidate bases and subscription-driven recruitment services.

The Story

The client wanted to move beyond keyword-based recruitment and build an AI-powered SaaS platform. They needed a system that could personalize job recommendations, streamline fragmented hiring workflows, support subscription monetization, and handle scheduling and notifications in one scalable ecosystem. FiftyFive Technologies was engaged as the engineering partner to design, develop, and scale the solution from MVP to a multi-portal recruitment platform.

The Challenges

- Matching job preferences beyond keyword search using AI/ML logic
- Fragmented hiring workflows requiring unified pipelines
- Subscription-based job posting with secure Stripe integration
- Smooth interview scheduling with Google Meet/Microsoft Teams integration
- Event-driven notifications across email, mobile, and admin dashboards
- Ensuring modular scalability across candidate, company, and admin portals

The Solution

- Developed three portals (Candidate, Company, Admin) using NestJS backend and Next.js frontend
- Built a Flutter mobile app for candidate access and engagement
- Implemented AI-powered job recommendation engine using OpenAI for preference-based matching
- Integrated Stripe for subscription plans, coupon handling, and recurring billing
- Enabled Google Meet/Microsoft Teams scheduling with automated reminders
- Designed event-driven notifications with cron jobs, push alerts, and email triggers
- Deployed on AWS with Docker, ensuring scalability, security, and modular infrastructure
- Delivered through a dedicated agile team of backend, frontend, mobile, QA, and DevOps engineers

The Outcome

The platform improved job-matching accuracy by 80%, reduced interview no-shows by 60% through automated scheduling, and generated a new revenue stream via recurring subscriptions. Recruiter efficiency increased by 40% with unified pipelines and notifications. Admin oversight reduced operational overhead, while AWS-based scalability supported rapid user growth. With a roadmap for ATS integrations, AI enhancements, and analytics dashboards, the platform now stands as a scalable, revenue-generating recruitment ecosystem—delivered end-to-end by FiftyFive Technologies.

103- Solana Automated Token Staking and Distribution System

A blockchain platform needed automated token staking; FiftyFive built secure smart contract workflows that delivered accurate distribution and scalable execution.

Industry - Blockchain / Web3.0

Services - Custom Software Development, Blockchain Engineering, Smart Contract Development, QA Testing, Performance Optimization

About the Client

The client is a blockchain-driven platform focused on automated token distribution, seeking a secure, scalable, and cost-efficient staking mechanism to validate an MVP for real-time token exchange and transparent reward allocation.

The Story

The client wanted to validate whether automated staking and token distribution could be executed reliably through smart contracts. They needed a Proof of Concept

demonstrating accurate token holding verification, secure execution, and rule-based distribution without manual intervention. FiftyFive was engaged as the technology partner to design and develop the full blockchain workflow, ensure fraud-resistant logic, and build a scalable foundation for future expansion. The engagement required end-to-end engineering, covering smart contract development, backend automation, and performance optimization.

Challenges

- Real-time wallet balance verification to prevent double-spending and ensure accurate token eligibility.
- Smart contract execution security, including fraud detection and minimal transaction cost handling.
- Enforcement of transparent, rule-based staking with time-locked withdrawal restrictions.
- High-volume transaction processing with low latency and congestion control.

The Solution

- Built a custom staking smart contract with gas-optimized execution, automated fraud checks, and secure state transitions.
- Implemented real-time wallet balance verification logic to authenticate staking actions.
- Configured time-locked staking rules ensuring transparent and tamper-proof token distribution.
- Architected parallel transaction processing for high-speed execution under load.
- Deployed blockchain engineers, smart contract developers, and QA specialists to deliver full-cycle development, testing, and validation.

The Outcome

The Proof of Concept successfully demonstrated automated staking-based token distribution with precise tracking, seamless execution, and zero manual oversight. Transaction speed improved significantly due to parallel processing, while optimized contract logic reduced operational costs and prevented fraudulent actions. The client gained a secure and scalable framework capable of supporting larger token ecosystems, improving stakeholder trust and enabling future revenue-generating distribution models. The engagement validated the feasibility of automated token exchange and established a strong technical foundation for the client's next phase of Web3 product development.

102- Quark Media Tech Cloud Infrastructure and Security Solutions

FiftyFive implemented secure hybrid cloud solutions to solve scalability, security, and cost challenges, reducing expenses and improving performance.

Industry - Cloud services and cybersecurity

Services - Cloud Engineering: Hybrid and poly-hybrid architecture deployment (AWS, Azure, VMware)

- Cybersecurity: API security, SOC monitoring, VAPT
- DevOps/DevSecOps: Jenkins, Terraform, Kubernetes, CI/CD automation
- Automation: Infrastructure as code, automated security testing

About the Client

The client is a global cloud services and cybersecurity provider delivering scalable, high-performance infrastructure solutions that enable enterprises across industries to optimize operations, reduce risk, and accelerate digital transformation.

The Story

The client partnered with FiftyFive Technologies to modernize and secure its cloud infrastructure while optimizing costs and scalability. As their business expanded from public to hybrid and poly-hybrid cloud models, they required a dedicated technology partner with expertise across hyperscalers, cloud security, and DevOps. FiftyFive deployed a specialized engineering team to design and implement solutions that aligned with the client's business objectives and compliance needs.

Challenges

- Infrastructure scalability without latency or performance degradation.
- Evolving cybersecurity threats and compliance with industry standards.
- High operational costs from underutilized cloud resources.
- Complex hybrid and poly-hybrid cloud environments requiring seamless orchestration.

Solutions

- Designed and deployed secure hybrid and poly-hybrid cloud architectures across AWS, Azure, and VMware.
- Integrated CDN services for optimized content delivery and reduced latency.
- Implemented advanced cybersecurity frameworks, including API security, SOC monitoring, and VAPT.
- Applied DevOps/DevSecOps practices with Jenkins, Terraform, Kubernetes, and CI/CD automation pipelines.
- Delivered infrastructure-as-code and automated security testing for continuous compliance.
- Provided a dedicated engineering team of cloud architects, DevOps specialists, and security engineers.

Outcome

The engagement resulted in a 40% reduction in operational costs through optimized cloud resource utilization and automation. Application performance improved by 35%, delivering faster response times and enhanced user experience. Robust cybersecurity measures ensured 100% compliance with regulatory standards, safeguarding enterprise data and strengthening client trust. With scalable infrastructure and automated DevSecOps pipelines, the client achieved faster go-to-market cycles for its cloud services, significantly boosting competitiveness and long-term business growth.

101- NorthWest nodes Develop a Scalable Blockchain Reward Automation System

FiftyFive built an automated system to process Layer 2 data, calculate verifier rewards accurately, and support scalable blockchain operations.

Industry - Blockchain / Web 3.0 / Ethereum Layer 2 ecosystem

Services - Custom Software Development

Backend Automation

Blockchain Data Processing

System Architecture & Optimization

About Our Client

The client is a blockchain-focused technology company working on Ethereum Layer 2 adoption, requiring real-time data processing, transparent verifier reward distribution, and scalable infrastructure to support ecosystem growth and operational accuracy.

The Story

The client partnered with FiftyFive Technologies to build a custom, fully automated system for retrieving Ethereum Layer 2 epoch data, linking historical dependencies, and calculating verifier rewards with high precision. With epochs generated every 6 minutes and 21 seconds, delays or missing data directly affected payout accuracy and network trust. FiftyFive acted as the technology partner—designing, engineering, and deploying the backend automation, data processing logic, and transparency layer. The goal was to create a scalable, tamper-proof solution that the client could operate independently while enabling future ecosystem expansion.

Challenges

- Real-time retrieval of Layer 2 epoch data without missed intervals or network-induced delays.
- Dependency mapping between historical and current epochs for reward accuracy.
- High-volume blockchain dataset management requiring optimized storage for speed and cost.
- Need for secure, verifiable, tamper-proof reward distribution with transparent activity logs.
- Requirement for scalable architecture capable of handling continuous data growth.

Solution

- Developed an automated data retrieval engine with retry mechanisms to eliminate missed epochs.

- Engineered reward-calculation logic linking past and current epoch datasets through dynamic dependency mapping.
- Implemented storage optimization using indexing, partitioning, and compression for large blockchain datasets.
- Integrated on-chain verification logs and a secure verifier dashboard for transparent earnings visibility.
- Architected a scalable backend system designed for continuous throughput and future adoption growth.
- Delivered full-cycle custom software development through a dedicated engineering team specializing in backend automation, blockchain data handling, and system performance optimization.

Outcome

The solution achieved 100% automated epoch retrieval, accurate reward calculation across all intervals, and a fully transparent verifier tracking experience. Storage performance improved significantly due to structured optimization, enabling the client to process and scale high-volume blockchain data without degradation. Operational efficiency increased through complete automation, reducing manual intervention while improving payout accuracy and verifier trust. The platform now supports long-term scalability, strengthening the client's market credibility and enabling faster ecosystem adoption.

100- Verifiable Credentials Blockchain Gate Pass Solutions

The client needed secure access control; FiftyFive built a blockchain-based NFT system delivering tamper-proof passes and real-time verification.

Industry - Access control / Security solutions leveraging blockchain

Services - Custom Software Development (full-cycle delivery)

- Blockchain Engineering (smart contracts, NFTs)

- Web & Mobile Development (cross-platform dashboards, QR verification)
- Software Testing & QA (functional reliability, availability)

About the Client

The client is an innovative organization seeking a decentralized gate pass system, requiring blockchain-powered digital credentials to ensure secure, tamper-proof, and role-based access management at scale.

The Story

The client engaged FiftyFive Technologies to replace traditional gate pass systems that were prone to forgery, loss, and misuse. Their goal was a blockchain-based model that could deliver tamper-proof credentials, automated role-based access, and seamless verification across platforms. FiftyFive provided end-to-end custom software development, blockchain engineering, mobile/web integration, and QA to deliver a scalable solution aligned with global security and compliance standards.

Challenges

- On-chain implementation of dynamic role-based access control with secure privilege modification
- Development of non-transferable Soulbound NFTs with time-bound validity and revocation mechanisms
- Simplified NFT issuance and usage for non-blockchain users
- Gasless transaction enablement to eliminate user transaction costs
- Multi-platform real-time verification across web, mobile, and entry systems

Solution

- Engineered smart contracts in Solidity to manage secure role assignment, modification, and revocation
- Deployed Soulbound NFTs with embedded expiration timestamps and batch minting capabilities
- Integrated managed wallet infrastructure to cover gas fees and enable user-friendly, gasless transactions
- Built cross-platform mobile (React Native) and web (React/Node.js) dashboards with QR-based verification modules
- Implemented continuous testing and QA for functional reliability and high availability

Outcome

The engagement delivered a fully decentralized, blockchain-powered gate pass system that reduced fraud risk by 100% through tamper-proof NFTs, cut administrative overhead by 40% with automated role management, and eliminated transaction costs for users. Real-time verification improved operational efficiency, enabling large-scale event onboarding with zero downtime. The project demonstrated FiftyFive Technologies' capability in delivering custom blockchain solutions that drive measurable security, compliance, and cost savings, positioning the client as a leader in secure access management.

99- Authentitas (Trustchain) Gas-Free Blockchain Platform for Digital Publishing

FiftyFive built a blockchain platform with digital passport verification and gas-free transactions, enabling secure journalist identity checks and article authentication.

Industry - Media technology, digital publishing

Services - Custom Software Development

- Blockchain Development (smart contracts, meta-transactions)
- Mobile App Development (Flutter, cross-platform iOS/Android)
- Web Development
- Cloud Engineering (AWS EC2, MongoDB)
- API & Third-Party Integrations (Web3.js)
- Security & Data Protection

About the Client

The client is a media-technology company building a blockchain-powered digital publishing platform that verifies journalist identities, authenticates articles, and enables secure, transparent, gas-free transactions across mobile and web applications.

The Story

The client partnered with FiftyFive to develop a blockchain-based MVP that verifies journalists through digital passport scanning and authenticates articles using immutable smart contracts. The goal was to create a secure, scalable, cross-platform system that improves trust in digital publishing. FiftyFive provided end-to-end engineering across blockchain, mobile, backend, integrations, and cloud to deliver the complete MVP.

Challenges

- Implementing NFC-based biometric passport scanning for journalist identity verification
- Designing Solidity smart contracts for article authentication and digital signatures

- Building a scalable backend for real-time verification and secure data retrieval
- Ensuring seamless functionality across iOS, Android, and web platforms
- Protecting biometric and identity data from security threats
- Enabling meta-transactions for gas-free blockchain interactions

Solution

- Developed a unified Flutter front-end for iOS, Android, and web
- Implemented Solidity smart contracts for identity and article verification
- Integrated NFC passport reading for biometric onboarding
- Enabled QR-based verification linked to blockchain records
- Used meta-transaction architecture with a client-funded wallet for gas-free usage
- Hosted backend on AWS EC2 with MongoDB for secure, scalable data management
- Integrated Web3.js, DMRTD, and APIs for blockchain communication and passport scanning
- Added Sentry for real-time error monitoring

Outcome

The MVP verified journalist identities within seconds and attached immutable blockchain seals to articles, ensuring trust and transparency in publishing. Gas-free transactions reduced operational friction, while cross-platform availability improved accessibility for authors and readers. The platform delivered a secure, scalable foundation for combating misinformation and positioned the client as an innovator in digital media technology.

98- Apica SaaS Modernization (Legacy)

Legacy SaaS platform modernized with microservices, cloud, and APIs, improving scalability, security, performance, and reducing downtime and costs.

About the Client

The client, a Sweden-based SaaS provider, specializes in digital performance monitoring, observability, and automated root cause analysis, ensuring seamless issue resolution and preventing revenue losses for businesses relying on their platform.

The Story

The client's monolithic architecture and outdated technology stack slowed down platform performance and made maintenance increasingly difficult. Their system struggled with scalability, integration issues, and high operational costs, affecting user experience and SLA commitments. They needed a modern, cloud-based architecture that could support seamless updates, better security, and enhanced system performance without disrupting existing functionalities.

Challenges

- **Outdated Monolithic Architecture** – Difficult to scale and maintain.
- **Performance Bottlenecks** – Slow system response and downtime risks.
- **Security Vulnerabilities** – Compliance and data protection concerns.
- **Complex Backend Integrations** – Difficulty in syncing with third-party services.
- **High Maintenance Costs** – Inefficient workflows and frequent bug fixes.

The Solution

FiftyFive Technologies migrated the legacy monolithic system to a microservices-based architecture, ensuring flexibility and scalability. The backend was modernized using .NET Core, Vue.js, and cloud-native technologies to improve performance and security. API-driven integrations streamlined backend processes, and a containerized deployment using Docker and Kubernetes enhanced system reliability. The transition to cloud-based infrastructure reduced maintenance overhead and improved automation through CI/CD pipelines.

The Outcome

The modernized platform became faster, more secure, and scalable, with 40% reduced downtime and 30% lower operational costs. Enhanced API integrations improved backend efficiency, and security upgrades ensured compliance with industry standards. The transformation allowed the client to deliver high-performance digital services, ensuring a seamless experience for end users.

97- Biote Healthcare Software Modernization

Client replaced the outdated monolithic system with FiftyFive's cloud and microservices solution, achieving 40% efficiency and full compliance.

Industry - Healthcare

Services - Legacy Modernization

- Cloud Engineering
- API Development & Integration
- Security & Compliance (HIPAA/GDPR)
- Software QA & Testing

About the Client

The client is a leading Hormone Replacement Therapy provider operating in the healthcare sector, aiming to modernize legacy systems to ensure scalability, compliance, and improved user experience across stakeholders.

The Story

The client relied on a monolithic healthcare system that was outdated, costly to maintain, and unable to scale with modern digital workflows. Manual processes, disconnected databases, and lack of secure integrations created inefficiencies for doctors, patients, and vendors. To address these challenges, the client engaged FiftyFive

Technologies to deliver a full-cycle modernization project, leveraging cloud infrastructure, microservices, and API-driven integrations. The engagement was executed through a dedicated development team specializing in backend, frontend, DevOps, and QA.

Challenges

- Monolithic architecture with limited scalability and high maintenance overhead
- Disconnected patient data across multiple databases causing operational delays
- No support for modern APIs or third-party digital workflows
- Absence of encryption and non-compliance with HIPAA/GDPR regulations
- High downtime and manual processes raising operational costs

The Solution

- Migrated from monolithic to microservices-based architecture using containerized deployments
- Implemented secure cloud infrastructure (AWS/Azure/GCP) with scalable storage and compute services
- Centralized patient records using a modern database system with real-time synchronization
- Developed and integrated REST APIs for telemedicine, e-prescriptions, and vendor management workflows
- Applied end-to-end encryption, role-based access, and compliance monitoring for HIPAA/GDPR standards
- Delivered through a cross-functional team: backend developers, frontend engineers, DevOps, and QA specialists

The Outcome

The transformation reduced operational inefficiencies by 40% and lowered maintenance costs by 30%. Doctors accessed patient records instantly, patients retrieved e-prescriptions without delays, and vendors managed inventory in real time. Compliance risks were eliminated, ensuring full HIPAA/GDPR adherence. The modernization not only improved user experience but also strengthened the client's ability to scale, adapt to future healthcare innovations, and enhance trust among patients, providers, and regulators.

96- Gelato Full-Cycle Software Development for Global Print Platforms

FiftyFive modernized the client's systems, built new digital solutions, and improved global performance, delivery efficiency, and product development predictability.

Industry -Ecommerce

Services -

- Custom Software Development
- Legacy Modernization
- AI/ML Engineering
- Cloud & DevOps
- Agile Consulting
- Team Deployment

About Our Client

The client is a Sweden-based unicorn operating the world's largest printer network, offering personalized print products across 30+ countries with extensive logistics partnerships and high-volume, on-demand production capabilities.

The Story

The client partnered with FiftyFive Technologies as a long-term engineering and consulting provider to modernize core systems, enhance global delivery efficiency, and build scalable digital products. The engagement covered custom software development, mobile app development, legacy modernization, AI/ML engineering, cloud-enabled scalability, DevOps automation, and agile transformation. FiftyFive deployed cross-functional engineering teams to improve performance, accelerate feature delivery, and strengthen the client's global operational capabilities.

Challenges

- Monolithic legacy architecture causing latency, instability, and complex debugging.
- Deprecated frameworks leading to security vulnerabilities and high operational risk.
- Limited scalability for growing global order volume and API integrations.
- Absence of a mobile platform for personalized product creation and editing.
- Inefficient delivery routing for capacity-constrained fleets across multiple countries.
- Unstructured documentation, sprint workflows, and story-pointing affecting development predictability.

The Solution

- Migrated legacy platform to microservices architecture for modular scalability and performance.
- Upgraded outdated frameworks, optimized database operations, and implemented security best practices.

- Built a React Native mobile application using JavaScript, Redux, React Navigation, Jest, PHP, Node.js, and CI/CD via CodeMagic.
- Developed ML models using Python, Solver Pulp, Graph Theory, Matplotlib, Google Maps API, and S3 for secure, scalable data processing.
- Integrated agile methodologies including sprint planning, grooming, documentation restructuring, and story-pointing frameworks.
- Deployed cross-functional teams including mobile engineers, backend engineers, data scientists, QA engineers, DevOps specialists, and business analysts.

The Outcome

The client achieved 50% faster system performance, 60% lower maintenance costs, and a scalable microservices foundation supporting increased global transactions. The mobile app saw widespread adoption across 30+ countries, improving user experience and personalization capabilities. The ML-driven route optimization reduced delivery costs by over 60%, improving fleet utilization and operational efficiency. Agile transformation improved development predictability by 50–60%, accelerating releases and strengthening product quality. Collectively, these initiatives enhanced customer satisfaction, operational resilience, and the client’s global market competitiveness.

94- ExpressVault Build a Secure Cross-Platform Content Management System

The client needed secure multi-device content management; FiftyFive built a cross-platform system that improved collaboration, synchronization, and scalability.

Industry -

Services -

Custom Software Development

Mobile App Development

Web Development

Cloud Engineering
QA & Testing
Third-Party Integrations

About the Client

The client is a technology-focused company building a secure, cloud-based digital content management platform enabling real-time collaboration, encrypted file sharing, and multi-device synchronization for users across mobile, web, and desktop environments.

The Story

The client engaged FiftyFive Technologies to build a unified digital content management platform capable of operating seamlessly across mobile, web, and desktop. They needed real-time collaboration, cross-platform synchronization, and enterprise-level security. The engagement required full-cycle custom software development, cloud engineering, integrations, and performance optimization. FiftyFive provided an end-to-end development team across mobile, backend, cloud, and QA to engineer a scalable, secure, and reliable platform.

Challenges

- Real-time file synchronization across multiple device types.
- Low-latency live collaboration with version control and document commenting.
- Secure backend architecture for large and sensitive data volumes.
- Maintaining consistent performance across mobile, web, and desktop clients.
- Implementing encryption, secure file transfer, and multi-factor authentication.

Solution

- Developed a unified cross-platform application using **Flutter** with selective native implementations.
- Engineered a secure backend with **MongoDB** hosted on **AWS EC2** for scalable storage and synchronization.
- Built real-time collaboration workflows including live editing, commenting, and versioning.
- Applied advanced encryption to secure stored data and file transfers, with two-factor authentication for access control.
- Integrated productivity tools such as **Microsoft Office** and **Google Workspace** for seamless document handling.
- Performed end-to-end QA, performance tuning, and optimization for consistent multi-platform responsiveness.
- Deployed a dedicated engineering team covering mobile, backend, cloud, and QA development.

Outcome

The platform now delivers secure, seamless, and highly responsive digital content management across devices, reducing file access and sync time significantly and improving collaboration efficiency for users. Enterprise-grade encryption strengthened data security posture while minimizing operational risks. Cross-platform consistency improved user adoption and retention, and scalable cloud architecture enabled the client to support growing data volumes without performance degradation. With a robust, integration-ready ecosystem, the client now offers a competitive, productivity-focused solution that enhances user workflows and strengthens long-term platform value, demonstrating FiftyFive's capability to deliver secure, high-performance custom software at scale.

93- Vagus Health AI-Driven Breath Monitoring Platform

AI breath monitoring platform built by integrating real-time ECG signals, personalized AI insights, and blockchain-backed data ownership to improve user engagement and platform scalability.

Industry - Health care

Services -

- Custom Software Development
- AI/ML Engineering
- Backend & Cloud Engineering
- Blockchain Engineering
- Security & Compliance

About the Client

The client is a digital health company developing an AI-driven breath monitoring platform with Apple Watch integration, real-time ECG analytics, and blockchain-enabled data ownership for global wellness users.

The Story

The client engaged FiftyFive Technologies to build a full-cycle wellness platform capable of capturing breath and ECG signals, processing them in real time, and generating personalized AI insights. They required a secure, scalable system connecting Apple Watch sensors, mobile interfaces, backend services, and blockchain-ledger components for user-owned data. FiftyFive provided end-to-end engineering across mobile, backend, AI/ML, blockchain, and cloud layers to deliver a production-ready solution. The engagement focused on transforming their concept into a robust, multi-layered AI-powered ecosystem.

Challenges

- Real-time ECG and breath signal acquisition via Apple Watch native APIs
- Low-latency integration between WatchOS, iOS native modules, Flutter app, and backend
- Accurate ECG preprocessing, filtering, and HRV computation on-device and server-side
- Scalable backend architecture for continuous data ingestion, AI analysis, and storage
- Development of AI-driven breath pattern detection and coaching logic
- Integration of an AI chat assistant for guided breath exercises
- Blockchain implementation for secure V-NFT generation and user-owned data
- Enforcement of strict data privacy, encryption, and secure identity management

Solution

- Implemented native WatchOS and iOS modules for ECG signal capture and preprocessing
- Built a Flutter mobile app for unified cross-platform UX, visualizations, and coaching flows
- Developed scalable backend services for real-time data pipelines, AI inference, and analytics
- Created AI/ML models for breath-flow classification, HRV insights, and personalized recommendations
- Integrated an AI assistant for interactive coaching and exercise guidance

- Implemented blockchain workflows enabling V-NFT generation, data ownership, and secure asset handling
- Applied end-to-end encryption, role-based access, and secure communication protocols

Outcome

FiftyFive delivered a fully engineered wellness platform that increased data processing speed, reduced signal interpretation latency, and improved accuracy of breath insights. The client gained a scalable, AI-driven ecosystem supporting new engagement models, blockchain-based data monetization, and higher user retention. The solution strengthened their product reliability, enabled faster feature rollout, and positioned the platform for wider market expansion and revenue growth.

92- NIBE myUplink Heating and HVAC IoT Applications Development

FiftyFive developed IoT, mobile, and web apps enabling remote heating control, streamlined maintenance, and global scalability.

Industry - Heating, HVAC, and IoT solutions

Services - Custom Software Development (mobile apps, web platforms, backend services)

- IoT Engineering (device connectivity, data handling)
- Cloud Engineering & DevOps (Azure IoT Hub, CI/CD pipelines)
- Software Testing & QA (automated UI testing, performance validation)

About the Client

The client is a Sweden-based global leader in heating and HVAC IoT solutions, serving homeowners, professionals, and enterprises with advanced energy-efficient technologies across multiple international markets.

The Story

FiftyFive Technologies partnered with the client to deliver a full-cycle digital ecosystem of mobile and web applications for homeowners, technicians, and administrators. The engagement involved custom software development, IoT integration, cloud deployment, and automated QA. A dedicated team of engineers, testers, and DevOps specialists worked in Agile sprints to ensure scalability, reliability, and global usability.

Challenges

- Cross-platform performance for iOS, Android, and web applications.
- Presenting complex heating and IoT data in user-friendly formats.
- Testing across multiple device brands with limited physical hardware.
- Scaling to support 100,000+ connected devices with real-time updates.
- Integrating with third-party APIs and smart home platforms.
- Reducing manual testing effort while maintaining release velocity.

Solution

- Developed native mobile apps (Swift, Kotlin) for homeowners and technicians.
- Built web portals (React, Redux) for service partners and administrators.
- Implemented backend with .NET Core, Node.js, and C# for scalability.

- Used Azure IoT Hub, Service Bus, Cosmos DB, PostgreSQL, and Redis Cache for cloud infrastructure.
- Automated UI testing with Java, Appium, and TestNG; integrated CI/CD pipelines with GitHub Actions.
- Deployed a dedicated team: mobile developers, backend engineers, QA automation specialists, and cloud architects.

Outcome

The client achieved 40% reduction in manual testing costs, 30% faster installation and commissioning for technicians, and accelerated release cycles by two weeks per sprint. Remote monitoring improved customer satisfaction, while multilingual rollout increased adoption in over 10 countries. Scalability supported 100,000+ IoT devices with <2s latency, ensuring reliable real-time control. The project delivered measurable cost savings, enhanced operational efficiency, and positioned the client for global market growth.

90- Husmus AI Platform Development for Smart Tenant Screening and Rental Management

Stabilized a multi-persona insurance SaaS platform with AI assessments and Stripe billing, improving accuracy by 35% and reducing manual effort by 80%.

Industry - PropTech / InsurTech / SaaS Platforms

Services - Custom Software Development

Payment & Integration Engineering

Backend & Platform Engineering

QA & Automation

About the Client

The client is a multi-persona PropTech and InsurTech SaaS platform enabling landlords, agents, tenants, and partners to purchase insurance products, run AI-driven tenant assessments, and manage subscriptions across multiple properties.

The Story

As the client expanded to support 8+ insurance products across 10+ insurance partners, its platform required architectural stabilization and scalable SaaS restructuring.

Disconnected insurance, assessment, and billing flows created operational complexity and regression risk. The client engaged FiftyFive as its engineering and QA partner to modernize role-based workflows, strengthen payment reliability, and enable subscription-led growth. The mandate included AI-driven assessments, Stripe billing, Rewardful referrals, and full regression protection.

Challenges

- Role-based access control across Landlord, Agent, Tenant, and Partner personas
- Unified commerce orchestration for 8+ insurance products and 10+ partners
- Stripe integration for one-time, monthly, yearly subscriptions with 3DS and failure scenarios
- Rewardful referral integration without disrupting revenue-critical payment flows
- AI-driven tenant assessment automation with account-level quota enforcement
- Complex pricing logic for bundled insurance and incremental rent protection charges
- Policy snapshot handling for legacy users after product updates
- UI state inconsistency and dashboard synchronization issues
- High regression risk due to frequent partner, pricing, and product changes

The Solution

- Delivered full-cycle custom software development using role-based architecture and metadata-driven persona handling (Clerk-based authentication and role storage)
- Engineered unified insurance purchase APIs with Stripe for one-time and subscription billing, including 3DS handling and automated failure testing
- Integrated Rewardful backend logic for referral attribution and rewards tracking
- Implemented AI-driven tenant assessment workflows with automated shortlisting and quota enforcement
- Refactored pricing and calculation engines to support multi-product bundles and incremental charges
- Built policy snapshot services to preserve backward compatibility
- Strengthened QA automation using Python, Selenium, and Pytest regression suites
- Stabilized dashboards and standardized UI terminology for consistent user journeys

The Outcome

The client achieved a 35% improvement in tenant assessment accuracy and reduced manual model management by 80%. Stripe-powered payment reliability increased revenue continuity across one-time and subscription flows, including secure 3DS handling. Unified multi-product journeys enabled scalable onboarding of 10+ insurance partners, while automated regression testing reduced release risk and improved platform stability, accelerating subscription growth and partner confidence.

89- Qopla Smart Restaurant Solution

The client needed to integrate order management and predictive analytics, and FiftyFive delivered a custom POS system, AI insights, and improved operational efficiency.

Industry - Food Tech

Services - Custom Software Development: Mobile & Web Apps, POS System, Hybrid Apps (React Native)

- DevOps: Infrastructure automation (Kubernetes, Jenkins)
- AI/ML: Predictive sales, inventory optimization, and personalized meal recommendations
- Quality Assurance (QA): End-to-end testing, integration testing, performance testing, security testing

About the Client:

The client is a fast-growing Swedish food-tech startup that provides a platform integrating order management, inventory optimization, and AI-driven insights to enhance restaurant operations and customer experiences.

The Story:

The client sought a solution to streamline restaurant operations by integrating third-party order platforms, predicting sales, optimizing inventory, and ensuring system reliability during peak times. They needed a custom platform that could provide predictive analytics, personalized recommendations, and robust system performance.

Challenges:

- **Order Management:** Difficulty in managing orders from multiple third-party platforms.
- **Sales & Inventory:** Inaccurate sales predictions and inefficient inventory management.
- **Personalization:** Lack of personalized meal recommendations for customers.

- **System Reliability:** Ensuring consistent performance under high demand.
- **Security:** Safeguarding sensitive customer data and ensuring compliance.

Solution:

- **Custom Software Development (Mobile & Web App/DevOps):** Built a Restaurant Management System with a POS platform and hybrid mobile apps (React Native). Integrated third-party order platforms and automated infrastructure management using Kubernetes and Jenkins for scalability and reliability.
- **AI-Driven Operations (Machine Learning):** Implemented predictive sales analytics and real-time cooking estimates using XGBoost. Integrated machine learning models to offer personalized meal recommendations and optimize inventory.
- **Quality Assurance (QA Testing):** Conducted comprehensive testing, including end-to-end, integration, performance, and security testing to ensure system stability and data security.

Outcome:

- **Cost Savings:** React Native development reduced costs by accelerating time-to-market and streamlining development.
- **Operational Efficiency:** AI-powered analytics improved inventory management, cutting food waste by 15%.
- **Customer Satisfaction:** Personalized recommendations and seamless POS integration enhanced the user experience, leading to a 20% increase in customer retention.
- **System Reliability:** Rigorous QA testing ensured 99% uptime, supporting high-volume usage during peak periods.

88- NIBE Ventilator **Smart Ventilation Systems**

Industry - Energy-efficient ventilation and heating systems.

Services - Custom Software Development

Software Testing & QA

Cloud & Connectivity Validation

CI/CD & DevOps

About the Client

The client is a Sweden-based global leader in energy-efficient ventilation and heating systems, employing over 17,000 people and serving cold-climate markets with advanced indoor comfort solutions.

The Story

The client required a complete software upgrade and QA validation for their ventilation systems to ensure reliability, seamless hardware–software integration, and improved user experience. They engaged FiftyFive Technologies to design mobile applications, strengthen system connectivity, and provide end-to-end functional testing. A dedicated engineering team was deployed, covering mobile development, embedded integration, and QA.

Challenges

- Inconsistent scheduling and data mismatches in trend view.
- Alarm failures under device error conditions.
- Cross-device UI discrepancies in plant view.
- Notes not syncing across user sessions.

- Connectivity issues over TCP, direct, and cloud modes.
- Role-based access control assigning incorrect permissions.
- Requirement for cost-efficient cross-platform apps using a single code base.

The Solution

- Built Android and iOS apps using Kotlin, Swift, Flutter (GetX) with clean architecture and shared repository.
- Developed core controller features with C#, integrating seamlessly with existing hardware.
- Conducted extensive QA using Scope tools, automated test scripts, and real-time validation.
- Verified scheduling, alarms, trend view, and notes syncing under multiple scenarios.
- Calibrated airflow and temperature sensors for accurate monitoring.
- Validated connectivity across TCP, direct, and cloud environments.
- Implemented secure role-based access control for multi-level user permissions.
- Established CI/CD pipelines for faster releases and reduced defects.

The Outcome

The project delivered a production-ready ventilation system with 99.8% functional accuracy across Android, iOS, and HMI platforms. Connectivity stability improved by 65%, defect rates dropped by 40%, and synchronization issues were fully eliminated. Time-to-market accelerated by 3 months, enabling the client to launch ahead of competitors. The enhanced UI/UX increased end-user adoption by 30%, while long-term maintenance costs were reduced by 25%. These results strengthened the client's market

position and demonstrated FiftyFive's ability to deliver scalable, high-quality engineering solutions.

87- NIBE Develop Your Unified Customer Portal

FiftyFive built and validated a unified portal that replaced manual workflows, improved partner operations, and enabled multi-country expansion.

Industry - Heat pump manufacturing (B2B)

Service -

Custom Software Development

Cloud Engineering / DevOps

Automation

Software Testing & QA

Analytics Integration

About the Client

The client is a Sweden-based B2B heat pump manufacturer serving multiple European markets, requiring a scalable digital platform to centralize spare parts, documentation, troubleshooting workflows, and multi-country partner operations.

The Story

The client needed to replace manual, phone-based ordering and fragmented tools with a unified Customer Portal capable of supporting high-volume service operations. They partnered with FiftyFive Technologies as their full-cycle engineering provider for architecture, development, DevOps, automation, and performance validation. A dedicated cross-functional team delivered the portal using a microservices-driven approach to ensure reliability, scalability, and localization accuracy across multiple regions. The engagement focused on building a production-grade platform while establishing an automation framework to accelerate global rollout.

Challenges

- Manual ordering flows dependent on phone coordination
- No centralized repository for spare parts, documentation, or software files
- Fragmented legacy tools requiring separate installations and manual updates
- Lack of automated error reporting and operational monitoring
- Multi-country workflow inconsistencies and localization differences
- Microservices interactions varying across regional configurations
- High-traffic performance risks during expansion to new markets

The Solution

- Built microservices-based Customer Portal with React.js frontend
- Developed GraphQL APIs integrated with PostgreSQL for structured product data
- Deployed Kubernetes clusters using Azure DevOps CI/CD for scalable releases
- Implemented cron-based error reporting with Microsoft Teams webhook alerts
- Integrated Microsoft Clarity for partner usage analytics and behavior tracking
- Created Cypress automation suites achieving 90% test coverage across countries
- Applied parameterized, region-specific test execution for language/routing validation
- Performed load and stress testing using Artillery for high-traffic readiness
- Optimized QA pipelines enabling continuous deployment across environments

The Outcome

The platform achieved 60% partner onboarding in the first year, exceeding the 40% target. Automated workflows reduced manual support load, improved spare parts discovery, and eliminated dependency on phone-based coordination. QA automation reduced manual testing effort by over 60%, enabling faster, safer releases during multi-country expansion. Performance validation ensured the portal could handle traffic spikes across new markets, giving the client a scalable, reliable foundation for long-term growth and stronger partner adoption.

86- NIBE JPI End-to-End QA Services for IoT Climate Control Platforms

We resolved subscription and payment issues through automated QA, reducing failures and improving operational efficiency across the client's IoT platform.

Industry - IoT Energy Technology

Services-

- Quality Assurance & Testing
- Automation Engineering
- Technical Consulting
- Platform Validation

About the Client

The client is a Sweden-based IoT energy technology company delivering smart climate control systems across Europe, supporting residential users, technicians, and administrators through a multi-region digital ecosystem.

The Story

The client expanded its IoT-driven climate control platform with multiple applications for consumers and service partners. As the product scaled across European markets, they required a structured QA partner to stabilize subscription workflows, streamline global

payments, and automate repetitive test cycles. FiftyFive Technologies was engaged as the dedicated QA and engineering partner to implement automated testing, validate subscription logic, and ensure reliable transactions across regions. The focus was to strengthen platform performance without altering existing architecture.

Challenges

- Complex multi-tier subscription logic requiring validation across consumer and professional interfaces.
- Google API-based address validation errors impacting user onboarding and requiring removal post-release.
- Migration from Nets to Stripe for secure global payment processing.
- Recurring payment failures in Norway and Germany requiring root-cause analysis and resolution.

The Solution

- Implemented automated API and UI testing utilizing Cypress and Postman for recurring validation.
- Conducted end-to-end subscription and e-commerce testing including trials, renewals, VAT rules, and regional pricing flows.
- Automated Service Partner workflows to reduce manual regression cycles and accelerate release readiness.
- Established continuous QA processes with structured smoke, regression, and sanity testing for each deployment.

The Outcome

FiftyFive's QA framework stabilized subscription operations, reducing support tickets by improving accuracy of billing and renewals. Payment reliability increased after Stripe migration, lowering regional failures and improving successful transactions across

Europe. Automated Service Partner workflows cut manual testing effort by over 40%, resulting in faster release cycles and reduced operational overhead. Earlier issue detection improved deployment quality, strengthening user trust and supporting scalable growth of the client's IoT climate control platform.

85- MyStudyWeb Student centric educational platform

The client's legacy EdTech platforms were modernized by FiftyFive with secure architecture, improved UI/UX, and higher scalability.

Industry - Educational Technology (EdTech)

Services - Custom Software Development: Backend modernization, code restructuring, API development

- UI/UX Design: Student-centric navigation, gamified learning, visual dashboards
- Cloud/Database Engineering: Database redesign for scalability and performance
- Software Testing & QA: Parallel legacy support during migration
- Security Engineering: JWT authentication, Spring Security implementation

About our client

The client is a nationwide EdTech provider supporting 400+ schools, offering digital platforms for mathematics, Swedish, and English to students, teachers, parents, and administrators.

The Story

The client managed two learning platforms with rich content but an outdated Java, Struts, and JSP codebase. Performance, usability, and scalability issues limited adoption across schools. They partnered with FiftyFive Technologies to modernize backend architecture, improve security, and redesign the UI/UX for student-centric engagement. A dedicated Agile team of full-stack developers, UI/UX designers, and QA engineers delivered full-cycle modernization while maintaining uninterrupted services on the legacy system.

Challenges

- Legacy Java, Struts, JSP framework with complex monolithic codebase
- Poor responsiveness and low scalability from inefficient database design
- End-of-life components creating ongoing security risks
- Limited API support restricting integrations with modern tools
- Complex multi-user hierarchy lacking structured navigation and access control
- Outdated UI/UX design reducing student engagement and learning outcomes
- Performance tracking system lacked clear analytics and easy accessibility

The Solution

- Backend modernization with Spring Boot, Spring JPA, and Next.js for scalable architecture
- Security upgrade using JWT-based authentication and Spring Security
- Code restructuring into modular components for maintainability and faster delivery
- Database redesign for high concurrency and faster query performance
- API layer development to enable seamless third-party integrations
- UI/UX redesign with subject-based navigation, gamified learning paths, and visual dashboards
- Progress tracking dashboards including rankings, analytics, and time-spent metrics
- Agile delivery by full-stack engineers, UI/UX designers, and QA specialists with parallel legacy support

The Outcome

The modernization achieved 40% faster load times, 60% lower maintenance costs, and 50% higher concurrent user capacity. Student engagement increased significantly through gamified dashboards and structured navigation, driving higher adoption across schools. Administrators and parents gained better oversight through structured access controls and performance analytics. Strengthened security and scalability enabled the platform's sustainable nationwide rollout. This engagement demonstrated FiftyFive Technologies' repeatable model for EdTech modernization, highlighting its ability to deliver secure, scalable, and student-first digital learning platforms worldwide.

84- Logical Contracts Redesigning a Legal Document Platform

We redesigned the client's legal document platform, improved workflows and AI integration, and delivered faster creation time with higher usability.

Industry - Legal Technology

Services - UI/UX Design, Front-end Consulting

About our Client

The client is a UK-based legal technology firm specializing in automated document generation, serving businesses that require streamlined, compliant, and user-friendly digital workflows for managing complex legal paperwork.

The Story

The client's existing document generation platform faced usability, workflow, and information architecture challenges that limited adoption and slowed internal operations. They engaged FiftyFive to redesign the platform's UX and build a scalable, consistent design system aligned with modern legal tech standards. A dedicated offshore UI/UX and front-end consulting team collaborated through agile sprints, user testing,

and iterative design improvements. The goal was to reduce cognitive load, enhance AI-assisted workflows, and prepare the product for future engineering scalability.

Challenges

- High cognitive friction due to fragmented information architecture and complex interface structures
- Inefficient admin workflows with low-affordance input patterns and unclear task hierarchy
- Poor visual hierarchy across collaboration states and document lifecycles
- Inconsistent microcopy, UI patterns, and interaction models violating established usability heuristics
- Weak integration between AI-assisted features and the UI, causing workflow interruptions
- Limited progress visibility due to absence of status indicators, wayfinding cues, and navigation logic

The Solution

- Rebuilt information architecture using card sorting, hierarchical layouts, and optimized content grouping
- Designed customizable admin panels with structured input flows and improved information density
- Implemented progressive disclosure in document templates to simplify complex legal inputs
- Integrated AI features through contextual microinteractions, guided input prompts, and structured user flows

- Developed a scalable atomic design system including components, states, and interaction patterns
- Added real-time collaboration indicators, status badges, and document lifecycle progress tracking
- Created engineering-ready UI specifications and component libraries to support future frontend development

The Outcome

The redesigned platform reduced document creation time by 60% and increased user engagement and dwell time by 45%. Administrator efficiency improved significantly through optimized workflows and clearer task structures. Enhanced information scent and streamlined navigation helped users manage complex legal requirements more confidently, resulting in an 85% improvement in System Usability Scale (SUS) scores. The platform is now structurally ready for scalable engineering development, faster feature rollout, and improved customer adoption, directly strengthening the client's product value and market competitiveness.

83- Resume Builder-Build a Centralized Resume Management System for Your Team

Enhancing resource management and proposal generation through intuitive UI/UX design and automated workflows.

Industry - Software Development / IT Services

Services -

- Custom Software Development
- UI/UX Design
- Web Platform Development
- QA & Software Testing

About our Client

The client is a global software development company providing outsourcing services, requiring a centralized platform to manage developer profiles, maintain project data, and generate consistent, client-ready resumes for proposal submissions.

The Story

The client relied on scattered documents and manual processes to maintain developer profiles and project histories, resulting in slow, inconsistent, and error-prone resume creation. Proposal teams struggled to update skills, technologies, and project information across multiple formats, affecting the speed and quality of client submissions. The client engaged FiftyFive Technologies as their engineering partner to design and develop a scalable, UI/UX-driven resume management system with automated workflows. A dedicated team of UI/UX designers, frontend engineers, backend developers, QA testers, and a project lead executed the project end-to-end.

Challenges

- Multiple resume versions stored across disconnected systems
- No unified database for developer skills, technologies, and project histories
- Manual formatting inconsistencies across resumes and proposal documents
- No real-time update mechanism for profiles and projects
- Lack of technology-based filtering for skill-specific searches
- High dependency on manual data entry, increasing errors and delays

The Solution

- Designed an intuitive UI/UX framework for resume creation and profile management

- Built a centralized database to store developer details, skills, and project metadata
- Developed dynamic, customizable templates ensuring consistent branding
- Implemented real-time editing and auto-sync capabilities across all profiles
- Created a technology tagging and filtering engine for skill-based search
- Added structured project management workflows and verification controls
- Integrated standardized input fields using dropdown menus and controlled vocabularies
- Established role-based access for HR, resource managers, and proposal teams
- Conducted full-cycle QA, including functional, UI, and data validation testing

The Outcome

The platform reduced resume creation and update time by **70%**, eliminated formatting inconsistencies, and ensured all developer profiles reflected verified, real-time project data. Proposal teams could now generate client-ready resumes within minutes, improving proposal turnaround speed and increasing the client's ability to respond quickly to outsourcing opportunities. The system enhanced operational efficiency, reduced manual errors, and strengthened the client's brand consistency across all submissions directly improving stakeholder confidence and proposal win rates.

82- The cobbler ULTRA- Build a Real-Time Padel Platform with AI Match Analytics

FiftyFive developed a unified padel platform integrating bookings, real-time communication, and AI analytics to streamline coordination and boost user activity.

Industry - Sports technology

Services -

- Full-cycle software development
- - Architecture design
- - Mobile app development (React Native)
- - Backend development (.NET)
- - Real-time communication integration (GetStream)
- - Cloud notifications integration (Firebase)
- - AI analytics integration (PlayByPoints)
- - iOS Live Activity implementation

About the Client

The client is a technology-driven sports platform offering court bookings, real-time player communication, AI match analytics, and reward-based engagement for padel players seeking performance tracking and seamless digital interaction.

The Story

The client partnered with FiftyFive Technologies to build a unified, data-driven padel platform capable of handling bookings, real-time interactions, and AI-powered performance insights. Their goal was to create a seamless user experience where players could coordinate matches, verify scores, receive live updates, and earn rewards. FiftyFive was engaged to deliver full-cycle engineering, from architecture to development and integration, ensuring the product met high standards of scalability, responsiveness, and real-time performance.

The Challenges

- Low-latency court booking flow requiring optimized request handling.
- Real-time in-app communication with stable, high-throughput message delivery.
- Integration of AI-based shot recognition and analytics workflows.

- Accurate score verification dependent on synchronized player inputs.
- Tier-based loyalty system requiring structured rules and progression logic.
- Continuous live updates using iOS Live Activity for real-time match visibility.

The Solution

- Engineered a cross-platform application using React Native for unified mobile delivery.
- Built a scalable backend with .NET to manage bookings, match data, and user operations.
- Integrated GetStream for low-latency chat and instant player communication.
- Implemented Firebase for real-time notifications and event triggers.
- Connected PlayByPoints to deliver AI-driven shot detection and analytics.
- Added iOS Live Activity for dynamic match status and real-time scoring updates.
- Delivered full-cycle development, architecture design, and integration aligned with best engineering practices

The Outcome

The platform achieved high user responsiveness, fast booking completion, and reliable real-time communication. AI-powered insights increased player engagement, while accurate score verification improved user trust. The loyalty system strengthened retention, and live updates boosted session activity throughout matches. Overall, the client gained a high-performing, scalable digital product that enhanced player experience, supported continuous usage, and strengthened their position as a tech-enabled padel platform.

81- Mondido Custom Payment Integration Development for Shopify & WooCommerce

FiftyFive Technologies built a secure, microservices-based payment integration for Shopify and WooCommerce, improving scalability, compliance, and transaction reliability by 35%.

Industry - FinTech

Services - Custom Software Development

Cloud Engineering

About the Client

The client is a next-generation payment solutions provider offering secure, modular, and compliant payment systems for Shopify and WooCommerce merchants, simplifying online transactions through advanced integrations and flexible architecture.

The Story

The client partnered with FiftyFive Technologies to enhance its e-commerce payment infrastructure by developing a redirection-based payment flow that strengthens security, scalability, and compliance. The goal was to build a modular, microservices-driven backend capable of supporting multiple payment methods while maintaining a seamless user experience. FiftyFive deployed a dedicated engineering team to design, develop, and integrate the system with Shopify and WooCommerce, ensuring real-time communication and fault-tolerant performance across platforms.

Challenges

- Implementing a secure redirection-based payment flow compatible with Shopify and WooCommerce APIs.
- Managing multiple payment gateways within a microservices architecture.

- Ensuring seamless synchronization between client-side UI and backend services.
- Maintaining scalability for new payment method integrations and merchant-specific customizations.
- Enforcing compliance with data security and transaction standards.

The Solution

- Developed modular microservices using **Node.js** and **Express.js** for backend architecture.
- Integrated Shopify and WooCommerce APIs for real-time transaction data exchange.
- Deployed **AWS Lambda** and **Docker containers** for scalable, serverless execution.
- Implemented **RESTful APIs** for communication between services and third-party gateways.
- Established CI/CD pipelines with **Jenkins** and **GitHub Actions** for automated deployment.
- Conducted comprehensive QA, load testing, and end-to-end security validation.

The Outcome

The project delivered a robust, future-ready payment system that improved merchant onboarding speed by **40%**, enhanced transaction reliability by **35%**, and reduced API downtime by **50%**. The modular design enabled faster integration of new payment methods, improved UI customization flexibility, and ensured compliance across all regions. Through this engagement, FiftyFive Technologies empowered the client with a scalable backend architecture that supports evolving payment demands, reinforcing their market competitiveness and customer trust.

80- Artefact **Enterprise Data Platform Development**

FiftyFive provided Azure DevOps engineers to build a centralized data platform, enabling secure governance, integration, and efficiency gains.

Industry - AI and Data Consulting

Services - Custom Software Development (data accelerator framework)

- Cloud Engineering (Azure PAAS/SAAS integration)
- Data Engineering (cataloging, governance, metadata standardization)
- DevOps (automated provisioning pipelines)
- Team Extension (Azure DevOps engineers)

About the Client:

The client is a Middle East–based AI and Data Consultancy, recognized as a global leader in enabling enterprise-wide data-driven transformation, governance, and accelerated AI adoption across industries.

The Story:

The client engaged FiftyFive Technologies to provide two Azure DevOps engineers for building a centralized data platform. The requirement was to unify sub-entities under one governance framework, enabling standardized data and metadata management. FiftyFive's role was to design and implement a Whole-of-Entity solution, ensuring secure, compliant, and automated data integration between on-prem sources and Azure services. The engagement showcased FiftyFive's expertise in custom software development, data engineering, and team extension for enterprise-grade projects.

Challenges:

- Rising demand for governed data handling across multiple sub-entities.
- Absence of a cloud-native platform for data collection and processing.
- Networking and connectivity issues between on-prem systems and Azure services.

- Limitations in Microsoft Purview, including collection caps.
- Compatibility conflicts among Azure PAAS services.
- Requirement for secure data cataloging, lineage, and quality management.

Solution

- Delivered a data accelerator framework integrating Azure PAAS and SAAS services.
- Implemented Microsoft Purview for enterprise-wide cataloging, lineage, and governance.
- Automated provisioning of Azure services using DevOps pipelines.
- Established private endpoints for secure connectivity between on-prem sources and Azure.
- Designed restricted access controls and role-based governance for compliance.

Outcome:

The client launched a fully governed enterprise data platform powered by the data accelerator. FiftyFive's engineers enabled end-to-end automated data flow, eliminating manual processes and reducing integration time by 40%. Secure private networking mitigated compliance risks, while standardized cataloging and metadata governance improved operational efficiency by 30%. The project demonstrated how FiftyFive's custom software development and team extension services help enterprises implement scalable, compliant, and business-impacting data solutions.

79- TTEK Ferry Booking Platform

The client needed a scalable ferry booking system, and FiftyFive rebuilt the platform end-to-end, improving performance, usability, and high-traffic event handling.

Industry - Transportation/ Travel

Services -

- Custom Software Development
- Frontend Development
- Backend Development
- UI/UX Design
- Database Engineering
- Quality Assurance

About the client

The client operates a large-scale ferry booking platform in Abu Dhabi, serving daily commuters, tourists, and event-driven travel, and required a modernized digital system to improve accessibility, performance, and user experience.

The Story

The client needed to upgrade their existing ferry booking system to support growing demand and event-specific traffic. They approached FiftyFive Technologies to redesign the platform, improve usability, streamline booking flows, and strengthen technical scalability. The engagement involved full-cycle custom software development across frontend, backend, database engineering, UI/UX, and QA. The goal was to build a future-ready system while ensuring minimal friction across user journeys, including Arabic-speaking passengers.

Challenges

- Designing a fully responsive, mobile-first interface with Arabic language and RTL support.
- Implementing flexible event-based booking workflows for high-traffic concerts.
- Optimizing booking steps to reduce navigation friction and click depth.
- Ensuring platform scalability to handle peak-load traffic efficiently.

The Solution

- Rebuilt the frontend using Angular 18 with optimized component rendering for high performance.
- Developed a modular .NET Core 8 microservices architecture enabling independent service scaling and maintainability.
- Implemented Microsoft SQL Server for structured data storage, high-volume transaction handling, and optimized query performance.
- Delivered a redesigned UI/UX system with simplified booking flows and enhanced accessibility for bilingual users.
- Applied continuous QA cycles covering functional testing, performance validation, responsiveness, and multilingual behavior.

The Outcome

The upgraded booking platform achieved significantly faster page response times, reduced booking steps, and improved conversion for both regular and event-based services. Scalability enhancements enabled the system to handle large concert-related traffic without downtime or performance drops. User satisfaction increased due to the intuitive interface and minimized booking friction. The modern architecture strengthened long-term maintainability and reduced operational bottlenecks, helping

the client improve service reliability, handle peak loads efficiently, and deliver a seamless booking experience across diverse user groups.

78- shoora Build Your Real-Time Fleet Management Platform

The client needed real-time fleet data at scale, and FiftyFive delivered a cloud-based system that improved accuracy, uptime, and throughput.

Industry - Transportation

Services -

Custom Software Development

Cloud Engineering

Data Engineering

DevOps

QA & Testing

About the Client

The client is a global fleet management platform focused on road safety and real-time vehicle monitoring, serving transport operators with cloud-based analytics, automated insights, and advanced driver-state monitoring capabilities.

The Story

The client engaged FiftyFive Technologies to scale and modernize their cloud-driven fleet management system using a dedicated engineering team. The goal was to improve global telemetry accuracy, reduce data latency, and handle massive real-time vehicle streams without performance loss. FiftyFive delivered full-cycle backend, cloud, data engineering, UI development, and DevOps support to enhance platform stability and readiness for large-scale adoption. The engagement focused on strengthening the system's reliability, usability, and ability to process continuous, high-volume fleet data across regions.

Challenges

- Achieving sub-second global delivery of real-time vehicle telemetry.
- Ensuring highly accurate geolocation tracking with continuous updates.
- Managing high-throughput streaming data without performance bottlenecks.
- Enabling auto-scaling for unpredictable traffic spikes across fleets.
- Maintaining system uptime during continuous deployments.

Solution

- Built backend services using **Python** and **Django** for optimized API execution and structured vehicle data processing.
- Implemented **PostgreSQL** for high-performance relational storage and geolocation data consistency.
- Deployed **Confluent Kafka** to stream, buffer, and process large-scale real-time events.
- Developed a responsive **React-based web interface** for operational dashboards and live driver-state monitoring.
- Deployed the platform on **AWS EKS** with auto-scaling, load balancing, and automated failover.
- Implemented DevOps workflows for CI/CD, zero-downtime deployments, and cluster-level optimization.
- Provided a **dedicated engineering team** across backend, frontend, data engineering, DevOps, and QA.

Outcome

The platform now delivers real-time fleet data with significantly improved precision and

sub-second latency. Streaming throughput increased by over 60%, ensuring smooth handling of continuous high-volume telemetry. Auto-scaling on AWS reduced downtime incidents by 90%, improving operational reliability for fleet managers. With enhanced dashboards and continuous monitoring, decision-making speed increased, safety compliance improved, and the client achieved higher platform adoption across their transport partners, strengthening both operational efficiency and market competitiveness.

75- PCMC Modernize Your Training Documentation System

FiftyFive built a digital documentation and approval platform that replaced manual workflows, improved collaboration, and reduced processing time by 50%.

Industry - EdTech

Services -

- Custom Software Development
Cloud Engineering
- Workflow Automation
- Quality Assurance
- Architecture & System Design

About the Client

The client is a leadership and organizational development firm delivering structured coaching programs using standardized learning models, serving multiple enterprise engagements requiring secure, compliant, and scalable documentation workflows.

The Story

The client relied on manual documentation, fragmented approvals, and unstructured workflows, creating delays and limited visibility across training engagements. They engaged FiftyFive Technologies to design and develop a fully digital platform that automated document generation, standardized approval flows, and enabled secure multi-stakeholder collaboration. Our engineering team implemented a scalable backend, cloud deployment model, and real-time communication features to support high-volume

coaching operations. The goal was to streamline operations, reduce manual dependency, and establish a scalable digital foundation for future growth.

Challenges

- Manual document processing causing inconsistent turnaround times and operational bottlenecks.
- Absence of a centralized repository for document tracking across engagements.
- Need for scalable, multi-tenant architecture with strong security and compliance controls.
- Lack of real-time, multi-user review and approval capability.
- Limited auditability due to missing activity logs and workflow traceability.

Solution

- Developed a full-stack platform using Node.js (backend APIs), React (frontend interface), MongoDB (document storage), and AWS cloud infrastructure.
- Implemented template-driven document generation with automated API workflows.
- Built hierarchical approval pipelines with role-based access control (RBAC) and granular permissions.
- Added real-time notifications, activity logs, and audit trails to support compliance and governance.
- Deployed scalable cloud architecture with secure authentication, automated backups, and multi-client support.
- Provided a dedicated engineering team including backend developers, frontend developers, QA analysts, and a cloud architect.

Outcome

The platform reduced documentation processing time by 50%, directly lowering operational effort and improving delivery speed across engagements. Centralized tracking and audit logging improved governance, compliance, and transparency for enterprise users. Scalable AWS deployment enabled the system to securely support multiple clients without performance degradation. Real-time collaborative workflows accelerated approvals and improved accuracy, while the structured UI and automated processes increased user adoption and reduced dependency on manual interventions. Through this solution, the client achieved faster turnaround, higher service reliability, and a stronger digital foundation for long-term growth.

74- Geco Intuitive Hiring Platform (UI/UX)

About the Client

Our client is a platform designed to streamline recruitment processes for organizations of all sizes. It simplifies workflows for recruiters and HR professionals by centralizing job postings, applicant tracking, and hiring activities into a single, user-friendly interface. The platform caters to diverse teams, ensuring seamless collaboration and compliance with data protection regulations.

The Story

The client aimed to address inefficiencies in recruitment caused by juggling multiple tools and disjointed workflows. They wanted to create an all-in-one solution that combines intuitive design with powerful features, enabling recruiters to track applicants, collaborate efficiently, and reduce errors. This platform not only serves as a job creation tool but also provides a robust Applicant Tracking System (ATS) to help hiring teams make faster, data-driven decisions.

Challenges

- **Disjointed Workflow:** HR teams relied on multiple systems, leading to inefficiencies.
- **Complex Interfaces:** Existing ATS platforms were confusing and required training.
- **Applicant Tracking:** Tracking candidates across stages was time-consuming and error-prone.

- **Collaboration Barriers:** Feedback and communication were fragmented and misaligned.
- **Data Security:** Ensuring compliance with GDPR and protecting sensitive information was critical.

Solutions

The platform introduced a centralized dashboard to manage job postings, track applicants, and oversee workflows. Streamlined job posting templates saved time, while intuitive applicant profiles offered a comprehensive view of candidates. Customizable workflows allowed teams to adapt the platform to their processes. Built-in messaging and feedback tools enhanced collaboration, and robust security protocols ensured GDPR compliance.

Outcomes

The platform improved efficiency by reducing time spent on applicant management and streamlining hiring workflows. Its intuitive design led to high user satisfaction, while collaborative tools enhanced team communication. Recruiters tracked candidates more effectively, enabling faster, transparent decision-making. The platform's compliance with data regulations ensured security and scalability for growing teams, making it a versatile solution for evolving hiring needs.

72- StratisfyPro AI Trading Platform

Enabled no-code trading strategy creation by developing an AI-powered platform with real-time backtesting and scalable cloud deployment.

Industry - Strategy management and financial technology

Services - Custom Software Development (end-to-end engineering, platform build)

- AI/ML Development (GPT-4o integration, domain-specific NLP, RAG implementation)

- Cloud Engineering (AWS EC2 deployment, scalability, low-latency infrastructure)
- Third-Party Integrations (broker APIs for execution)

About the Client

The client is a global strategy management platform combining AI software, expert coaching, and proven methodologies to help enterprises align teams, execute faster, and achieve measurable growth outcomes.

The Story

The client aimed to simplify algorithmic trading for non-technical users by creating a no-code trading assistant. They partnered with FiftyFive Technologies to design and deliver an AI-powered platform that could translate natural language inputs into executable trading strategies, provide real-time backtesting, and scale to meet high user demand. FiftyFive deployed a dedicated engineering team to build, integrate, and optimize the end-to-end solution.

Challenges

- Non-technical users lacked coding expertise for algorithmic trading model creation.
- Stock market terminology required domain-specific natural language processing.
- High concurrency demands required real-time backtesting under low latency.
- Query resolution needed instant, context-specific responses.
- Backend infrastructure required scalable, fault-tolerant deployment.

The Solution

- Built AI-powered trading assistant on Flask framework.
- Integrated GPT-4o for automated strategy generation.

- Deployed RASA NLU for domain-specific stock market queries.
- Applied Retrieval-Augmented Generation (RAG) with vector database for instant knowledge retrieval.
- Enabled real-time backtesting engine with a library of pre-configured strategies.
- Enabled dual-environment deployment allowing users to test strategies in demo mode before executing in live market.
- Scaled infrastructure on Amazon EC2 to ensure high availability and low latency.
- Delivered end-to-end custom software development, AI/ML integration, and cloud engineering through a dedicated team.

The Outcome

The platform empowered non-technical users to design and execute trading strategies without coding, reducing entry barriers and accelerating adoption. Real-time backtesting increased efficiency by 60%, while scalable infrastructure ensured uninterrupted performance during high traffic. Dual-environment deployment gave users flexibility to validate strategies safely in demo mode before switching to live markets, boosting confidence and retention. Broker integrations enabled seamless trade execution, improving satisfaction. By leveraging FiftyFive's expertise in AI/ML development, cloud engineering, and custom software delivery, the client accelerated go-to-market, reduced operational costs, and enhanced competitive positioning in the financial technology space.

71- JBots- AI-Powered Conversational Solutions

FiftyFive developed a scalable AI-powered conversational assistant for real-time user interactions, enhancing engagement and personalization.

Industry - Creative Industries / AI

Services - Custom Software Development: Full-stack backend pipeline.

- AI/ML Development: Integration of AI-driven responses with phonetic data.

- Cloud Engineering: Google Cloud Platform (GCP), Firebase.

- Automation: Terraform for deployment.

About the Client

The client is a UK-based singer and AI enthusiast, seeking to integrate cutting-edge technology into creative fields by developing a conversational assistant that mimics their unique voice and speech patterns.

The Story

The client engaged FiftyFive Technologies to develop a custom AI-driven conversational assistant capable of handling multi-modal inputs like text and audio. The goal was to create a system that mimicked the client's voice while providing a seamless, personalized user experience. Over time, the initiative evolved from an experimental project into a robust backend pipeline that integrated Google Cloud Platform (GCP) and Firebase to handle real-time, scalable conversational interactions.

Challenges

- Converting text and audio inputs into structured data while preserving context and intent.
- Handling high-frequency queries with low-latency performance.
- Maintaining integrity across Firebase Realtime Database transactions.
- Efficient retrieval and formatting of phonetic data.
- Ensuring secure and scalable deployment with Terraform automation.
- Tuning the Google Gemini API for personalized, context-aware responses.

Solution

- Modular backend architecture powered by **Google Cloud Run** for flexible, scalable deployment.
- Asynchronous communication using **Pub/Sub** for handling high-frequency queries.
- Secure real-time data synchronization with **Firebase** Realtime Database.
- A **Flask-based framework** to manage service interactions and data routing.
- Integration with a third-party **phonetic API** to enhance AI-generated responses.
- Automated deployment using **Terraform** for infrastructure scalability and security.

Outcome

The solution delivered a high-performance, scalable backend pipeline capable of processing real-time conversational queries. By integrating AI-driven responses with phonetic metadata, the system enhanced user interaction, offering a more personalized and engaging experience. The modular architecture ensures future adaptability and scalability, providing the client with a flexible foundation for ongoing enhancements. This solution enabled the client to deepen their engagement with users while maintaining low-latency performance at scale.

70-Contemi Custom RPA Solutions

FiftyFive built a custom RPA system replacing costly tools, automating workflows, and reducing operational costs by 40%.

Industry - Insurance, Wealth Management, Banking, Capital Markets

Services - Custom Software Development (RPA development, workflow automation, parser building)

- DevOps & CI/CD (orchestration with Airflow, Jenkins integration)
- Quality Assurance & Testing (reliability and secure deployment)

About the Client

The client is a global digital solutions provider in insurance, wealth management, banking, and capital markets, delivering compliance-driven technology services to improve efficiency, scalability, and operational agility worldwide.

The Story

The client partnered with FiftyFive Technologies to replace costly off-the-shelf RPA tools with a secure, customizable, and scalable automation system. Their objective was to automate repetitive tasks such as data extraction, form-filling, and file management through a Python-script-based platform. The engagement was delivered by a dedicated FiftyFive team comprising backend developers, RPA specialists, DevOps engineers, and QA experts. The goal was to unify workflows across applications, reduce licensing costs, and ensure complete data privacy.

Challenges

- High licensing and operational costs of market RPA platforms.
- Data privacy risks associated with third-party RPA solutions.
- Requirement for script-based workflow automation instead of GUI-driven tools.
- Complex handling of diverse input file types and formats.

The Solution

- Designed and implemented a custom RPA system optimized for cost efficiency and security.
- Developed logic framework to map and process input files into standardized data structures.
- Built robust parser to unify SQL and JSON into consistent formats.
- Delivered Python-script editor with pre-defined UI templates for workflow creation.
- Orchestrated workflows and scheduling using Apache Airflow.
- Integrated FastAPI, Asyncio, Selenium, and Jenkins for scalable automation and CI/CD pipelines.
- Applied dedicated QA and DevOps practices to ensure reliability and secure deployment.

The Outcome

The client achieved a 40% reduction in RPA licensing and operational costs, with processing time for repetitive tasks reduced by 60%. Standardized data handling minimized errors across departments and improved compliance confidence. The custom platform enabled greater agility, allowing new workflows to be deployed 2x faster compared to previous tools. This engagement demonstrates how FiftyFive Technologies delivers secure, business-impacting automation through tailored software development services, helping global enterprises reduce costs while scaling operations efficiently.

69- qx/5Tech Lab Astro Dating App (MVP- UI/UX)

The client needed a secure astrology-based dating MVP, FiftyFive delivered custom development, compatibility engine, and faster market launch.

Industry - Dating technology sector

Services - Custom Software Development (MVP development, backend engineering)

- UI/UX Design (mobile-first design, intuitive workflows)
- Cloud & Security (encryption, authentication)
- Software Testing & QA (iterative sprints, validation)
- Community Management Tools (moderation, abuse-prevention)

About the Client

The client is a mobile-first platform in the dating technology sector, leveraging Vedic astrology and modern software to deliver compatibility-driven matchmaking for global users seeking meaningful relationships.

The Story

The client engaged FiftyFive Technologies to design and develop a Minimum Viable Product (MVP) with intuitive UI/UX and secure architecture. The goal was to merge traditional astrology with advanced technology, enabling compatibility-driven partner discovery. FiftyFive deployed a dedicated team covering UI/UX, backend engineering, and QA to deliver a scalable solution within a rapid development cycle.

Challenges

- Designing a mobile-first interface integrating complex Vedic astrology workflows.
- Engineering an accurate algorithm for compatibility using planetary datasets and zodiac parameters.
- Securing sensitive user data through enterprise-grade encryption.
- Establishing credibility with validated astrological insights.

- Building community features with moderation and abuse-prevention mechanisms.

Solutions

- Delivered an MVP using Flutter for cross-platform mobile development and Node.js + MongoDB for backend scalability.
- Designed a compatibility engine incorporating astrological calculations and real-time data processing.
- Implemented AES-256 encryption and OAuth 2.0 authentication for user data security.
- Integrated an admin dashboard for astrologer validation and algorithm oversight.
- Developed moderation tools with automated flagging and role-based community management.
- Followed Agile methodology, executing iterative sprints for rapid deployment and feedback integration

Outcomes

The client launched a secure, scalable MVP that achieved a 40% faster time-to-market compared to standard app builds. Strong encryption and transparent moderation increased user trust and retention by 35%, while validated compatibility insights drove higher daily active usage by 28%. FiftyFive's full-cycle development approach enabled the client to establish market credibility, attract early adopters, and position itself for revenue growth through premium subscriptions and partnerships.

68- DUBAI TECHNOLOGIES (Yas Island) Island Tourism (Mobile app)

Client needed a unified tourism app so FiftyFive built a cross-platform solution, improving visitor engagement, transport efficiency, and operations.

Industry - Tourism / Hospitality / Mobility

Services - Custom Software Development – Mobile App Development (cross-platform).

Backend Integration – Real-time data synchronization.

AI & Personalization – Recommendation engine.

UI/UX – Interactive maps, multilingual support.

Security – Facial recognition for contactless access.

Consulting & Offshore Team Extension.

About the Client

The client is a globally recognized island destination offering large-scale tourism services, attracting millions of international visitors annually with diverse attractions, events, and mobility experiences.

The Story

The client wanted to eliminate fragmented visitor journeys by integrating attractions, transportation, bookings, and access management into one mobile platform. FiftyFive Technologies was engaged as the dedicated offshore partner to design, develop, and deploy a full-cycle mobile solution. Our team provided cross-platform expertise, backend integration, and advanced personalization to deliver a seamless and scalable digital experience.

Challenges

- Consolidating multiple tourism services into a unified platform.
- Real-time shuttle tracking with synchronized bookings.
- Multi-language support, including Arabic (right-to-left rendering).
- Facial recognition for secure, contactless entry across devices.
- Personalization engine for visitor-specific recommendations.
- High-frequency data synchronization with low battery consumption.
- Offline functionality for low-connectivity areas.

Solution

- Built a cross-platform app using Flutter and Dart for iOS and Android.
- Integrated Firebase and Hive for secure user data and real-time updates.
- Applied Socket Connection for precise shuttle tracking.
- Enabled facial recognition for seamless authentication and access.
- Developed interactive maps with live shuttle and service updates.
- Implemented personalization engine for AI-driven recommendations.
- Delivered multi-language support including Arabic RTL.
- Enabled offline mode to ensure usability in low-network zones.

Outcome

The application reduced visitor wait times by 40%, improved transportation efficiency by 35%, and streamlined bookings across multiple services. Personalized itineraries increased user engagement by 50%, while contactless access reduced manual interventions by 60%. For the client, the platform enhanced revenue potential through higher visitor satisfaction, optimized resource allocation, and data-driven operational decisions. By delivering a full-cycle mobile solution, FiftyFive demonstrated its capability to provide scalable offshore development services, positioning the client as a leader in digital tourism innovation.

67-Authentitas (Trust Chain) Journalism with blockchain (Mobile)

Subhead- A blockchain-based MVP app ensured journalist verification, article authenticity, and transparency through **e-Passport** NFC, biometric scanning, and real-time verification.

The Story

The client wanted a platform that could combat misinformation and enhance trust by verifying authors and their articles. With features like NFC-based biometric passport verification, blockchain integration, and a unique verification seal, the app aimed to guarantee the authenticity of published content. Their goal was to provide a seamless user experience for authors and agencies alike, while ensuring robust security and scalability across iOS, Android, and web platforms.

Challenges

- Implementing NFC-based biometric passport scanning for accurate journalist authentication.
- Developing smart contracts for unique digital signatures and seamless blockchain registration.
- Creating a backend system to retrieve and display verification details dynamically.
- Ensuring app functionality across Android, iOS, and web platforms.
- Safeguarding sensitive user data, including biometric and identity details.

Solution

We developed the MVP app using Flutter for a unified front-end experience and Solidity for blockchain integration. Features included biometric passport verification via NFC, a unique seal for article authentication, and QR codes linked to blockchain records. The backend, hosted on EC2, retrieves and displays real-time verification details through MongoDB. APIs like DMRTD ensured seamless passport scanning, while Sentry enabled real-time error tracking.

Outcome

The app successfully streamlined journalist verification, ensuring transparency and trust in digital publishing. Authors gained a reliable platform to authenticate their work, while readers benefited from verifiable content. The app's intuitive design, multi-platform support, and robust security established it as a groundbreaking solution in media technology.

Quote- Our client is a media-tech startup dedicated to enhancing trust in journalism by leveraging blockchain technology for secure journalist verification and authenticating digital content with innovative tools and features.

64- Kriva.ai/ FemHealth: AI-Powered Mobile App for Personalized Menstrual Health

FiftyFive developed a cross-platform AI-powered app to help women manage menstrual health, offering personalized insights and secure data tracking

Industry - Health Tech

Services - Custom Software Development

- Mobile App Development (cross-platform)
- AI Integration (OpenAI's LLM)
- Cloud Database Management (Firebase)
- Fraud Prevention (Local_Captcha)

About the Client

The client is a health tech company focused on revolutionizing menstrual health management for women by providing AI-driven, personalized health insights and symptom tracking through a mobile application.

The Story

The client sought to develop a comprehensive mobile app that would empower women to track their menstrual cycles and manage related health issues using AI-powered, personalized recommendations. FiftyFive Technologies was engaged to deliver a custom software solution, building an app that could address these challenges through cutting-edge technology and ensure data security. The app was designed to offer real-time health insights and holistic recommendations, tailored to each user's needs.

Challenges

- No effective, scalable solution for tracking menstrual health and managing gynecological conditions.
- Need for a cross-platform mobile app that supports both iOS and Android.
- Requirement for AI-driven recommendations based on personalized user data.
- Ensuring secure storage of sensitive health data and preventing fraud.
- Scalability and integration with external technologies for enhanced functionality.

Solution

- Developed a cross-platform mobile application using Flutter for seamless iOS and Android support.
- Integrated OpenAI's LLM to provide AI-powered, personalized health insights based on user data.
- Implemented symptom logging and tracking capabilities to monitor menstrual cycles and related symptoms.
- Utilized Firebase for secure, scalable cloud storage and data management.
- Incorporated Local_Captcha for fraud protection and enhanced data security.
- Applied Flutter Riverpod for state management, Go Router for efficient navigation, and Node.js for backend development.

Outcome

The app, currently in MVP development, is expected to revolutionize menstrual health management by providing AI-driven insights and personalized recommendations. Once fully launched, the app will empower women to better manage their menstrual health, improving user engagement and retention. The client is poised to capture a significant share of the health tech market, enhancing their brand reputation and user base through this innovative solution.

63- GecoMVP : Custom Green Job Platform Development

A dedicated green-industry hiring platform built with cross-platform development, AI matching, and secure workflows to improve recruitment efficiency for the client.

Industry - Green industry / Sustainability-focused recruitment technology

Services -

- Custom Software Development
- Web Development
- Mobile App Development
- AI Integration
- Cloud Engineering
- QA & Testing

About the Client

The client is a sustainability-driven organization building a specialized digital recruitment platform for the green industry, requiring secure, scalable, cross-platform software to connect environmentally focused professionals with hiring companies.

The Story

The client needed a dedicated hiring ecosystem tailored to the green sector, where traditional job platforms lacked sector-specific workflows and personalization. They partnered with FiftyFive Technologies to design and build a full-cycle, cross-platform recruitment solution with strong security, AI-driven matching, and GDPR-compliant infrastructure. Our team delivered end-to-end development across web, mobile, backend, cloud, and QA, ensuring a scalable product that could support future growth. The objective was to create a seamless experience for job seekers and employers while enabling data-driven hiring efficiency in the sustainability domain.

Challenges

- No sector-specific digital infrastructure for green-industry recruitment workflows.
- Traditional platforms lacked technical workflows for pipeline management and AI-driven matching.

- Need for GDPR-aligned data security, multi-provider authentication, and protected user management.
- Requirement for unified performance across web, iOS, and Android platforms.
- Demand for scalable architecture to support high-volume job postings and candidate activity.

Solution

- Developed the web application using Next.js and mobile apps using Flutter for iOS and Android.
- Built a Node.js backend with MongoDB for scalable, schema-flexible data management.
- Implemented AI-driven job matching and personalized recommendation models.
- Designed employer tools including Kanban-style application pipelines and analytics dashboards.
- Integrated Firebase Cloud Messaging for cross-platform push notifications.
- Configured AWS SES for secure transactional email delivery.
- Enabled multi-provider authentication with GDPR-compliant data protection and access control.
- Delivered a dedicated engineering team across frontend, backend, mobile, cloud, QA, and UI/UX.

Outcome

The platform launched successfully across all platforms, increasing candidate–employer matching efficiency by enabling automated AI recommendations and structured hiring workflows. The client reduced manual screening time, improved hiring pipeline visibility,

and expanded outreach to sustainability-focused professionals. The scalable, secure infrastructure supported rapid early adoption, strengthened the client's market position, and accelerated recruitment processes within the green industry.

63- CL2API - Groundwater Analysis Application Development

A scientific script was transformed into a secure, scalable groundwater analysis system that reduced processing time and improved operational efficiency.

Industry - Environmental science / groundwater analysis

Services -

- Custom Software Development
- Backend Engineering
- Cloud Engineering

About the Client:

The client is a scientific innovator in Sweden developing groundwater-level analysis tools using sedimentation data, requiring a reliable, scalable, and secure digital system to deliver accurate environmental insights to authenticated users.

The Story:

The client had a Python-based scientific script for calculating groundwater levels but needed a production-grade, cloud-native application capable of handling heavy computations securely and efficiently. They partnered with FiftyFive Technologies to transform the script into a scalable service with optimized performance, automated data workflows, and authenticated access. Our role was to provide custom software development, backend engineering, and cloud deployment, ensuring the solution could support continuous scientific analysis without resource bottlenecks. The engagement

focused on building a robust architecture that improved speed, optimized compute usage, and delivered reliable results for end users.

Challenges:

- Enforcing user authentication before executing groundwater-level calculations.
- Managing high-load Python computations without exhausting runtime resources.
- Preventing API Gateway timeouts caused by long-running execution cycles.
- Automating cleanup of outdated sedimentation result files from storage.
- Ensuring scalable compute capacity during peak calculation requests.

The Solution:

- Deployed the Python groundwater analysis script on AWS Lambda for event-driven, auto-scaling execution.
- Implemented an API service using AWS Lambda to manage authentication, request routing, and execution triggers.
- Optimized the Python code to reduce execution time and computation overhead.
- Configured Lambda concurrency scaling to handle variable workloads efficiently.
- Used a scheduled cron job (EventBridge) to automatically delete outdated groundwater-level data.
- Added monitoring and performance logging through CloudWatch for ongoing visibility.

The Outcome:

The optimized architecture reduced groundwater-level processing time to 2–3 minutes, enabling faster environmental assessment and improved decision-making for end users. Automated resource scaling lowered compute overhead, while secure authentication

ensured controlled access. Scheduled cleanup eliminated unnecessary storage costs, improving long-term system efficiency. The client now operates a stable, production-grade application that transforms scientific logic into a reliable digital service, demonstrating how FiftyFive's engineering expertise adds measurable operational value and accelerates scientific workflows.

62- Prospexs Modernize Your Training Documentation System

FiftyFive built an AI-powered SDR platform automating outreach and scheduling, delivering 6x more meetings and 8x higher conversions.

Industry - SaaS

Services - Custom Software Development (MERN-based platform development)

- AI/ML Engineering (personalization, NLP-driven automation)
- Cloud Engineering (AWS deployment, GDPR compliance)
- API & CRM Integration (CRM systems, calendar tools)
- Security & Compliance (GDPR/ISO, encryption, audits).

About the Company

The client is a SaaS innovator delivering AI-driven sales automation solutions, targeting global enterprises with hyper-personalized outreach, seamless CRM integrations, and GDPR-compliant access to 700M+ B2B contacts.

Story

The client partnered with FiftyFive Technologies to design and build an autonomous Sales Development Representative (SDR) platform. The goal was to automate prospecting, outreach, and scheduling while ensuring compliance, scalability, and market differentiation. FiftyFive deployed a dedicated engineering team specializing in MERN stack, AI/ML, and cloud solutions to deliver a platform integrating OpenAI models, real-time data sources, and CRM ecosystems. The engagement focused on achieving business growth by aligning advanced AI with secure, enterprise-grade architecture.

Challenges

- Training AI models for accurate business profile interpretation and personalized email generation.
- Scaling and maintaining a 700M+ B2B contacts database with real-time query performance.
- Enabling seamless integrations with diverse CRM and calendar tools.
- Implementing continuous analysis of LinkedIn profiles and external business datasets.
- Ensuring GDPR/ISO compliance with end-to-end encryption and security audits.
- Building trust in autonomous SDR automation while balancing hyper-personalization with automation.

The solution

- Applied MERN stack with Next.js , Node.js, Express, and MongoDB for scalable, modular development.
- Integrated OpenAI APIs for NLP-driven email personalization and lead analysis.
- Built microservices-based architecture with load balancing and real-time data streaming.
- Deployed AWS (EC2, S3, RDS) in European data centers for GDPR compliance and scalability.
- Developed secure APIs for CRM and calendar integrations.
- Conducted regular penetration testing and encryption audits to ensure compliance and data security.
- Delivered project through a cross-functional team of MERN developers, AI/ML engineers, and cloud specialists.

Outcome

The platform delivered measurable business impact, enabling up to 6x more meetings and 8x higher conversions. By combining AI-driven personalization with scalable architecture and compliance-first engineering, FiftyFive empowered the client to increase outbound efficiency, build trust in global markets, and achieve faster revenue growth.

61- NUG: Custom Cannabis E-Commerce Platform Development

FiftyFive developed a centralized e-commerce system that solved regulatory, data, and multi-store challenges while driving measurable revenue growth.

Industry - E-commerce

Services -

- Custom Software Development
- Mobile App Development
- Backend Engineering
- UI/UX Design
- Data Migration
- Platform Integration

About the Client

The client is a major U.S.-based cannabis enterprise managing end-to-end product lifecycle operations, including research, cultivation, extraction, distribution, and retail across multiple company-owned dispensaries in California and New York.

The Story

The client engaged FiftyFive Technologies to build a unified e-commerce and mobile platform capable of supporting multi-state operations, regulatory workflows, and cross-location customer experiences. Their existing systems were fragmented, store-specific, and unable to synchronize customer data or support central compliance. FiftyFive was tasked with developing a centralized, scalable, and regulation-ready architecture that could consolidate identity verification, streamline purchases, and introduce a multi-store rewards engine. Our role was to deliver full-cycle custom software development, backend engineering, UI/UX design, and platform integration.

Challenges

- Integrating multi-store sales flows governed by varying taxation and regional cannabis regulations.

- Consolidating customer identity verification and profile data into a unified, compliant system.
- Enabling real-time synchronization of customer and regulatory data across all retail locations.
- Migrating existing users into a centralized identity and rewards platform without downtime.
- Designing an intuitive shopping and reorder interface optimized for high-frequency mobile users.
- Implementing a multi-tier reward system with cross-category configuration logic.

Solution

- Built a unified e-commerce and mobile purchase platform enabling cross-location ordering via a single app.
- Implemented centralized identity verification workflows supporting in-store and online compliance checks.
- Engineered real-time regulatory data synchronization across stores for zero-gap compliance.
- Migrated all existing customer accounts to a centralized database with seamless continuity.
- Designed a simplified mobile-first UI/UX for browsing, ordering, tracking, and reordering.

- Developed a configurable points-based reward engine with store-level acceleration controls.
- Added personalized shopping logic and real-time order tracking to enhance user engagement.
- Initiated development of an automated notification engine for targeted customer communication.

Outcome

Within six months, online sales increased by 40% and customer retention improved by 25%. Administrative workload decreased by 30%, and inventory efficiency improved by 35%. Compliance incidents dropped to zero due to synchronized verification workflows. The rewards system generated a 50% rise in repeat purchases and a 20% increase in average order value, demonstrating the direct business impact of FiftyFive's engineering and platform modernization.

60- SustainLab: SaaS Dashboard Modernization and Performance Improvement

FiftyFive upgraded the client's sustainability platform by improving dashboard performance, modernizing the UI, and enabling scalable future-ready integrations.

Industry - Sustainability Tech / SaaS Platform

Services -

Custom Software Development

UI/UX Implementation

Cloud Engineering

Architecture Modernization

Performance Optimization

Migration Services

About the Client:

The client is a SaaS-based sustainability data platform helping organizations automate environmental data collection, processing, and visualization to accelerate sustainability initiatives and improve operational efficiency across growing enterprise users.

The Story:

The client engaged FiftyFive Technologies to modernize their sustainability data platform, which struggled with slow dashboard performance, limited scalability, and outdated UI/UX. Their existing tech stack restricted the addition of new visualization features and caused delays in rendering charts, drag-and-drop actions, and widget resizing. FiftyFive provided a dedicated engineering team across frontend, backend, UI/UX, and cloud to rebuild core components, implement the new Figma-driven interface, and migrate the platform to a modern, scalable architecture. The goal was to deliver a faster, intuitive, future-ready dashboard experience while enabling the client to onboard more enterprise customers.

Challenges:

- Slow dashboard rendering with >10 chart components
- Unresponsive drag-and-drop and widget resizing operations
- Complex UI overhaul aligned with new Figma components
- Outdated tech stack: Angular v9, Chart.js, legacy architecture
- Limited scalability, weak integration readiness, and security gaps
- Need for new chart settings: treemap, donut, theme-based configuration

Solutions:

- Formed a dedicated engineering team: Angular frontend, Node.js backend, UI/UX, DevOps, QA

- Implemented modular architecture supporting future micro frontends
- Rebuilt dashboard with grid-based layout for alignment and responsive interactions
- Added advanced chart capabilities: treemaps, donut charts, theme selectors
- Migrated platform to Angular v14, Node.js, Firebase, and AWS for higher performance and scalability
- Optimized rendering engine to support real-time interactions and reduced load times
- Applied complete UI/UX rebuild based on updated Figma design system

Outcomes:

The platform now renders complex dashboards up to 60% faster, with drag-and-drop and resizing responsiveness improving by over 40%. The modular architecture enables the client to roll out new micro-frontend-based features without disrupting existing workflows. The upgraded tech stack improved scalability and security, supporting onboarding of larger enterprise environments. The enhanced UI/UX increased user adoption and reduced configuration time, enabling the client to deliver a more competitive, future-ready sustainability product strengthening their market position and improving operational efficiency for their customers.

58- Wade Insights SaaS Development for Data Management

FiftyFive developed a unified SaaS data platform, integrating diverse sources, automating processing, and reducing release cycles by 60%

Industry - Data platform management / SaaS

Service - Custom Software Development (SaaS platform build)

- Cloud Engineering (Azure infrastructure)
- Data Engineering (ingestion pipelines, processing workflows)
- UI/UX Development (frontend redesign with React)
- DevOps & Automation (release management, automated testing)
- Security & Compliance (enterprise-grade frameworks).

About the Client

The client is a leading data platform management company providing enterprise-grade SaaS solutions that centralize and integrate diverse data sources for organizations across multiple industries worldwide.

The Story

FiftyFive Technologies partnered with the client to design and develop a unified SaaS platform for end-to-end data pipeline management. The goal was to consolidate diverse data sources, enable advanced processing, and strengthen governance. This engagement was delivered through a dedicated development team including frontend engineers, backend developers, data engineers, cloud architects, and QA specialists. By leveraging our expertise in custom SaaS development, cloud engineering, and data management, we built a scalable, future-ready platform aligned with the client's growth needs.

Challenges

- Integrating heterogeneous data sources (data lakes, warehouses, APIs, cloud storage).
- Maintaining flexibility for future scalability and modular expansion.
- Enabling seamless interoperability across pipeline processes.

- Delivering intuitive, device-agnostic frontend navigation.
- Ensuring robust data security, compliance, and governance.

Solutions

- Designed modular, scalable architecture supporting multiple data sources.
- Built ingestion pipelines for automated, high-throughput data integration.
- Implemented advanced data transformation and processing workflows.
- Revamped frontend with React for responsive, user-centric experience.
- Introduced data catalog system for searchability and governance tracking.
- Automated deployment/testing with release management pipelines.
- Applied enterprise-grade security and compliance frameworks.
- Tech stack: React (frontend), Python with FastAPI (backend), SQL Server (database), Microsoft Azure (cloud).

Outcome

The platform reduced data integration and processing time by 40%, improving efficiency across business units. Automated deployments cut release cycles by 60%, enabling faster feature delivery. Enhanced governance and security compliance increased organizational trust and reduced operational risks. With an intuitive interface and scalable infrastructure, the client achieved cost efficiency, streamlined data management, and future-ready growth capability. This engagement showcases FiftyFive's expertise in custom SaaS development, data engineering, and cloud solutions, delivering measurable business impact for global enterprises.

57- CAIT - AI Tutoring Platforms

The client needed a scalable AI tutoring platform, FiftyFive built and deployed it, boosting engagement and reducing educator workload.

Industry - Education Technology

Services - Custom Software Development (full-cycle backend and platform build)

- AI/ML Engineering (modules for question answering, course delivery, conversational support)
- Cloud Engineering (Google CloudRun deployment, scalability)
- System Integration (Intent Recognizer, Context Engine)
- Logging & Tracking (local file storage, custom logging)

About the Client

The client is a global online learning provider delivering AI-driven tutoring solutions, focused on creating interactive, scalable, and personalized digital education experiences for students and institutions worldwide.

The Story

The client engaged FiftyFive Technologies to address the limitations of traditional e-learning platforms, which often lack personalization, real-time responsiveness, and measurable engagement. Our mandate was to design, build, and deploy a scalable AI tutoring platform capable of replicating classroom-like interactivity while reducing manual workload for educators. FiftyFive provided a dedicated engineering team to deliver full-cycle custom software development, AI/ML integration, and cloud deployment.

Challenges

- Designing adaptive learning paths for diverse student requirements.
- Enabling real-time responsiveness with minimal latency.
- Achieving scalable, fault-tolerant system performance.
- Integrating conversational AI modules into a seamless architecture.
- Maintaining session continuity for personalized experiences.
- Reducing manual workload for educators without compromising quality.

Solution

- Built backend architecture with **FastAPI** for lightweight, high-performance API development.
- Applied **asyncio** to enable concurrent, asynchronous execution for speed.
- Implemented **WebSockets** for low-latency, two-way real-time communication.
- Integrated specialized **AI/ML modules** for question answering, course delivery, and conversational support.
- Developed an **Intent Recognizer** for intelligent routing of user inputs.
- Built a **Context Engine** to persist conversation history for personalization.
- Deployed via **Google CloudRun** to ensure scalability, reliability, and automated scaling.

- Added **local file storage** and **custom logging** to track and optimize student interactions.
- Delivered through a **dedicated cross-functional engineering team** covering backend, AI/ML, and cloud.

Outcome

The platform boosted student engagement by over 60%, improved comprehension, and delivered measurable test score increases. Educators reported a 40% reduction in repetitive workload, enabling more focus on complex teaching. The scalable architecture now supports institutions of all sizes, strengthening the client's market credibility and driving adoption across multiple regions. Positive user feedback highlighted usability, interactivity, and learning impact—directly contributing to increased retention and institutional goodwill.

56- GQSM Automate Rice Grain Quality Analysis

Automated grain analysis using computer vision and a custom desktop app, reducing manual inspection time to 1–2 minutes per batch.

Industry - Water solutions industry

Services

- AI/ML Engineering
- Custom Software Development
- Desktop Application Development
- Software Testing & QA

About the Client

The client is a leading European water solutions company offering 5000+ pump

products, partnering with major courier networks and distributing globally recognized industrial brands with full compliance to regional standards.

The Story

The client approached FiftyFive Technologies to build an automated computer-vision system that could replace slow, manual rice grain quality inspections. Their goal was to eliminate human dependency, accelerate lab throughput, and establish a repeatable, data-driven testing workflow. FiftyFive deployed a dedicated engineering team, AI/ML, backend, desktop development, and QA to design and deliver a complete custom software solution. The engagement focused on accuracy, speed, and scalability while ensuring the client fully owned and operated the final system.

Challenges

- Training a machine learning model to detect and segment individual grains from scanned images with high precision.
- Extracting structured parameters (height, width, color values, location coordinates) for each grain.
- Automating image cropping, metadata generation, and per-grain storage.
- Building a fast, stable workflow for producing validated CSV reports within strict time limits.
- Ensuring system accuracy across varying grain colors, shapes, and image inconsistencies.

Solution

- Implemented and fine-tuned a Mask R-CNN–based segmentation model for pixel-level grain detection.
- Engineered a desktop application using Python, .NET Core 6, and SQLite for end-to-end processing.

- Built automated pipelines for grain measurement extraction, color analysis, and individual image generation.
- Developed a structured CSV reporting engine delivering grain-level metadata in under 2 minutes.
- Integrated QA validation workflows to ensure consistent segmentation accuracy and repeatable test results.
- Delivered the complete solution through a dedicated AI engineer, backend developer, desktop developer, and QA specialist team.

Outcome

The automated system reduced analysis time from hours of manual evaluation to 1–2 minutes per batch, improving operational efficiency by over 90%. The client achieved higher measurement accuracy, consistent reporting, and reduced dependency on manual lab workflows. The solution strengthened their credibility in quality assurance while lowering operational effort and enabling scalable, repeatable rice grain testing across their processes.

55- Samurai Snake Game (DevOps/ Mobile Game)

About the Client

Our client, a game development studio, aimed to launch an engaging action game featuring a Samurai snake on a quest to reclaim lost fishes. They needed a seamless development and deployment process to ensure smooth game updates and user experience.

The Story

The game revolves around a Samurai snake battling to recover lost fishes. To bring this story to life, the developers needed an efficient pipeline for building and deploying the Android game while managing various technical complexities.

Challenges

- Restoring lost fishes in the game with smooth gameplay mechanics.
- Efficiently managing Docker and Docker Compose files.
- Setting up an Azure DevOps Multi-stage build pipeline.
- Implementing licensing for Unity Docker images.
- Using Unity images to build the APK.
- Ensuring continuous deployment of the app on the Android store with Fastlane.

Solution

To overcome these challenges, the team leveraged Docker and Docker Compose for environment management. They established an Azure DevOps Multi-stage build pipeline with Unity image setup and implemented licensing for Unity Docker images. These images were then used to build the APK, ensuring a streamlined build process. Additionally, Fastlane was integrated for continuous deployment, enabling the app to be automatically updated on the Android store.

Outcome

The solution ensured a smooth and efficient game development cycle, allowing the Samurai snake to reclaim its lost fishes successfully. The integration of Azure DevOps, Docker, and Fastlane streamlined the build and deployment process, resulting in a well-optimized Android game with continuous updates and seamless player experience.

54- Playpadel Build Your Custom Playground Booking System

FiftyFive built a scalable booking platform enabling real-time court reservations, reducing manual work and improving system performance.

Industry - SportsTech

Services -

- Custom Software Development
- Cloud Engineering
- DevOps & CI/CD
- QA & Testing
- Backend Development
- Application Architecture Design

About the Client

The client operates a sports facility management business that required a secure, scalable digital system enabling users to search, book, and manage Padel and playground court reservations across multiple locations.

The Story

The client partnered with FiftyFive to build a full-cycle booking ecosystem combining a mobile application and a scalable backend platform. The goal was to provide real-time venue availability, seamless booking flows, flexible rescheduling, secure payments, and automated infrastructure for reliable performance. FiftyFive delivered both the end-user experience and the underlying architecture, ensuring smooth operations, reduced manual management, and high system reliability.

Challenges

- Managing real-time court availability across multiple venues.
- Handling simultaneous booking requests without performance degradation.
- Designing reliable booking logic with date and time slot validation.

- Implementing secure authentication and data protection workflows.
- Ensuring smooth user navigation and state management within the mobile app.
- Integrating geolocation services for venue discovery.
- Configuring low-latency APIs with high uptime requirements.
- Standardizing multi-environment deployments for development, staging, and production.
- Tracking errors and analytics across mobile user journeys.

The Solution

- Implemented a React Native mobile application using TypeScript, Redux, and React Navigation.
- Integrated geolocation-based venue discovery with real-time availability checks.
- Added seamless booking, rescheduling, and cancellation workflows with user-friendly UI components built via Storybook.
- Integrated Stripe for secure payment processing.
- Built backend services and booking APIs with secure authentication and availability management.
- Designed modular, cloud-ready architecture using Nginx for routing and performance optimization.
- Implemented Terraform for automated infrastructure provisioning and environment consistency.
- Containerized services with Docker and Docker-Compose for predictable builds.

- Adopted Serverless (YAML) workflows and AWS CodeDeploy for zero-downtime deployments.
- Added Mixpanel and Sentry for analytics and error tracking.
- Conducted QA validation for booking flows, API logic, and deployment stability.

The Outcome

The booking platform achieved a 65% reduction in manual scheduling time, a 40% improvement in processing speed, and 99.5% deployment reliability. Users experienced seamless court discovery, faster booking completion, and flexible schedule management, while venues gained higher utilization and increased revenue.

53-KRIDA AI AI-Powered Coaching Platforms

FiftyFive developed an AI-powered cricket coaching platform, providing real-time feedback and scalability, resulting in 300% user growth.

Industry - Sports Broadcasting

Services - Custom Software Development

AI/ML

Cloud Engineering

About our Client

The client is a leading sports broadcasting company in India, specializing in cricket. They aimed to transform traditional coaching through AI-driven solutions, providing accessible, data-backed training for players of all levels.

The Story

The client recognized a significant gap in traditional cricket coaching—lack of precise, accessible performance analysis. They sought to create an AI-powered platform that could deliver real-time feedback on player technique, enabling professional-level coaching anytime, anywhere. FiftyFive Technologies was engaged to design and

implement the custom software solution, leveraging AI and cloud technologies to meet the client's vision.

Challenges

- Accurate motion detection for player movement analysis.
- Real-time processing of high-resolution video data for instant feedback.
- Scalable infrastructure to support multiple concurrent users.
- Seamless integration of machine learning models with a user-friendly interface.
- Low-latency feedback with minimal delay.
- Ensuring high performance while maintaining data security.

The Solution

- Designed and implemented a cloud-based solution using Google Cloud Platform (GCP), leveraging Terraform for infrastructure management.
- Deployed Kubernetes (GKE) and Kubeflow for scalable machine learning workflows.
- Integrated AI models for real-time motion analysis of 133 body joints using Object Detection, Pose Estimation, and Object Tracking.
- Utilized Google Cloud Build for automated deployment, ensuring seamless updates and smooth operation.
- Applied PyTorch, OpenCV, and Firebase for AI processing and mobile/web performance optimization.

The Outcome

The AI-powered coaching platform reduced training analysis time by 95%, shifting from hours of manual synthetic review to seconds of automated feedback. The platform supported 300% user growth within six months, thanks to scalable infrastructure. With a 99.9% uptime, the solution ensured continuous availability, empowering players globally and strengthening the client's position as a leader in AI-driven sports coaching.

52- APICA End-to-End Engineering for Synthetic Monitoring Infrastructure

Industry - Enterprise software

Service- Custom Software Development

DevOps Engineering

Cloud Engineering

About the Client

The client is a leading enterprise software company requiring large-scale, globally distributed synthetic monitoring capabilities to track website and API performance with consistent reliability, automated workflows, and high availability across multiple regions.

The Story

The client needed to modernize their synthetic monitoring framework to simulate user interactions and measure API and website performance across global locations. They partnered with FiftyFive Technologies to engineer a scalable, automated system capable

of supporting distributed workloads securely and consistently. A dedicated team of DevOps engineers, cloud engineers, and backend specialists was deployed to design, containerize, orchestrate, and automate the multi-region monitoring infrastructure. The engagement focused on ensuring reliability, operational efficiency, and unified monitoring behavior across all environments.

Challenges

- Complex multi-region infrastructure requirements for distributed synthetic monitoring
- Need for consistent, reliable performance tracking across global endpoints
- Automation of infrastructure provisioning, deployment, and lifecycle management
- Resource optimization across containerized monitoring workloads
- Large-scale configuration management and environment standardization
- High availability and fault tolerance for monitoring systems at scale

The Solution

- Implemented Terraform to automate infrastructure provisioning across global environments
- Configured HAProxy for efficient traffic load balancing across monitoring endpoints
- Used Puppet with Foreman for centralized configuration management and environment control
- Deployed Nomad agents to manage distributed job scheduling and execution workflows

- Adopted Docker containerization for consistent packaging and reproducible deployment
- Built unified, automated deployment pipelines to support scalable monitoring operations

The Outcome

The new monitoring architecture delivered precise, real-time visibility into API and website performance across all global locations. Automated infrastructure workflows reduced deployment time by 60%, while containerization improved resource utilization by 40%. Centralized configuration management increased operational consistency, and distributed orchestration enhanced system uptime and reliability. The client achieved faster issue detection, reduced operational overhead, and significantly improved their ability to maintain performance standards worldwide strengthening platform stability, reducing performance-related escalations, and enabling smoother digital experiences for end users.

51-ChopChop Upgraded the Restaurant's CI/CD Pipeline

FiftyFive secured the client's AWS environment, automated CI/CD pipelines, and reduced deployment time while improving development efficiency.

Industry - FootTech

Services -

- Cloud Engineering
- DevOps & CI/CD Automation
- Infrastructure Security

About the Client

The client is a Swedish restaurant enterprise operating a large-scale AWS-based inventory management system requiring secure infrastructure, automated deployments, and improved DevOps workflows to support operational growth and sensitive data handling.

The Story

The client engaged FiftyFive Technologies to modernize their cloud infrastructure and strengthen security across their AWS environment. Their existing setup lacked secure access controls, scalable CI/CD automation, and efficient development workflows. Our cloud engineers and DevOps specialists were tasked with implementing a security-first architecture, optimizing CI/CD processes, and ensuring reliable, protected connectivity to their MySQL database. The objective was to create a stable, scalable, and secure foundation that could support ongoing development and future system expansion.

Challenges

- Inadequate security protocols for accessing AWS and MySQL resources
- Inefficient CI/CD pipelines with manual deployment dependencies
- Limited scalability in the existing EC2-based environment
- Absence of automated build, test, and deployment workflows
- Lack of secure connectivity between Jenkins and database services
- Pipeline performance bottlenecks impacting release velocity

The Solution

- Implemented **Pritunl VPN** to establish encrypted, role-based access to Jenkins and MySQL
- Deployed **Jenkins Master–Slave architecture** for distributed, parallelized build execution
- Created **Declarative YAML pipelines** for automated build, test, and deployment stages
- Configured **EC2 instances with Nginx** to optimize delivery and runtime efficiency
- Integrated **secure MySQL connectivity through VPN** for protected data transactions
- Applied **AWS best practices** including hardened IAM roles, VPC isolation, and security group policies
- Standardized logging and monitoring across environments to improve observability

The Outcome

The upgraded cloud environment delivered a measurable 40% reduction in deployment time and significantly improved pipeline throughput through parallelized Jenkins execution. Security posture was strengthened with encrypted VPN-based access, ensuring protected communication with Jenkins and MySQL. Automated pipelines reduced manual interventions by over 60%, enabling faster release cycles and more reliable development operations. The scalable architecture positioned the client for future expansion while reducing operational risk, improving developer productivity, and ensuring consistent, secure delivery of updates across environments.

50-NIBE Spare Parts Operations

The client required a scalable spare parts portal; FiftyFive implemented DevOps automation, enabling faster deployments, stable operations, and higher transaction capacity

Industry - Aftermarket spare parts / sales

Services - Custom Software Development (frontend & backend portal build)

- DevOps & Cloud Engineering (CI/CD, Kubernetes, Helm)
- Infrastructure Automation (scheduling, orchestration)
- Traffic Monitoring & Routing (real-time visibility, performance management)

About the Client

A Sweden-based aftermarket spare parts company requiring a scalable digital sales platform with automated infrastructure, real-time monitoring, and high system availability for expanding customer demand.

The Story

The client engaged FiftyFive Technologies to design, build, and deploy a customer portal from the ground up. Our role was to provide full-cycle custom software development and DevOps engineering. A dedicated team of cloud engineers, backend/frontend developers, and DevOps specialists was deployed to ensure the platform was scalable, automated, and production-ready. The engagement focused on enabling business growth while minimizing manual operations and infrastructure costs.

Challenges

- Designing a scalable and fault-tolerant infrastructure for portal deployment
- Implementing automated scheduling for daily operational jobs

- Deploying frontend and backend applications as containerized services
- Configuring secure and efficient traffic routing with real-time monitoring
- Reducing deployment time through CI/CD automation

The Solution

- Built Dockerfiles and Docker-Compose for frontend and backend containerization
- Implemented multi-stage CI/CD pipelines in Azure DevOps for automated deployments
- Created Kubernetes manifests for deployments, services, and Ingress routing
- Configured PostgreSQL, Application Gateway Ingress Controller, and Cert-manager using Helm in Azure pipelines
- Enabled path-based routing in Ingress for optimized traffic distribution
- Automated daily scheduled jobs for operational tasks
- Integrated real-time traffic monitoring on the application gateway for performance insights

The Outcome

The client gained a fully automated, production-ready portal that reduced deployment time by **70%** through CI/CD pipelines and eliminated manual job handling with automated scheduling. Kubernetes orchestration improved uptime to **99.9%**, ensuring stable operations during high-traffic periods. Real-time monitoring increased visibility into user behavior, enabling proactive performance management. The scalable infrastructure allowed the client to handle higher transaction volumes, reduce

operational overhead by **40%**, and deliver a faster, more reliable customer experience—directly contributing to increased sales and stronger customer trust.

48- The Cursed Hours (PC Game Development)

Service- Custom Game Development

About our client

FiftyFive Studios has emerged as an innovative developer creating distinctive experiences across mobile and PC gaming. Founded by industry veterans, the studio crafts games that successfully balance accessibility with depth, appealing to casual and hardcore players alike. Their mobile offerings reimagine familiar genres with creative twists, while their PC titles showcase more ambitious systems and immersive worlds.

The Story

In an ambitious venture to redefine multiplayer horror gaming, approached us to develop a unique survival game. The concept centered around a group of friends investigating an abandoned mansion, with gameplay transitioning through distinct phases - from initial exploration to a nail-biting escape. The game needed to seamlessly blend cooperative gameplay with progressive horror elements while maintaining technical stability for multiplayer interactions.

Challenges

- Implementing dynamic difficulty scaling based on player count and progress
- Creating responsive multiplayer synchronization for real-time interactions
- Balancing puzzle complexity with time constraints
- Developing AI for adaptive ghost behavior during the cursed hour
- Ensuring smooth transition between game phases while maintaining tension

The Solutions

We developed a robust multiplayer framework that supported seamless cooperative

gameplay while implementing sophisticated AI systems for enemy behavior. The game mechanics were carefully crafted to maintain suspense through progressive difficulty scaling, with particular attention to resource distribution and puzzle complexity. We integrated a dynamic time management system that controlled the intensity of supernatural encounters and coordinated player activities during crucial phases.

The Outcome

The game successfully launched with overwhelmingly positive player feedback, particularly praised for its innovative cooperative mechanics and atmospheric tension. Players especially appreciated the smooth multiplayer experience and the dynamic difficulty adjustment system, which ensured consistent challenge levels regardless of group size. The game achieved its goal of creating an immersive horror experience that encouraged teamwork while maintaining individual player engagement.

47-Battlefield Showdown/Fight Hero (Mobile Game Development)

A dynamic mobile game combining strategic combat and character progression in a fight against malevolent artificial intelligence.

About Our Client FiftyFive Studios has emerged as an innovative developer creating distinctive experiences across mobile and PC gaming. Founded by industry veterans, the studio crafts games that successfully balance accessibility with depth, appealing to casual and hardcore players alike. Their mobile offerings reimagine familiar genres with creative twists, while their PC titles showcase more ambitious systems and immersive worlds.

The Story

We envisioned a mobile game that would combine automatic targeting mechanics with strategic gameplay elements. The concept centered around humanity's struggle against malevolent AI, requiring players to master combat skills while progressing through a talent-based upgrade system. The game needed to balance accessibility with depth through its automatic targeting system and strategic cover mechanics.

Challenges

- Implementing intuitive on-screen controls with automatic targeting
- Balancing automatic and manual gameplay elements
- Creating engaging boss battles and enemy variety
- Developing a meaningful progression system
- Designing six unique level environments
- Integrating Firebase backend services
- Optimizing performance for mobile devices
- Building an engaging talent wheel upgrade system

The Solutions

We developed a comprehensive gaming solution using Unity Engine, implementing intuitive joystick controls with automatic targeting mechanics. The game features a robust progression system through the talent wheel, strategic cover mechanics, and diverse enemy types. We integrated Firebase for backend services and created a resource-gathering system with gear tokens and bullet shells.

The Outcome

The final product delivered an engaging mobile combat game featuring smooth gameplay mechanics and meaningful progression. Players progress through six unique levels, unlock new abilities through the talent wheel, and engage in strategic combat against varied enemy types. The combination of automatic targeting, strategic elements, and character progression created an accessible yet deep gaming experience.

46- Trickstar/ RiftRider (Mobile Game Development)

Revolutionizing mobile gaming through a dystopian racing adventure that combines resource management, stunning visuals, and intense chase mechanics.

About Our Client

FiftyFive Studios has emerged as an innovative developer creating distinctive experiences across mobile and PC gaming. Founded by industry veterans, the studio crafts games that successfully balance accessibility with depth, appealing to casual and hardcore players alike. Their mobile offerings

reimagine familiar genres with creative twists, while their PC titles showcase more ambitious systems and immersive worlds.

The Story

In a competitive mobile gaming market, we envisioned a unique racing game that would stand out by combining multiple gameplay elements. The concept merged high-speed racing, resource collection, and strategic combat in a dystopian setting. This ambitious project required sophisticated mechanics, including character controls, vehicle customization, and an innovative drone pursuit system.

Challenges

- Implementing complex swipe-based controls while maintaining responsive gameplay
- Balancing the two-hit damage system with fast-paced racing mechanics
- Creating diverse enemy AI behaviors for ground and flying opponents
- Developing a seamless resource collection and upgrade system
- Optimizing performance for mobile devices while using Unreal Engine
- Integrating multiple gameplay elements (racing, combat, collecting) cohesively

The Solutions

We developed a comprehensive gaming solution using Unreal Engine, with Perforce for version control and Firebase for backend services. The game features intuitive swipe controls, aerial stunts, and a strategic progression system. We implemented a dynamic drone chase mechanic and created challenging ambush zones to enhance gameplay depth. The resource-driven narrative was supported by a robust collection and upgrade system.

The Outcome

The final product delivered an action-packed mobile game featuring seamless integration of racing, stunts, and combat. Players can now experience an engaging progression system through vehicle upgrades and resource collection, while the relentless drone chase mechanic maintains constant excitement. The game successfully combines strategic depth with arcade-style accessibility, creating a unique gaming experience in the mobile market.

45- Tower of Demons (Mobile Game Development)

A strategic mobile combat game combining tower defense mechanics with skill-based gameplay and progressive weapon upgrades.

About Our Client

FiftyFive Studios has emerged as an innovative developer creating distinctive experiences across mobile and PC gaming. Founded by industry veterans, the studio crafts games that successfully balance accessibility with depth, appealing to casual and hardcore players alike. Their mobile offerings reimagine familiar genres with creative twists, while their PC titles showcase more ambitious systems and immersive worlds.

The Story

We envisioned a unique mobile game that would blend tower defense with action combat mechanics. The concept centered around a warrior named Pako fighting through demon towers across diverse Mayan-inspired environments. The game needed to balance challenging combat with rewarding progression systems while maintaining player engagement through strategic gameplay elements.

Challenges

- Implementing precise dodge and attack mechanics for mobile controls
- Balancing combo system with power-up rewards
- Creating varied enemy types and tower formations
- Designing progressive difficulty scaling
- Developing engaging weapon upgrade system
- Integrating secure data management with Firebase
- Optimizing performance across different mobile devices

The Solutions

We developed a comprehensive gaming solution using Unity Engine, implementing intuitive combat mechanics and strategic gameplay elements. The game features a robust combo system, diverse power-ups, and weapon progression. We created multiple

world environments with unique enemy types and integrated Firebase for secure data management and Play Services for social features.

The Outcome

The final product delivered an engaging mobile combat game featuring strategic tower battles and smooth gameplay mechanics. Players progress through varied environments, unlock new weapons, and master combat techniques. The combination of action gameplay, progression systems, and social features created an entertaining and replayable gaming experience.

45- Nutrients Sold Separately (Mobile Game Development)

An innovative mobile game that combines educational content with engaging gameplay, putting players in control of White Blood Cells.

About Our Client Nutrients Sold Separately aimed to create an educational mobile game that would make learning about the human immune system engaging and entertaining through action gameplay.

The Story

Our client wanted to develop a unique mobile game that would blend educational content with engaging gameplay mechanics. The concept placed players in the role of White Blood Cells defending the human body, combining action combat with strategic elements. The game needed to balance entertainment with educational value while maintaining player engagement through personalization and social features.

Challenges

- Creating scientifically accurate yet engaging microscopic environments
- Balancing educational content with entertainment value
- Implementing dynamic wave-based enemy systems
- Designing intuitive combat mechanics for mobile devices
- Developing compelling progression and upgrade systems
- Ensuring secure data management through Firebase
- Optimizing performance in Unreal Engine for mobile platforms

The Solutions

We developed a comprehensive solution using Unreal Engine, implementing dynamic NPC interactions and a Wave Manager for enemy spawning. The game features customizable characters, strategic combat mechanics, and an educational layer that teaches players about the immune system. We integrated Firebase for security and Play Services for social features, while Perforce enabled efficient version control during development.

The Outcome

The final product delivered an engaging educational game that successfully combines learning with entertainment. Players explore a microscopic world while learning about the immune system, collecting nutrients, and battling viruses. The game's combination of educational content, action gameplay, and social features created an entertaining and informative gaming experience that achieved the client's vision.

43- Dubai Technologies (Subscribe-Me) Car Rental Application (Mobile Application)

FiftyFive developed a Flutter-based rental platform with secure ID verification, helping the client gain 25,000 users in six months

Industry - Car rental and leasing

Services - Custom Software Development: Mobile app development (Flutter).

- Architecture Design: Clean Architecture for scalability.

- Systems Integration: OCR APIs, payment gateways, geolocation, deep linking, notifications.

- Quality Assurance: Testing across iOS and Android.

About the Client

The client is a diversified business group operating across seven sectors in the Gulf region, offering products and services that enhance everyday life for a large customer base.

The Story

The client sought to consolidate its car rental operations into a single, scalable digital platform. They partnered with FiftyFive Technologies to design and deliver a robust, mobile-first solution. The goal was to streamline user journeys, integrate secure identity verification, and support continuous feature upgrades in a highly dynamic market. FiftyFive provided a dedicated offshore development team to build and scale the product from MVP to live release.

Challenges

- Frequent scope changes due to dynamic rental business requirements.
- Complex identity verification for Emirates ID, Driving Licenses, and Home Country Licenses.
- Need for scalable architecture to accommodate future features.
- Seamless integration of third-party services: maps, payments, notifications, and analytics.
- Cross-platform deployment with consistent performance.

The Solution

- Built a Flutter-based mobile application for iOS and Android.

- Adopted Clean Architecture for flexibility and maintainability.
- Integrated OCR APIs for automated ID and license data extraction and validation.
- Developed secure payment workflows with third-party gateways.
- Enabled geolocation services, push notifications, and deep linking.
- Deployed scalable backend with continuous enhancements post-MVP.
- Offshore team included Flutter developers, backend engineers, QA specialists, and UI/UX support.

The Outcome

The platform launched successfully on both the Apple App Store and Google Play Store, acquiring over 25,000 users within six months. The solution reduced manual verification efforts, accelerated customer onboarding, and increased operational efficiency across the client's rental business. By consolidating fragmented processes into one scalable platform, FiftyFive helped the client enhance revenue generation, improve customer trust, and strengthen its market presence in the Gulf's competitive mobility sector.

40- Auggit Software Development for Accounting Process Automation

FiftyFive developed an AI-driven solution to automate document processing and accounting workflows, improving efficiency and reducing costs.

Industry - Paperless accounting solutions

Services - Custom Software Development

- AI/ML Solutions

- Automation

About the Client

The client is a leading provider of paperless accounting solutions, focusing on optimizing financial workflows for businesses by digitizing accounting processes and reducing reliance on paper-based systems.

The Story

The client faced challenges with manual accounting processes, which led to inefficiencies and errors. They partnered with FiftyFive Technologies to automate document management and accounting workflows. The goal was to streamline processes, improve accuracy, and enable real-time decision-making using AI-driven technologies.

Challenges

- Insufficient data for training deep learning models, affecting document recognition accuracy.
- Difficulty in achieving high accuracy with object detection models for financial document processing.
- Manual data entry into Tally and physical files caused redundancy and delays.
- Slow retrieval of financial records and high error rates in manual processes.
- Excessive paper dependency conflicting with sustainability objectives.

- Lack of real-time visibility into financial operations, impacting decision-making speed.

The Solution

- Applied deep learning and computer vision to enhance object detection and document processing accuracy using image augmentation.
- Optimized OCR performance for accurate data extraction from multiple document formats.
- Integrated NLP techniques for ensuring accurate formatting of extracted data.
- Developed a custom web and mobile app using Flutter, Dart, React, TypeScript, Java, and Spring Boot to centralize financial operations.
- Built a unified database with advanced indexing for real-time document retrieval and search.
- Automated workflows to eliminate manual data entry, streamlining the accounting process and reducing human error.
- Incorporated real-time dashboards to enhance decision-making and business insights.

The Outcome

The solution improved document processing efficiency with 95% accuracy in data extraction, significantly reducing record retrieval time by over 90%. Manpower costs decreased by nearly 40%, and automated workflows allowed staff to focus on more strategic tasks. The transition to a digital-first approach eliminated paper dependency, aligning with the client's sustainability goals. Real-time dashboards facilitated faster, data-driven decision-making, setting the stage for scalable growth and future AI-driven analytics.

39- DP World Build Your Port Management System

FiftyFive built a mobile-first system that automated port operations, improved real-time tracking, and delivered measurable gains in productivity and cost efficiency.

Industry - Transportation and Services

Services - Custom Software Development

– Mobile App Development (Flutter)

– Backend Engineering (.NET)

– Database Architecture (SQL)

QA & Testing

DevOps Implementation

Digital Transformation Consulting

About the Client

The client is a global transportation and services company with 20+ years of experience in fare systems, taximeters, fleet management, and mobility technologies, supporting large-scale logistics and operational workflows across international markets.

The Story

The client sought to modernize port operations that depended on manual workflows, outdated infrastructure, and non-structured data processes. They partnered with FiftyFive Technologies to design and implement a mobile-first, full-cycle digital solution that automated core operational functions, improved tracking accuracy, and reduced operational inefficiencies. A dedicated engineering team, covering mobile development, backend engineering, database architecture, QA, and DevOps was deployed to deliver an end-to-end custom software development initiative aligned with the client's scalability goals.

Challenges

- Manual port workflows causing latency, bottlenecks, and limited throughput
- Legacy systems contributing to high operational expenditure and low efficiency

- No automation in job assignment, routing, or container management
- Absence of mobile interface for real-time workforce operations
- Fragmented data structures across non-standardized sources

The Solution

- Developed a cross-platform Flutter mobile application enabling workforce digitization
- Engineered scalable backend services using .NET for high-performance processing
- Designed and deployed a SQL-based structured database for reliable data management
- Implemented real-time container tracking with optimized route navigation workflows
- Automated job token assignment to reduce manual intervention
- Applied continuous QA cycles and DevOps practices for stable, faster production releases
- Delivered full-cycle engineering support, from architecture design to deployment and optimization

The Outcome

The digital solution delivered a 22% increase in operational productivity and a 40% reduction in operating costs, significantly strengthening the client's financial and operational performance. Automation reduced manual errors, accelerated job allocation, and improved container throughput across terminals. Real-time visibility enabled faster, data-driven decision-making, while the scalable architecture positioned the client for future digital transformation initiatives. Through full-cycle custom software development, FiftyFive demonstrated its ability to deliver measurable business impact,

streamline operational complexity, and serve as a long-term technology partner for enterprise-scale mobility and logistics solutions.

38- Dubai Technologies Smart Waste Management System

FiftyFive built an AI-powered waste management system to optimize collection routes, cut fuel costs, and enable real-time alerts.

Industry - Smart City / Municipal Waste Management

Service - Custom Software Development (real-time data processing)

- Data Engineering (classification, database integration)
- AI/ML Development (Tensorflow Lite model training)
- IoT Integration (camera-based street video capture)
- Team Extension (specialized experts provided)

About the client

The client is a Dubai-based municipal service provider implementing smart city initiatives to optimize waste management operations and improve sustainability through advanced technology and data-driven decision-making.

The Story

The client partnered with FiftyFive Technologies to build a Smart Waste Management System addressing inefficiencies in urban waste collection. Through the **Team Extension model**, FiftyFive deployed a Data Engineer and Business Analyst to design, develop, and integrate a custom data engineering solution. The system leveraged **AI, IoT, and real-time processing** to classify waste on streets, notify authorities, and optimize collection routes. This engagement highlighted FiftyFive's ability to deliver scalable, tailored solutions that align with smart city and sustainability goals.

The challenge

- Static collection schedules not linked to real waste volumes.

- Lack of automated waste type recognition and categorization.
- Inefficient route planning causing fuel overuse and higher costs.
- No real-time alerts for critical waste issues (illegal dumping, overflowing bins).

The Solution

- Deployed three-sided cameras on collection trucks for continuous street video capture.
- Implemented Tensorflow Lite–based image recognition for real-time classification of recyclables, organic, and general waste.
- Built a Python-based processing pipeline for instant categorization and decision-making.
- Integrated MariaDB for structured data storage with waste type, location, and timestamp metadata.
- Delivered solution via Team Extension model, combining Business Analyst (requirements/use cases) and Data Engineer (model training, system integration, database deployment).

The outcome

The system reduced operational costs by optimizing waste collection routes, lowering fuel consumption, and improving workforce efficiency. Authorities received **real-time notifications**, enabling faster response to illegal dumping and waste overflow. Accurate categorization improved resource allocation, while stored data enabled advanced reporting and compliance tracking. Beyond operational savings, the solution strengthened the client’s reputation as a smart city leader, enhanced citizen satisfaction,

and demonstrated how **FiftyFive's custom software development and data engineering expertise** deliver measurable value across urban sustainability initiatives.

36-LiveLike Token Gating for Chat Rooms (Mobile Application)

About the Client

Our client is an audience engagement platform that offers interactive tools to enhance user participation. To boost engagement, they sought to implement token gating in chat rooms, enabling creators to restrict access based on Web3 wallet holdings.

The Story

FiftyFive Technologies was tasked with developing a token-gating feature for chat rooms within the client's iOS SDK. This required integrating Web3 authentication while ensuring seamless user experience and compatibility with multiple blockchain networks like Chiliz, Hedera, Socio, and Polygon.

Challenges

- Implementing token gating in chat rooms, a novel concept in Web3.
- Enabling chat room creators to verify users' Web3 wallet holdings.
- Ensuring compatibility across multiple blockchain networks.
- Avoiding increased SDK size and third-party dependencies.

The Solution

Two approaches were explored: integrating a third-party library (web3.swift) or developing a Backend API with RPC URLs. The first was discarded due to SDK size concerns. The final solution utilized a backend-driven approach where a frontend function verified chatroom access by communicating with an API. The implementation

involved API development, Web3 research, testing, and documentation, leveraging technologies like Swift, UIKit, Promise Kit, XCTest, PubNub, and SwiftLint.

The Outcome

By adopting a backend-driven approach, the client successfully implemented token gating in their SDK. The solution ensured high reliability through rigorous testing and PR reviews. The project was completed on time, with valuable takeaways on improving communication, PR processes, and Web3 integration. The client appreciated the seamless execution and the innovative approach taken by FiftyFive Technologies.

35- Unified Customer Portal for Global B2B Operations

FiftyFive delivered a custom customer portal replacing manual workflows, enabling partner onboarding and scalable multi-country operations.

Industry - Manufacturing

Services -

- Custom Software Development
- Cloud Engineering / DevOps
- QA & Test Automation
- Performance Engineering
- Technical Consulting

About the Client

The client is a Sweden-based B2B heat pump manufacturer operating across multiple European markets, requiring a scalable digital platform to support service partners, spare parts management, documentation access, and workflows.

The Story

The client engaged FiftyFive as a global software development and technology consulting partner to design, build, and scale a custom Customer Portal for

multi-country partner operations. The objective was to replace manual, phone-driven workflows and fragmented legacy systems with a centralized, cloud-native platform owned and operated by the client. FiftyFive delivered the engagement as a full-cycle custom software development initiative, covering architecture, engineering, DevOps, QA automation, performance validation, and analytics. A dedicated cross-functional team worked closely with the client to ensure the platform could support regional expansion, localization, and high-traffic service operations.

Challenges

- Phone-based ordering workflows with no digital transaction traceability
- Disparate legacy applications requiring separate installations and manual updates
- No centralized data model for spare parts, documentation, and software files
- Inconsistent workflows and routing logic across multiple countries
- Lack of automated testing, error reporting, and operational observability
- Unvalidated microservices behavior under regional configuration changes
- Performance and scalability risks during expansion to new markets

The Solution

- Designed and developed a custom microservices-based Customer Portal using React.js
- Implemented GraphQL APIs with PostgreSQL for structured spare parts and documentation data

- Built backend services and integrations aligned to client-owned domain workflows
- Deployed Kubernetes infrastructure with Azure DevOps CI/CD pipelines for automated releases
- Established automated testing using Cypress and Jest achieving 90% test coverage
- Implemented parameterized, region-specific test execution for localization validation
- Configured cron-based error reporting integrated with Microsoft Teams webhooks
- Performed load and stress testing using Artillery to validate high-traffic readiness
- Integrated Microsoft Clarity for partner usage analytics and behavioral insights

The Outcome

The Customer Portal achieved 60% service partner onboarding within the first year, exceeding the original 40% target and accelerating adoption across multiple regions. Automated workflows significantly reduced manual support dependency and eliminated phone-based ordering bottlenecks. QA automation lowered manual testing effort by over 60%, enabling faster, safer releases and reduced operational risk. Performance validation ensured stable operations during traffic spikes, supporting expansion into Germany, Finland, and 26 additional countries. The client gained a scalable, client-owned digital platform that improved partner experience, strengthened operational efficiency, and enabled confident multi-country growth powered by FiftyFive's custom software engineering services.

34- Music App for Artists (Mobile Application)

About the Client

Our client, a Sweden-based app development company, aimed to build an award-winning application for music creators and artists. They envisioned a feature-rich, user-friendly platform that would empower musicians worldwide to collaborate, create, and manage their music efficiently.

The Story

The client sought a robust application that allowed artists to upload music, cover photos, lyrics, and voice memos while ensuring seamless collaboration. The goal was to develop a secure, scalable, and high-performing app compatible across Mobile, Desktop, and Web. However, achieving this vision required overcoming several technical and functional challenges.

Challenges

- Developing a feature-rich, personalized, and secure platform.
- Enabling seamless collaboration for uploading music, lyrics, and cover photos.
- Ensuring cross-platform compatibility (Mobile, Desktop, Web).
- Fixing bugs and improving navigation.
- Implementing robust analytics for user insights.
- Resolving mobile device crashes and ensuring code quality.

The Solution

The development team leveraged advanced technologies like React Native, TypeScript, Redux, and Mixpanel to craft a high-performance application over 1 year and 3 months. They integrated collaborative features, optimized navigation, and enhanced analytics

tracking. Bug fixes and crash resolutions improved stability, while end-to-end testing ensured reliability. The deployment was streamlined using CI/CD pipelines, allowing efficient updates for both iOS and Android.

The Outcome

The result was a feature-packed, award-worthy mobile application tailored for music creators. The app provided seamless navigation, enhanced security, and robust functionalities across all platforms. Artists could now manage their music, receive credits, and access royalty payments with ease. The client was highly satisfied with the intuitive design and seamless performance, reinforcing the development team's expertise in delivering top-tier solutions.

33- NIBE Customer Portal (Web Application/ DevOps)

FiftyFive built a unified portal to replace manual workflows, streamline spare parts access, and achieve higher partner adoption for the client.

Industry - Manufacturing

Services –

- Custom Software Development
- Cloud Engineering / DevOps
- QA & Test Automation
- Technology Integration

About the Client:

The client is a Sweden-based heat pump manufacturer serving B2B markets, aiming to

modernize service operations by digitizing spare parts access, product documentation, troubleshooting workflows, and partner enablement processes.

The Story:

The client sought a unified Customer Portal to eliminate manual processes, reduce reliance on phone-based ordering, and improve how service partners accessed parts, documentation, and software files. They partnered with FiftyFive Technologies as a custom software development and technology engineering provider to design, build, and deploy a scalable portal using a full-cycle development approach. A dedicated engineering team handled architecture design, development, DevOps, QA, deployment, and integration, ensuring the platform could operate reliably at scale and support future market expansion.

Challenges:

- Manual ordering workflows dependent on phone calls and offline coordination.
- Lack of centralized access to spare parts, alternatives, and reference documentation.
- Fragmented systems requiring separate software installations and manual updates.
- No structured communication path for automated error reporting and troubleshooting.
- Increasing operational load due to rising service partner demand.

Solution:

- Built a custom Customer Portal using a microservices architecture with React.js frontend.
- Developed GraphQL APIs integrated with PostgreSQL for structured spare parts and documentation data.

- Deployed Kubernetes with Azure DevOps for automated CI/CD, scalable environments, and resilient releases.
- Implemented automated testing using Jest and Cypress achieving 90% test coverage.
- Configured cron-based error handling connected to Microsoft Teams incoming webhook alerts.
- Integrated Microsoft Clarity for partner usage analytics and behavior tracking.

Outcome:

The portal achieved 60% onboarding of service partners within the first year—exceeding the 40% adoption target and significantly reducing manual order-handling time. Automated workflows decreased dependency on phone support, improved spare parts discovery, and reduced operational inefficiencies. With a stable architecture and clear usage insights, the client gained a competitive advantage and is now expanding the platform into Germany, Finland, and 26 additional countries.

32- DT Mobile App Development(Mobile Application/UI/UX/QA)

Service- Custom software development and QA

About the client

Our client is a well-known mobile app development company specializing in creating high-quality, feature-rich applications for a diverse range of clients. They sought our expertise to develop a complex mobile app for a high-profile client, aiming for a seamless user experience and tight deadline adherence.

The story

The client approached us with a challenging project that required developing a robust mobile app within a limited timeframe. The app had to meet specific requirements, handle scope creep, undergo rigorous testing, provide an exceptional user experience, and integrate seamlessly with third-party APIs. The app also had to be prepared for submission to app stores with no issues.

Challenges

- Ensuring clear communication to understand the client's requirements.
- Tight deadlines and resource allocation for effective time management.
- Managing scope creep due to evolving client needs.
- Rigorous testing across multiple devices and platforms.
- Balancing aesthetics, usability, and performance.
- Smooth third-party API integration and app store submission.

Solution

We used the Flutter framework, Dart programming language, Riverpod for state management, and Go Router for routing to meet the client's needs. Regular meetings ensured clear communication, while an agile development approach allowed us to manage scope changes. Rigorous testing and QA processes helped deliver a bug-free app, while UI/UX designers crafted a visually appealing, user-friendly interface. We collaborated with API providers for seamless integration and ensured smooth app store submission by addressing all guidelines.

Outcome

The mobile app was successfully delivered, meeting all client requirements, timelines, and scope changes. The app provided a great user experience, was rigorously tested, and integrated smoothly with third-party APIs. It was ready for app store submission, receiving approval without any issues, making the project a complete success.

31- Nokia Jira Integration for Third-Party Issue Automation

Services - Custom Software Development

Integration & Automation

Atlassian Development

About the Client:

The client is a large, multi-department organization using Jira as its central project management system and required seamless integration of diverse third-party issue types into existing workflows.

The Story:

The client relied on Jira across departments such as Asset Management, HR, and Sales, each operating with unique workflows, issue categories, and custom fields. Importing third-party issue types created operational delays and inconsistencies across teams. They engaged FiftyFive Technologies as their software development partner to engineer a custom integration layer within the Atlassian ecosystem. The goal was to streamline ingestion, unify processes, and enhance overall project visibility without disrupting existing workflows.

Challenges:

- Importing heterogeneous third-party issue types into Jira.
- Mapping department-specific workflows and custom fields.
- Normalizing issue categories across Asset Management, HR, and Sales.
- Automating ingestion without breaking existing Jira configurations.

Solution:

- Engineered a custom Jira plugin using Java for backend development.
- Integrated third-party issue ingestion via REST APIs and webhook-based triggers.
- Mapped department-specific workflows and custom fields across Jira Core, Jira Agile, and Jira Service Desk.
- Implemented JQL-based filters and automated rules for issue classification and tracking.
- Connected Confluence and Atlassian Cloud for documentation, process alignment, and cross-team collaboration.
- Deployed a dedicated development team including Atlassian specialists, backend engineers, and QA testers.

The Outcome:

The custom plugin reduced manual issue handling time by streamlining the ingestion of all third-party issue types into Jira. Department-specific workflow mapping improved process consistency, reducing cross-team misalignment by enabling standardized issue structures. Automated API-driven ingestion eliminated repetitive manual tasks, improving operational efficiency across Asset Management, HR, and Sales. Integration with Confluence and Atlassian Cloud strengthened documentation workflows and communication, resulting in faster task resolution and better project visibility. The engagement demonstrated FiftyFive's ability to build scalable, enterprise-grade Atlassian solutions that improve productivity, reduce operational delays, and optimize project management at scale.

30- ZipApply Custom Software Development for Automated Hiring Platforms

FiftyFive built automated data pipelines with HPCC and ECL to streamline hiring, improve job matching, and reduce costs.

Industry - Recruitment technology

Services - Custom Software Development (platform engineering, backend development, job ranking algorithms)

- Data Engineering (ETL automation, distributed data processing, API integrations)
- Quality Assurance (testing, performance validation, reliability checks)

About the Client

The client operates a digital recruitment platform designed to simplify job discovery and candidate selection through data engineering, automation, and quality-of-life-based job ranking.

The Story

The client collaborated with FiftyFive Technologies to build a scalable, data-driven platform that redefined how job seekers and employers connect. The goal was to deliver real-time job insights, automate data extraction, and improve hiring efficiency using advanced engineering and analytics. FiftyFive deployed a dedicated team of data engineers, backend developers, and QA specialists to design and implement the platform's data ecosystem using HPCC Systems and the ECL language.

Challenges

- Extracting large-scale job and candidate data from multiple API sources.
- Automating data extraction and transformation pipelines.
- Delivering faster service response and real-time updates.

- Matching job seekers with positions based on lifestyle and preference parameters.
- Providing hiring managers with deep, structured candidate insights for accurate shortlisting.

The Solution

- Deployed **HPCC Systems** for distributed data processing and scalable architecture.
- Utilized the **ECL language** to build automated data extraction and transformation workflows.
- Developed **Quality of Life update modules** to deliver real-time insights to job seekers.
- Built **Profile Insights dashboards** enabling hiring managers to filter and evaluate candidates efficiently.
- Designed **Custom Job Ranking engines** based on candidate-defined lifestyle parameters.
- Streamlined **end-to-end data pipelines** for speed, accuracy, and reliability across job and profile databases.
- Conducted continuous **testing and optimization cycles** for performance stability and uptime.

The Outcome

The solution enhanced job matching accuracy by 45% and reduced manual data handling by 60%. Hiring managers experienced a 35% faster candidate selection process with improved match quality. Automated API integration ensured real-time updates and reduced operational delays by 50%. By combining HPCC Systems and ECL-driven

automation, the platform achieved higher scalability, faster service delivery, and a measurable improvement in user engagement—transforming how talent acquisition platforms operate.

29- NIBE PCB Identification(Machine Learning)

About the Client

The client is a technology-driven organization seeking to streamline their PCB (Printed Circuit Board) identification process. They required a robust, efficient, and scalable solution to identify PCBs from a dataset with diverse images, varying in angles, scales, and environmental conditions.

The Story

The client needed a machine learning model capable of identifying PCBs within a dataset despite challenging variations. The project aimed to ensure reliable performance for real-world applications while addressing image variability. Real-time or near-real-time predictions were essential to meet their operational demands.

Challenges (Crisp Pointers)

- Handling variability in PCB images due to angles, scales, and environmental conditions.
- Achieving high accuracy for practical applications.
- Optimizing computation time for real-time or near-real-time predictions.

The Solution

The solution incorporated transfer learning using the VGG 16 CNN architecture for feature extraction, leveraging pre-trained weights for accuracy and efficiency. Implemented in Python with Keras, the model used Kubeflow on Google Cloud Platform for orchestration and deployment. Scikit-learn supported data preprocessing and evaluation, while OpenCV handled image manipulation. A user-friendly interface was developed using Flask, ensuring seamless interaction.

The Outcome

The project achieved a remarkable accuracy of over 85%, effectively identifying PCBs across variations in angle, scale, and conditions. Predictions were optimized to less than 2 seconds, meeting the requirement for real-time performance. The project exemplified the power of transfer learning and a modern tech stack in addressing complex image recognition challenges.

28- NIBE Component Classification (Machine Learning)

About the client

Our client sought a solution to efficiently classify components from images, with potential applications across various industries. The objective was to create a system that could handle images of components captured at different angles, scales, and backgrounds, and provide recommendations for similar components from a dataset.

The story

The project aimed to build an advanced component classification system using object detection. The challenge was to ensure high accuracy despite the variability in images, including different angles, scales, and artificial backgrounds. Achieving over 87% accuracy was key to the success of the project, with the system also recommending similar components from the dataset.

Challenges

- Handling image variations (angle, scale, and background).
- Ensuring robust performance with artificial images.
- Achieving high classification accuracy (above 87%).

The solution

A comprehensive tech stack was employed to develop the system. Python, Keras, and TensorFlow were used for deep learning, while SSD Mobilenet CNN provided the object detection framework. Kubeflow and GCP enabled scalable cloud-based deployment and training. scikit-learn was integrated for machine learning tasks, and Flask and OpenCV were used to build user-friendly interfaces and handle image processing.

The outcome

The solution successfully met the objectives by accurately classifying components, even under varying conditions. The model achieved an accuracy rate of over 87%, demonstrating its robustness to different image conditions and artificial backgrounds. This project highlighted the effectiveness of deep learning and a well-integrated tech stack in solving complex challenges, with scalable applications across industries.

27- Winnio NIBE IoT and Edge Engineering for HVAC Optimization

FiftyFive integrated IoT sensors and edge-based ML to automate HVAC control, enabling real-time adjustments and reducing energy costs.

Industry - IoT

Service-

- Custom Software Development
- AI/ML Engineering

- IoT Engineering
- Cloud & Architecture
- Quality Assurance

About the Client:

The client is an organization focused on modernizing large-scale building climate systems, seeking to implement IoT, edge computing, and data-driven automation to reduce energy costs and improve occupant comfort.

The Story:

The client engaged FiftyFive Technologies to transform its traditional HVAC infrastructure into an intelligent, fully automated, and energy-efficient system. The objective was to redesign climate control using IoT sensors, edge-based machine learning, and real-time device-level decision-making. FiftyFive deployed a dedicated engineering team covering IoT development, ML engineering, backend APIs, cloud architecture, and QA. The engagement focused on full-cycle delivery—from architecture, model deployment, and integration to testing and optimization.

Challenges:

- Static HVAC logic lacking data-driven and adaptive control mechanisms.
- No device-level capability for real-time temperature and humidity adjustments.
- High operational energy consumption due to inefficient system behaviour.
- Limited visibility into room-level environmental patterns and optimization opportunities.
- Need for edge-ready ML models capable of running directly on HVAC-connected devices.

Solution:

- Engineered predictive temperature and humidity models and optimized them for edge deployment using Ekkono's ML framework.
- Integrated IoT sensors to stream real-time room-level climate and device data for continuous monitoring.
- Converted ML outputs into device-level control logic enabling autonomous HVAC adjustments with minimal latency.
- Developed backend services and data pipelines to process environmental data, trigger optimizations, and maintain system scalability.
- Performed Exploratory Data Analysis (EDA) to identify actionable operational patterns and refine HVAC decision rules.
- Conducted iterative QA and system validation to ensure reliability across diverse building conditions.

Outcome:

FiftyFive delivered a fully automated, IoT-enabled HVAC optimization system capable of real-time climate control. The edge-powered decision models reduced energy consumption by enabling targeted room-level adjustments, resulting in significant operational cost savings. Occupant comfort improved through stable temperature and humidity regulation, increasing productivity and overall facility satisfaction. Continuous data insights supported ongoing optimization, giving the client a scalable and future-ready climate control foundation while demonstrating clear ROI through reduced energy expenditure and higher system efficiency.

25- Connectel AI-Powered Customer Service Solutions

The client needed AI to improve customer service; FiftyFive built and deployed custom solutions that reduced call abandonment and improved efficiency.

Industry - Service provider / Customer service operations

Services - Custom Software Development (tailored AI-driven tools)

- AI/ML Engineering (queue analysis, weighted scoring models, NLP-powered reporting)
- Cloud Engineering (AWS-based automation pipelines)
- Data Visualization & BI (real-time dashboards)
- Automation (email workflows, alerts)

About the Client

The client, a leading global service provider, manages large-scale customer interactions and required AI-powered optimization to improve efficiency, agent performance, and customer satisfaction across multiple service operations.

The Story

The client partnered with FiftyFive Technologies to modernize customer service operations through advanced AI, automation, and data-driven insights. Our role was to design, develop, and deploy a tailored solution integrating queue analysis, NLP-powered reporting, and performance monitoring. A dedicated engineering team specialized in custom software development, AI/ML, and cloud deployment ensured the solution was robust, scalable, and business-driven.

Challenges

- Data variability across multiple queues impacting benchmark consistency.
- Handling time differences caused by varying query complexity.
- Lack of unified performance metrics for actionable recommendations.

- Manual reporting delays reducing operational visibility for administrators.

The Solution

- Built adaptive queue analysis algorithms dynamically adjusting benchmarks using evolving datasets.
- Implemented weighted scoring models ensuring fair evaluation across diverse agent query workloads.
- Integrated OpenAI-driven natural language processing for automated, detailed performance reports.
- Deployed visualization dashboards using BI tools for real-time monitoring of agent metrics.
- Configured cloud-based automation pipelines (AWS) for secure, scalable processing and reporting.
- Automated email workflows delivering concise metric updates directly to administrators.
- Dedicated development team deployed (AI/ML engineers, cloud specialists, and software developers) to ensure scalability and continuous optimization.

The Outcome

The client achieved a 35% reduction in abandoned call rates and a 28% improvement in agent efficiency. Handling times decreased significantly, improving first-response rates and boosting overall customer satisfaction scores. Real-time dashboards and automated reporting shortened decision-making cycles by 40%, while AI-driven alerts enabled proactive training interventions, reducing performance gaps among agents. These measurable improvements directly translated into higher customer retention, stronger brand goodwill, and cost savings through streamlined operations. The engagement highlights how FiftyFive delivers custom AI-powered solutions that transform service delivery while driving long-term business value.

22- LoopKitchen Data Engineering Solutions for Restaurant Operations

Industry - Food Technology / Restaurant Operations Platform

Service- Custom Software Development (frontend & backend engineering)

- Data Engineering (pipeline design & ingestion)
- Workflow Automation (dispute management, task orchestration)
- Business Intelligence & Analytics (Looker Studio dashboards)
- Cloud/DevOps Support (monitoring, deployment optimization)

FiftyFive enabled a food tech client to streamline onboarding, automate workflows, and improve efficiency with measurable business results.

About the client

A US-based food tech startup, recognized among the Top 10 AI startups in 2022, offering a co-pilot platform to streamline restaurant operations and revenue optimization.

The story

The client aimed to transform restaurant management by simplifying onboarding, handling large-scale multi-platform data, automating dispute resolution, and delivering actionable insights. They partnered with FiftyFive Technologies to engineer a scalable solution, deploying a dedicated six-member team comprising frontend developers, backend engineers, data engineers, and business analysts. The engagement focused on accelerating onboarding, automating workflows, and enabling data-driven decision-making through modern engineering practices.

The challenge

- Complex onboarding of new delivery platforms delaying restaurant adoption.

- High-volume data ingestion from multiple sources requiring scalable pipelines.
- Manual resolution of order disputes creating operational inefficiencies.
- Fragmented workflows across platforms reducing process visibility.
- Limited analytics restricting data-driven decision-making for restaurateurs.
- Continuous need to monitor evolving consumer trends and food preferences.

The solution

- Built CSV-based onboarding pipeline for seamless, error-free store integration.
- Designed scalable data pipelines with Python and SQL for ingestion and processing.
- Automated order dispute submission and tracking with integrated workflow management.
- Deployed Looker Studio dashboards for advanced BI and real-time analytics.
- Orchestrated cohesive workflows across platforms using Linear for task management.
- Optimized development lifecycle with BQ Console and VS Code for monitoring and deployment.

The Outcome

FiftyFive accelerated onboarding by 60%, reducing time-to-market for new restaurants. Automated workflows decreased manual dispute handling effort by 70%, directly improving customer satisfaction. Enhanced data ingestion pipelines and Looker

dashboards increased reporting accuracy and speed by 55%, empowering restaurateurs with actionable insights. Platform-wide workflow integration improved overall operational efficiency by 40%, boosting user adoption. This engagement demonstrates how FiftyFive's expertise in custom software development, data engineering, and workflow automation delivers measurable business impact for global food technology enterprises.

20- PolarBears Data Engineering and BI Development for Content Platforms

FiftyFive used data engineering and BI to resolve stagnant engagement, implement scalable solutions, and deliver measurable revenue growth.

Industry Online content / Digital media / Entertainment platform

Services - Data Engineering

- Business Intelligence & Analytics
- Cloud Engineering

About the Client

The client is a global online content platform with a large subscriber base, operating across multiple regions and seeking to optimize engagement, viewership, and revenue in competitive digital markets.

The Story

The client engaged FiftyFive to address stagnating engagement and revenue for their YouTube 'Official Channel' in the UK and US markets. Despite a loyal audience, growth had plateaued, traffic sources were declining, and revenue targets were not met. FiftyFive deployed its data engineering and software development expertise to build scalable analytics pipelines, optimize content strategies, and enhance decision-making through advanced BI solutions.

Challenges

- Stagnant viewership and flat growth in key geographies (UK, US).
- Declining traffic from YouTube Browse Feature.
- Plateaued UK views (4.14% growth) and modest US watch-time (7.12%).
- Below-target revenue growth: UK 23.11%, US 17.19%.
- Lack of scalable infrastructure for traffic spikes and demand surges.
- Limited actionable insights from fragmented historical datasets.

The Solution

- Implemented **data engineering pipelines** for large-scale data collection, cleaning, and preprocessing.
- Designed and optimized **DAX queries** to enable advanced BI reporting and real-time performance dashboards.
- Integrated **Power BI** for visual analytics and executive-level reporting.
- Deployed **statistical analysis and predictive modeling** to identify content demand and forecast viewership.

- Built **personalized recommendation algorithms** based on behavioral data.
- Applied **adaptive pricing models** with dynamic ad placements and subscription strategies.
- Delivered **cloud-based scalable infrastructure** to handle demand spikes and ensure uptime.
- Assembled a cross-functional team of **data engineers, BI developers, and cloud specialists** to deliver end-to-end execution.
- Conducted **stakeholder workshops** with automated reporting and actionable insights delivery.

The Outcome

The engagement delivered measurable growth across both markets. In the UK, viewership increased by 14.23%, adding 4,979,788 new views, while revenue rose by \$13,703.37. In the US, watch-time surged by 15.87% (87,324,537 minutes) and revenue increased by \$26,323.92. Beyond financial impact, the client gained a scalable analytics ecosystem, improved personalization, and stronger decision-making capability—showcasing how FiftyFive’s data engineering and software development expertise directly drives growth and operational efficiency.

19- MyTripDesk Travel Platform Development for Agents and Consumers

FiftyFive built dual travel apps to digitize agent–consumer interactions, streamline offer sharing, and improve booking efficiency.

Industry - Travel technology

Services -

- Custom Software Development
- Cloud & Backend Engineering
- Product Consulting
- UI/UX Design
- Quality Assurance & Testing
- Team Deployment

About the Client

The client is a fast-growing global travel platform providing digital booking services, multilingual travel content, and agent-led travel assistance to users across India and international markets.

The Story

The client wanted to modernize the traditional travel agent ecosystem, where agents lacked a digital interface to showcase offers or engage consumers. They engaged FiftyFive Technologies to design, architect, and build two applications, one for travel agents and another for consumers—to streamline communication, centralize offer sharing, and create a scalable digital infrastructure. Our engineering team handled full-cycle development, from product consulting and UI/UX to backend, mobile apps, cloud setup, and QA. The goal was to deliver a unified travel-tech solution that could support high traffic, multilingual features, and multi-currency travel services.

Challenges

- No centralized digital interface for listing, updating, or managing agent travel offers.
- Lack of real-time communication channels for agent–consumer interaction.

- No invite-based onboarding, deep linking, or location-aware agent discovery.
- Absence of scalable backend services for handling multilingual and multi-currency data.
- No unified workflow for group chats, offer publishing, or agent availability tracking.

The Solution

- Developed Flutter-based mobile applications for Agents and Consumers with shared component architecture.
- Implemented Firebase Authentication, Firestore, and Cloud Messaging for secure access, real-time sync, and event-based notifications.
- Added deep linking for agent invite flows, group chat joining, and agent–consumer connection.
- Built a Node.js backend with scalable APIs for offer management, agent discovery, and chat workflows.
- Integrated location tracking, multi-language content support, and role-based access across the platform.
- Executed full-cycle delivery with dedicated Flutter developers, Node.js engineers, QA testers, and cloud specialists.

The Outcome

The platform enabled travel agents to digitize service offerings and reach a wider customer base, increasing consumer engagement by improving discovery and communication workflows. Real-time offer updates reduced manual coordination and accelerated booking decisions. The multilingual, multi-currency architecture expanded accessibility for global travelers while strengthening the client’s digital presence. By

modernizing offline travel planning, the platform helped the client boost agent adoption, increase operational efficiency, and create a scalable revenue-enabling travel ecosystem.

16- AddEq_t End-to-End FinTech Platform Development and Modernization

FiftyFive Technologies enhanced a Swedish FinTech platform by improving scalability, integrating new services, and reducing defects through full-cycle development and QA.

Industry - FinTech

Services - Main: Custom Software Development, QA & Testing

Subcategories:

- Backend Development
- Frontend Development
- API Integration
- QA Automation
- Cloud Deployment & Configuration
- Documentation & Maintenance

About the Client

The client is a Swedish FinTech company providing secure, high-performance digital investment solutions for Private Banking, Wealth Advisory, and Asset Management across the Nordic financial ecosystem.

The Story

The client partnered with FiftyFive Technologies to modernize their investment platform, aiming to improve user experience, strengthen system performance, and accelerate feature deployment. The engagement involved enhancing the web application, integrating backend services, and implementing QA automation to ensure high availability and compliance. A dedicated FiftyFive team worked alongside the client's internal tech lead under an agile delivery model to execute full-cycle development and testing.

Challenges

- Legacy codebase requiring reverse-engineering and documentation restructuring.
- Complex backend built on PHP, Node.js, TypeScript, Python, and MariaDB with SOAP APIs hosted on Google Cloud.
- Inconsistent frontend architecture using React, Redux, and Webpack.
- Gaps in integration between CRM and investment management modules.
- Missing services for user notifications and NordNet brokerage linking.
- Limited QA coverage and undefined validation workflows.

Solution

- Refactored backend logic for modularity, scalability, and efficient API communication.
- Enhanced frontend with React and Redux for dynamic data rendering and localization.
- Developed and deployed Notification Service and NordNet Linking Service to streamline communication and brokerage synchronization.

- Integrated CRM with COB system for unified client data management.
- Implemented automated QA test cycles covering regression, validation, and performance.
- Configured third-party tools for seamless local integration and updated technical documentation for long-term maintainability.

The Outcome

The upgraded platform achieved faster response times, improved data synchronization, and 99.9% system uptime. User onboarding speed increased by 35%, and post-launch defect rates dropped by 40%. The collaboration enhanced operational efficiency, reduced maintenance overhead, and enabled faster rollout of new financial products—strengthening client trust and end-user satisfaction. The project reflects FiftyFive’s expertise in delivering scalable, secure, and compliant FinTech software solutions through dedicated development and QA teams.

15- AutoVisa Automated Visa Processing Platform Development

A LegalTech company automated manual visa checks with a scalable platform built by FiftyFive, enabling real-time tracking, document processing, and timely deadline alerts.

Industry - LegalTech / Immigration Automation

Services -

- Custom Software Development
- Automation & Data Extraction
- Cloud Engineering
- Quality Assurance
- Team Extension

About the client

The client is a USA-based LegalTech company building automated immigration solutions

for individuals, employers, and attorneys, aiming to streamline visa monitoring, document processing, and deadline tracking at national scale.

The story

The client relied on slow, manual visa status checks that required over 30 days and lacked automated deadline alerts, creating operational delays and legal risks. They partnered with FiftyFive Technologies to build a fully automated, high-scale system capable of real-time extraction from government portals. Our role was to deliver end-to-end custom software development, cloud engineering, automation pipelines, and a dedicated engineering team. The objective was to replace manual workflows with a reliable, event-driven platform that seamlessly integrated with existing immigration processes.

Challenges

- Manual visa status retrieval requiring repeated navigation of government systems.
- No automated alerts for visa, document, and compliance deadlines.
- Need for high-throughput architecture supporting millions of concurrent users.
- Real-time integration stability across government platforms.
- Lack of scalable workflows for document extraction and structured data processing.

The solution

- Developed a responsive web application using React Native Web for unified UX.
- Built backend services in Golang and Python for high-performance processing.
- Implemented Redis caching and PostgreSQL for structured, scalable data storage.
- Integrated AWS Textract for automated document scanning and data extraction.

- Used Selenium automation for real-time status collection from government portals.
- Designed event-driven microservices to support high system throughput and reliability.
- Implemented subscription-based status monitoring, auto-scan modules, and automated email alerts.
- Deployed a dedicated engineering team covering frontend, backend, cloud, and QA.

The outcome

The platform eliminated manual visa checks, reducing processing time from 30+ days to near real-time automation. Attorneys and users received timely email alerts, improving compliance accuracy and reducing missed deadlines. Workflow efficiency increased significantly, lowering operational overhead and enabling higher user throughput for the client's LegalTech business. Delivered in seven months, the solution strengthened the client's product offering, improved customer experience, and supported scalable growth across their immigration services.

14- Apica Upgrading .NET to Vue 3 (Migration/.NET)

Swift Vue 3 Migration

About the Client

The client, a technology-driven enterprise, aimed to modernize its legacy platform built on the .NET framework. To future-proof its application, the company decided to migrate to Vue.js, leveraging Vue's seamless integration with .NET for improved performance and scalability.

The Story

The client initially adopted Vue 2 for application views but faced obsolescence when Vue 3 was released. To stay updated and leverage the latest features, they decided to migrate the entire platform from Vue 2 to Vue 3. The process, however, posed significant technical challenges due to breaking changes, compatibility issues, and the complexity of refactoring a large-scale application.

Challenges

- Migrating over 20 screens and hundreds of components from Vue 2 to Vue 3.
- Addressing compatibility issues and ensuring seamless functionality.
- Adapting the existing codebase to Vue 3's syntax and features.
- Performing rigorous testing to maintain stability and prevent disruptions.
- Executing the transition within a tight deadline.

The Solution

The development team executed a structured migration plan, systematically upgrading the codebase to Vue 3. They utilized Vue 3's advanced features while maintaining compatibility with existing components. Technologies such as Vuex, Storybook, and Jest were deployed for frontend optimization, while Java, Spring Boot, OPA/OPAL, KeyCloak, and Postgres ensured backend stability. Rigorous testing was conducted at each stage to ensure a seamless transition without compromising performance.

The Outcome

Within a month, the migration was successfully completed, modernizing over 20 screens and hundreds of components. The transition to Vue 3 enhanced the platform's longevity, security, and performance, enabling future UI enhancements. With a unified technology stack, the client significantly reduced downtime, minimized revenue loss, and improved overall operational efficiency.

11- LiveLike GetSet **Multi-Organization CMS and Web Platform Development**

A unified CMS and web platform built to manage multi-organization content, automate data workflows, and deliver scalable user engagement.

Industry -

Services - Custom Software Development

Web Application Development

CMS Development

DevOps

Quality Assurance (QA)

About the Client

The client builds interactive digital experiences for live events and operates a web-based learning and engagement platform for older adults, requiring scalable, secure, and user-friendly software systems to support diverse audiences.

The Story

The client engaged FiftyFive Technologies to build a unified digital ecosystem combining a CMS, a multi-organization web application, real-time analytics, and interactive engagement modules. They needed a platform that could scale with growing users and data volume while remaining accessible to older adults. FiftyFive provided full-cycle custom software development, DevOps, and QA, delivering an enterprise-grade system aligned with the client's technical and operational goals.

Challenges

- Integrating CMS, web application, data pipelines, and analytics services into a single, cohesive architecture.
- Ensuring horizontal scalability for rising traffic, data imports, and organization-specific workloads.
- Designing an accessible interface optimized for older adults across varied devices.
- Managing continuous stakeholder feedback while maintaining sprint velocity and delivery timelines.

The Solution

- Engineered a modular CMS using Strapi to support multi-organization content creation and structured data management.
- Developed a React and TypeScript web application with organization-specific routing and unified CMS integration.
- Implemented bulk data import workflows using Node.js, Postgres, and MongoDB for large-scale dataset processing.
- Integrated real-time analytics pipelines to track engagement patterns and application usage.
- Added widget-based interactive components embedded directly within the application interface.
- Containerized services using Docker and established CI/CD pipelines for automated deployment and environment consistency.
- Delivered the project through a dedicated engineering team including frontend developers, backend developers, QA engineers, a Scrum Master, and a DevOps specialist.

The Outcome

The platform improved operational efficiency by streamlining content management across multiple organizations and reduced manual workflows through automated imports and analytics. Scalability enhancements supported growing traffic without performance degradation, while optimized UX significantly improved adoption among older adult users. The client achieved faster content rollout cycles, stronger engagement insights, and a more reliable infrastructure. FiftyFive continues to provide post-deployment support, ensuring ongoing stability and enabling the client to scale their platform confidently.

10- Connectel Contact Center Platform Development for Real-Time Customer Service Optimization

We built a custom contact center platform that enabled real-time monitoring, streamlined workflows, and helped the client handle over 85% of calls promptly.

Industry - Customer Service Technology / Contact Center Solutions

Services - Custom Software Development

Backend Engineering

Frontend Engineering

Cloud Engineering

DevOps

About the Client

The client is a Sweden-based customer service technology company founded in 2008, focused on modernizing support operations through digital platforms that enhance agent performance, process efficiency, and real-time service delivery.

The Story

The client aimed to build a modern Contact Center as a Service (CCaaS) web application to streamline customer support workflows and improve responsiveness. They engaged

FiftyFive Technologies as their dedicated software engineering partner to design, architect, and develop a scalable platform capable of real-time monitoring, intelligent queue handling, and data-driven reporting. FiftyFive delivered full-cycle custom software development covering backend engineering, frontend development, cloud deployment, DevOps integration, and performance optimization ensuring the client retained full ownership of the product while relying on our engineering expertise.

Challenges

- Need for real-time agent performance monitoring with live data streams
- Requirement for analytical dashboards with processed metrics
- Complex organization and user management workflows
- Rating model for agent evaluation and quality scoring
- High-load backend requirements with responsive UI performance
- Queue optimization to reduce latency during peak request volumes

The Solution

- Engineered a full-stack web application using Node.js, React.js, and AWS services
- Built a real-time dashboard powered by RabbitMQ for live agent activity tracking
- Developed an Admin Portal for multi-level user, notification, and B2B purchase management
- Integrated a reporting module to generate structured, data-driven performance reports
- Implemented a secure chat application for internal communication workflows

- Added an FAQ module enabling self-service support across internal and external users
- Optimized queuing logic using message queues and cloud-native scaling configurations

The Outcome

The platform significantly improved operational efficiency and service responsiveness. Real-time monitoring increased agent productivity and reduced queue delays, enabling the client to respond promptly to over 85% of incoming calls. Processing speed improved across workflows, administrative tasks became more automated, and system stability increased under load. By delivering a fully custom, scalable CCaaS solution, FiftyFive strengthened the client's service reliability, enhanced customer experience, and positioned them for sustained operational growth.

9- NIBE Smart Home Appliance Management System

The client needed unified appliance control, and FiftyFive built a mobile and metaverse platform that centralized device management.

Industry - Smart Home / IoT

Services -

- Custom Software Development
- IoT Integration
- UI/UX

- Metaverse & 3D Experience Development

About the Client

The client builds smart home management solutions, enabling homeowners to control IoT-enabled household appliances through unified digital interfaces, with a focus on automation, connectivity, and future-ready interaction experiences

The Story

The client sought a centralized way to manage multiple smart home appliances that previously required separate apps, remotes, and workflows. They partnered with us to engineer a unified mobile control system supported by a scalable backend architecture. After delivering the core mobile platform, we extended the engagement by integrating the solution into the metaverse using Unreal Engine and Meta Quest 2 to create an immersive interface for appliance interaction. This project demonstrated how full-cycle custom software development, IoT integration, and 3D experience engineering can modernize home automation.

Challenges

- Fragmented appliance ecosystems with limited interoperability across brands and protocols.
- No unified control layer for real-time device commands and status monitoring.
- Inefficient user workflows due to switching between multiple interfaces and control systems.
- Absence of immersive or advanced interaction models for smart home management.

The Solution

- Engineered a unified mobile application consolidating all appliance commands into a single interface.

- Built backend services to enable consistent device connectivity, control routing, and data synchronization.
- Implemented structured UI/UX flows to streamline multi-device management and reduce operational friction.
- Integrated the solution into the metaverse using Unreal Engine for interactive 3D appliance control.
- Added support for Meta Quest 2 to enable VR-based home environment navigation and device operation.

The Outcome

The centralized platform reduced multi-app dependency by 100%, enabling homeowners to manage all appliances through one unified experience. Daily control workflows were simplified, lowering user effort and improving operational efficiency. The metaverse integration introduced a next-generation interaction layer that strengthened the client's product innovation capability and market differentiation. By delivering an engineered, scalable, and future-ready system, the project improved user engagement, enhanced product competitiveness, and positioned the client to expand into advanced IoT and immersive-tech markets.

8- LiveLike Multi-Network Access Control (Web3)

Service- Token Gating and NFT minting

Transforming live events with token gating and NFT capabilities across blockchain networks

"Our client specializes in creating interactive, immersive experiences for live events, integrating features like polls, chats, VR/AR elements, and social media to foster community engagement during broadcasts."

The Story

This project aimed to enhance our client's service by implementing token gating and NFT minting with custom metadata, including airdropping NFT collections. The goal was to create a robust system for controlling access to services and minting NFTs across multiple blockchain networks, making digital experiences more secure and engaging.

Challenges

- **Contract Address Validation:** Ensuring contract addresses existed on selected networks
- **Token Ownership Verification:** Verifying user access for NFTs with specific attributes
- **NFT Minting Configuration:** Dynamically configuring network types and contract instances
- **Multi-NFT Minting Functionality:** Addressing the absence of bulk minting capabilities
- **Testing Limitations:** Navigating scarcity of test network coins, deprecated test networks, and high costs of using real coins

The Solution

Our solution implemented contract address validation through Web3, preventing the creation of chat rooms with invalid contracts. We enhanced token ownership verification by adding Enumerable support to the smart contract, streamlining token ID retrieval. For NFT minting, we crafted a provider using ethers for network switching and contract creation. We developed a custom "bulkMintNfts" function for efficient mass minting with custom metadata, and implemented off-chain collection creation to mitigate gas fees. Finally, we integrated a custom event filter to retrieve specific collection token IDs.

The Outcome

The project successfully delivered secure token gating and efficient NFT minting capabilities across multiple networks. This implementation ensured robust access control for service owners while enhancing user engagement through NFT ownership. The solution's resilient design overcame testing challenges, resulting in a cost-effective, production-ready system that significantly elevated our client's blockchain capabilities and user experience value proposition.

6- Quickbit Custom Crypto Transaction Platform Development

FiftyFive built secure affiliate and merchant crypto payment systems, integrating AML workflows and third-party services to improve transactions and operational efficiency.

Industry - FinTech

Services - Custom Software Development

- Blockchain Engineering
- System Architecture
- Integrations
- QA & Testing
- DevOps / Deployment
- UI/UX

About the Client

The client is a Sweden-based financial services company offering secure cryptocurrency payment solutions for e-merchants and consumers, focused on scalable blockchain-driven transactions across the Nordic market.

The Story

The client partnered with FiftyFive to build a secure, scalable cryptocurrency payment platform comprising two core components: an Affiliate Flow system for user onboarding

and trading, and a Merchant Payment system for transaction execution. FiftyFive provided full-cycle custom software development, system architecture, integrations, QA, and optimization. A dedicated engineering team delivered backend, frontend, and DevOps capabilities to build a compliant, high-performing fintech solution.

Challenges

- Developing a secure cryptocurrency transaction engine supporting multi-flow processing.
- Integrating Affiliate Flow and Merchant Payment modules with shared logic and independent scaling.
- Implementing AML/KYC verification workflows with robust authentication logic.
- Enabling seamless integration with external financial and operational third-party services.
- Designing a modular, scalable architecture supporting white-label deployment.
- Ensuring responsive UI, low-latency transactions, and stable merchant/admin back-office operations.

The Solution

- Built Affiliate Flow system using PHP, Laravel, and MySQL with modular service components.
- Developed Merchant Payment system using Python, Django, and React with API-driven architecture.
- Implemented AML/KYC workflows with secure profile verification and compliance logic.
- Integrated required third-party services for identity checks, payment routing, and operational workflows.

- Added white-label capabilities, including casino-focused configurations and customizable merchant flows.
- Applied test-driven development, code optimization, and secure deployment practices for performance stability.
- Deployed a dedicated engineering team including backend developers, frontend developers, QA engineers, and a project lead.

The Outcome

FiftyFive delivered a fully functional, secure, and scalable crypto payment platform that improved transaction speed by enabling optimized payment routing, reduced verification time through automated AML/KYC workflows, and strengthened platform reliability with modular architectures. Merchant onboarding became faster, operational overhead decreased, and both admin and merchant teams gained better visibility through optimized dashboards. The platform's white-label model expanded revenue opportunities, while improved system performance and security enhanced the client's market credibility and customer trust.

3- LiveLike Event Platforms with Web3 Cross Platform Compatibility (Web3)

About the Client

The client is a leader in creating interactive and immersive experiences for live events and broadcasts. They focus on enhancing user engagement by integrating innovative technologies, including blockchain, to create more dynamic experiences for their audiences.

The Story

The client aimed to revolutionize their event platform by leveraging Web3 technologies to boost user engagement. Their vision was to integrate blockchain support, introduce

token gating, and enable NFT features like minting and airdrops, enhancing the overall event experience and ensuring a secure, scalable system.

Challenges

- Integrating Web3 technologies (blockchain, token gating, NFT minting) into an existing platform.
- Ensuring scalability to handle increased user engagement.
- Addressing security concerns, including profile verification and anti-money laundering.
- Maintaining a user-friendly experience while adding new features.
- Ensuring cross-platform compatibility (Android, iOS, and Web).

Solution

Our team expanded blockchain network support to include Polygon, Chiliz Chain, Hedera Network, and Ethereum. We integrated flexible token-gating options and added features like polls, quizzes, and quests for increased interactivity. NFT support was expanded with metadata and multi-network minting, and we developed an efficient token airdrop system. We ensured all features were compatible across Android, iOS, and Web platforms, delivering a smooth, immersive experience.

Outcome

The platform now supports multiple blockchain networks, enhancing user access and engagement. Security features, including profile verification and AML protocols, were successfully integrated. Cross-platform compatibility was ensured, leading to a more dynamic, interactive user experience during live events. The client expressed high satisfaction, with positive feedback from stakeholders, as the platform successfully transformed the live event experience.

2- Auto Slider AI Presentation Tool Development

The client needed faster presentation creation, FiftyFive built an AI-powered web app that automated slides, scaled globally, and cut creation time by 70%.

Industry - Presentation technology / Digital tools

Service - Custom Software Development (web app, frontend/backend)

- AI/ML Engineering (content generation with NLP)
- Cloud Engineering (deployment, storage, scalability)
- Software Testing & QA (validation, CI/CD pipelines)

About the Client

The client is an entrepreneur in the presentation technology sector, focused on building scalable, user-friendly digital tools for students, professionals, and business owners worldwide.

The Story

The client envisioned a platform to simplify presentation creation by combining automation with customization. They partnered with FiftyFive Technologies as their software development provider, leveraging a dedicated engineering team covering backend, AI/ML, frontend, and cloud deployment. The goal was to design and deliver a scalable, AI-powered web application that could generate professional presentations quickly, reduce manual effort, and ensure reliability for a growing user base.

Challenges

- Automating content generation with natural language processing while supporting flexible customization
- Designing scalable backend architecture with efficient storage and retrieval
- Managing presentation file storage and secure access at scale
- Integrating external data and content sources into generated presentations
- Delivering an intuitive frontend for diverse user segments
- Ensuring deployment, monitoring, and scalability of infrastructure

The Solution

- Backend built with FastAPI for high-performance APIs
- OpenAI + LangChain integrated for AI-driven content creation
- python-pptx applied for automated slide generation and formatting
- MongoDB implemented for structured user and presentation data storage
- Google Cloud Storage deployed for file management and access control
- Google Cloud Run for scalable, containerized deployment
- React frontend developed with signup, creation flow, and admin dashboards
- Agile methodology with iterative sprints, CI/CD pipelines, and QA validation

The Outcome

The platform reduced presentation creation time by over 70%, enabling users to generate customized, professional slides in minutes. It successfully scaled to support thousands of users, with infrastructure designed for global growth. The intuitive interface drove strong adoption, while positive feedback boosted the client's credibility in the presentation technology space. This engagement highlights FiftyFive Technologies' expertise in delivering custom software development, AI/ML engineering, and cloud solutions, translating complex ideas into scalable digital products that accelerate business growth.

1- Run Viable **Building Custom AI-Powered Finance Assistants**

AI finance assistant built to handle multi-source financial data, automate queries with GPT integration, and deliver faster, accurate insights for the client.

Industry - FinTech

Services - Custom Software Development

- AI/ML Integration
- Backend Engineering
- Cloud Deployment
- Data Engineering

About the Client

The client provides intelligent finance tools for consumer brands, offering Connected P&L, cash flow insights, projections, and a conversational AI interface for financial data across e-commerce ecosystems such as Shopify and Amazon.

The Story

The client needed an AI-powered finance assistant that allowed users to query complex financial datasets through natural language. They engaged FiftyFive Technologies to architect a scalable backend, integrate GPT-based intelligence, and enable secure access to multi-source financial data. The goal was to improve user interaction, streamline data retrieval, and support the platform's funding and product growth plans. FiftyFive deployed a dedicated engineering team to deliver full-cycle development—from backend design to deployment.

Challenges

- Designing a flexible database schema supporting multi-source financial inputs.
- Implementing automated ETL pipelines for data cleaning and transformation.
- Integrating ChatGPT API with structured query execution and permission enforcement.

- Building concurrent request handling with low-latency response formatting.
- Implementing API-level error recovery, rate-limit management, and security controls.
- Ensuring backend–frontend alignment for seamless conversational workflows.

The Solution

- Developed a Python + FastAPI backend optimized for scalable, concurrent processing.
- Integrated OpenAI GPT-3 for natural language interpretation and financial query mapping.
- Built a secure AI Assistant API with structured output, access control, and audit layers.
- Deployed on Heroku with environment-specific configurations for high availability.
- Implemented ETL pipelines to normalize financial data for model consumption.
- Added monitoring, exception handling, and performance tuning for stable production usage.
- Provided a dedicated team of backend engineers, AI engineers, and QA specialists for end-to-end delivery.

The Outcome

The AI Assistant enabled users to retrieve financial insights up to 60% faster, improving overall product usability and reducing manual data navigation time. The enhanced functionality strengthened the client’s investor presentations, contributing to successful funding discussions and increased platform credibility. Post-launch engagement rose significantly, with users reporting smoother query handling and improved trust in financial data accuracy. The project also boosted the client’s market perception, earning

industry recognition and positive social media traction, demonstrating the value and impact of FiftyFive's custom AI engineering capabilities.